V1.8

v1.8

ClassTechStar

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1. (0-7)

0:

? ( )

“ ” “ ”

“ ”

“ ”

:

- [ ]: [ ], [ ]
- >: ” [ ] > [ ]”
- : !
- :

iOS

: ( ) App (iOS), ( ), ( ) (iOS)

: 128G 110G?

: 1TB = 1024GB, 1GB = 1024MB, 1MB = 1024KB... :
1000 1024 “128G” “ ”

3 :

1. : 10-20GB
2. :
3. :

128GB 256GB 512GB

: ” ”

[ ] ( )

“ ”

:

QR ...

?

—

3 :

- :
- : /
- : + + ” ”

: “ ”

“ ”

:

- : —” ”
- : —
- : ” ” —

1:

“ ”

“ ” :

SoC (System on Chip): “ + ”

SoC System on Chip “ ” CPU ( ) CPU, GPU ( ), NPU ( ), ISP ( ), ( )

CPU:

CPU SoC CPU ARM x86
ARM (RISC)
CPU - : CPU “ - ” (big.LITTLE ARM)
:

- / :
- :
- / :

: —

:

- **Qualcomm Snapdragon 8 Elite**: CPU Oryon (Nuvia) 2 (4.32GHz) + 6 (3.53GHz) TSMC 3nm 2 ARM single-core multi-core GPU Adreno

830 sliced 40% NPU Hexagon 45 TOPS
AI 45% (: Samsung Galaxy 4.47GHz)

- **MediaTek Dimensity 9400**: 1 (Cortex-X925, 3.62GHz) + 3 (Cortex-X4, 3.3GHz) + 4 (Cortex-A720, 2.4GHz) All-Big-Core
  TSMC 3nm 2 GPU Immortalis-G925 (12) 41% Ray Tracing 40%
  AI NPU 890 8 Large Model LoRA

- **Apple A18 Pro**: 2 (~4.04GHz) + 4 (~2.4GHz)
  TSMC 3nm 2 GPU 6 Ray Tracing NPU 16 (35 TOPS)
  AI Apple Intelligence Apple
  single-core Unified Memory AI

- **Huawei Kirin 9020**: Mate 70 Series CPU Taishan NPU
  ISP 5.5G

GPU:

GPU (Graphics Processing Unit) CPU
( )
:

1. : 3D
2. **Rasterization**: 3D 2D
3. **Pixel Shading**:
4. :

GPU : ! AR GPU
GPU

GPU : Qualcomm Adreno GPU ; MediaTek ARM Mali GPU; Apple
GPU

NPU: AI

NPU (Neural Processing Unit) CPU GPU
NPU AI

: NPU —
NPU CPU

AI:

1. :

2. :

3. : NPU forward propagation

4. : AI

:

- : HDR AI
- :
- : AI
- : (OCR)
- **Large Model** : AI Large Model (7B) AI Snapdragon 8 Elite NPU Hexagon 45 TOPS Dimensity 9400 NPU 890 Large Model (LLM 80%) Apple A18 Pro NPU 16 35 TOPS
- **AI**: AI Eraser ( ) AI Expand ( ) AI Style Transfer

ISP:

ISP

? ! Apple A18 Pro, Qualcomm Snapdragon 8 Elite, MediaTek Dimensity 9400, Huawei Kirin 9020 Snapdragon 8s Gen3/7+ Gen3 Dimensity 8400 ( ) 3nm 5nm 4nm

RAM:

RAM 12GB 16GB 24GB

RAM : LPDDR5 LPDDR4X RAM “ ”

RAM ? WeChat

(ROM/Flash):

UFS 3.1 UFS 4.0 eMMC

?

:

- : OLED ( ) LCD
  OLED
- : 1080P 2K ” ” :
- : 60Hz 90Hz/120Hz
  (LTPO)
- :
- : DC PWM PWM

HDR

HDR (High Dynamic Range)

- : (SDR) HDR (1,000)
- : HDR10 ( ), HDR10+ ( ), Dolby Vision
  ( )
- : HDR HDR HDR

HDR

HDR HDR HDR “ ”

- : ( )
- : ( ) /
- : ”HDR ” HDR Smart
  HDR HDR

:

“ ” 2-3 5500-7500mAh

- : 100W 200W 20-30W ?

- :
- :

:

3-4 :

- :
- :
- :
- :

! ( ) (f/1.8 f/2.4) (OIS)
(AI)

90
5 10

- : 90
- : (5x-10x)
- :

OIS vs EIS

:

- **OIS** ( ):
- **EIS** ( ):
  ( )
- : OIS
  (Sensor-Shift)

vs

- : (2x, 3x, 5x, 10x)
- : + 100
- :

:

vs :

- : -
- ( ):
- : ” ”

:

:

- : 30fps 60fps 120fps GPU SoC
- : AI
- : SoC

GPS:

GPS

- **GPS** : L1 3-5
- **GPS** : L1 L5 L5 1-3
- : GPS BeiDou ( ) GLONASS ( ) Galileo ( )

:

:

- + :
- :
- :
- :
- : Face ID 3D 2D
- **NFC**:
- :
- :

: Android VS iOS VS HarmonyOS

:

- :

- **iOS**:

- :

:

Micro USB

- : ( )
- : USB 2.0 480Mbps 10W
- :

Lightning ( )

- : 8 iPhone 14
- : Lightning USB 2.0 480Mbps USB-C Lightning Lightning
- : 2024 USB-C Type-C iPhone 15 Lightning
- : MFi iPhone Lightning

USB Type-C

- : 24 2024 USB-C
- : Type-C USB 2.0 (480Mbps), USB 3.0/3.1/3.2 (5-20Gbps) USB4 (40Gbps)
- : Type-C USB 3.0! Type-C USB 2.0 iPhone 16/17 USB 2.0 Pro USB 3.0 (USB-C to USB-C)
- : USB PD 240W
- : DisplayPort Alt Mode

3.5mm

- : 3 4 3 4
- :
- : 3.5mm Type-C

:

- **Apple**:iPhone ( ) / iPhone Plus ( ) / iPhone Pro ( ) / iPhone Pro Max ( ) / iPhone SE ( ) / iPhone 16e (AI A18 C1 Apple) 2026 iPhone 17 A19 Pro AI

- **Huawei**: Mate ( ) / P/Pura ( ) / Mate X ( ) / nova ( ) / Enjoy ( )
- **Xiaomi**: Xiaomi ( ) / Xiaomi MIX (/) / Redmi K ( ) / Redmi Note ( ) / Redmi ( )
- **OPPO**: Find X ( ) / Reno ( ) / A ( ) / OnePlus ( )
- **vivo**: X ( ) / iQOO ( ) / S ( ) / Y ( )
- **Samsung**: Galaxy S ( ) / Galaxy Z Fold/Flip ( ) / Galaxy A ( )

:

- **Pro**: Professional
- **Max**: Pro
- **Ultra**: Samsung S24 Ultra, Xiaomi 14 Ultra
- **SE**: Special Edition iPhone SE
- **Plus**:
- **Lite/** :
- **Note**: Redmi Note
- **Neo**:

:

:

Qualcomm MediaTek
C1 iPhone 16e
C1 4nm iPhone 16/16 Plus/16 Pro/16 Pro Max Qualcomm (Snapdragon X75 5G) iPhone XS/11 Intel

:

, :

- **MIMO**: 4×4 MIMO 2×2 MIMO

- : (Wi-Fi NFC GPS)
- : ( )

:

- **4G LTE**: 1Gbps / 500Mbps (Cat.18) 50-150Mbps
- **5G NR**: 10Gbps+ Sub-6GHz mmWave
- **5G-A(5G Advanced)**: 5G 5.5G 10Gbps 1Gbps 2024 China Mobile, China Telecom China Unicom 5G-A Huawei Mate 70/80 , Xiaomi 15/16 5G-A 5G-A " "
  ( )

:

:

- **FDD-LTE Band(B)**: B1, B3, B5 4G B1/B3/B5/B8/B34/B38/B39/B40/B41
- **TDD-LTE Band**: 4G B38/B39/B40/B41
- **5G NR Band(N)**: n1, n3, n28, n41, n78, n79 n78 (3.5GHz) 5G n41 (2.5GHz) China Mobile
- :

?

:

1. : SMS WeChat ……
2. : ……
3. : ……
4. : ……
5. : 3A ……

6. : ……

7. : ……

8. : ……

9. : ……

10. : ……

!

:

[ ] RAM ROM “AnTuTu” “DevCheck”

:

“12GB+8GB RAM” RAM 8GB ?

? “ ”

“ ” (, , RAM Plus) **RAM**

: (ROM/UFS)

(RAM)

: RAM “ ” “ ”

2 :

zRAM

RAM 2:1 RAM

CPU

swap

UFS RAM

“12GB+8GB” RAM 12GB + 8GB zRAM

zRAM

UFS 4.0 4200MB/s 1/10 - 1/15
RAM LPDDR5X zRAM RAM

:

- : RAM 6GB + UFS 3.1 ≥30GB 3GB 3-4
  68%
- : WeChat
- : 4K
- : 8GB + UFS 2.2
- : 16GB 1 CPU 12% 2-3℃
  RAM 12GB

: Xiaomi “ ” Huawei “ ” Samsung “RAM Plus” vivo “Extended RAM” OnePlus “Memory Extension”

: [ ][ ] [ ] [ ] “ ” (2GB-7GB) “8GB+4GB”

- 20-30GB
- UFS
- RAM (4GB)

-
RAM “ ” RAM

:

1. [ ] ” ” ” ”RAM Plus” ”Extended RAM”
2.
3. : RAM ? ? ?

4.

1 :

HarmonyOS *2019* “ ”

HarmonyOS ? ?

HarmonyOS “ ” HarmonyOS

HarmonyOS “ ”
“ ”

: HarmonyOS “ ”

HarmonyOS 4.3: AOSP

HarmonyOS 4.3 Huawei HarmonyOS 4.3
(AOSP) —App AppGallery
Huawei HarmonyOS HarmonyOS 4.3
Ark Huawei AOSP

:

- : AOSP HarmonyOS 4.3
- **Ark**: Huawei
- : HarmonyOS

- : iOS
- : AppGallery
- :

:

- :
- :

- :
- :

HarmonyOS NEXT (HarmonyOS 6): Huawei

HarmonyOS NEXT (HarmonyOS NEXT 5.0) HarmonyOS 4.3
— HarmonyOS Huawei Linux AOSP
HarmonyOS “ ” “ ”

:

- : HarmonyOS NEXT Huawei AOSP
- : HarmonyOS NEXT HarmonyOS APK HarmonyOS
- : Huawei HarmonyOS WeChat Alipay Taobao HarmonyOS
- **Star Shield**: ” ” —
- **AI** : AI AI AI AI
- :
- **Meta Service ( )**:

HarmonyOS NEXT

HarmonyOS NEXT : (1) HarmonyOS (2) HarmonyOS 4.3

:

- :
- :
- **MeeTime**: Huawei

HarmonyOS:

, 1

:

- : ,
- :
- ( ):
- :
- : HarmonyOS
- :

:

AppGallery HarmonyOS Huawei HarmonyOS 2026 WeChat, Alipay, Taobao, Douyin, Meituan, JD HarmonyOS

HarmonyOS

- : ” ”
- : / / ” ”
- :
- :
- : >

HarmonyOS HarmonyOS 4.3 6

: HarmonyOS

1. ” ” HarmonyOS
2. ” ”:
3.
4. >

2: —

:

1. :
2. **SD** ( ): : SD
3. : + iCloud, Baidu Netdisk, OneDrive

“File Manager”

[ ] [ ] iOS [ ] “ ”

:

- **DCIM**:
- **Pictures**: ( )
- **Download**:
- **Music**:
- **Movies**:
- **Documents**:
- **Android**:

:

Linux

:

- **/system**: root
- **/data**:
- **/sdcard /storage/emulated/0**:

App :

- :
- :

:

,

:

1. : iCloud, Huawei Cloud, Xiaomi Cloud
2. :
3. : + ,

?

- :
- :
- **WeChat**: WeChat
- :

:

- : ” ” ” ”
- :
- :

?

- : >
- : ” ”
- :
- :
- : WeChat

:

JPG/JPEG			/
PNG			/
MP3		JPG	
FLAC			/
WAV		CD MP3 3-5	
M4A		AAC MP3	Apple Music/
MP4			/
MKV			VLC/nPlayer
DOC/DOCX		Word DOCX DOC	WPS/Office
PPT/PPTX		PowerPoint PPTX	WPS/Office
XLS/XLSX		Excel XLSX	WPS/Office
PDF			Adobe Reader/WPS
ZIP			/
RAR		ZIP	
ISO			/
HTML			

: ” ”

> “ ” ? (“ ”“ / ”
)

:

“ ”

.

- ” + ”: 202606_ ,2025_ ,2024_

- :
  - : (/ / /)
  - : (/ /)
  - : (”2026Q2_ A”)
- : (1) ” ” :

:

1. 2026Q2_ A
2. ” ”
3. 2025 ,2026

“ ”

- **72** : App WeChat QQ (/)
  [WeChat] > [ ] > [ ] > [ ] ” ” 72
- : [ ] > [ ] > [ ] iOS [ ] > [ ] > [iPhone]
  ” / / / ” : (1) (2) ( )
  (3) ( )
- : : ” > ”
  ” ” ” ” .

“ ” __ “ ”

:

/

1. ( ): (Meitu, Snapseed)
2. : ” ” ” ” ” ” GPS
3. ( ): ” XX ”

(EXIF) (6 “ ”):

- : iOS [ ] > [ ] > [ ] ” ” ” ” **EXIF** .
- : ” ” ” ”
- : ExifTool ( ) Metapho (iOS) Photo Metadata Remover ( )

( ) :

- iPhone: [ ] > > (Face ID)
- Huawei/Xiaomi: [ ] > > ” ”
- :**360** ,**Tencent Album Manager** ( ),**KeepSafe**

3: “ ” —

“ ”

— Wi-Fi “ ” “ ”

Wi-Fi : **Wi-Fi**
— “ ”

:

“ ” “ ” Wi-Fi ( )

: vs

Wi-Fi **(LAN)**
“ ” ( ) 4G/5G

: vs

Wi-Fi ( )

:

[ ] > [ ] > [ ]

: “ ”

“ ”
: HTTPS ( ) **HTTP**
WeChat HTTPS —
SSL

(“Free_WiFi_2024”)

: **“ ”**

? Huawei WLAN

iPhone : Wi-Fi Wi-Fi
“5G” “4G”

(4G/5G)

2G 5G

:

- **4G LTE**: (100Mbps+)
- **5G**: (10Gbps)
- **5G SA vs NSA**: SA , NSA 4G SA

:

- :
- : ” ” Wi-Fi
- ≠ :

: [ ] **!**
! /

Wi-Fi:

Wi-Fi Wi-Fi

Wi-Fi:

[ ] > [WLAN]

:

iOS iOS

Wi-Fi:

Wi-Fi IEEE :

Wi-Fi 4	802.11n	2.4/5GHz	600Mbps	
Wi-Fi 5	802.11ac	5GHz	3.5Gbps	
Wi-Fi 6	802.11ax	2.4/5GHz	9.6Gbps	
Wi-Fi 6E	802.11ax	2.4/5/6GHz	9.6Gbps	6GHz
Wi-Fi 7	802.11be	2.4/5/6GHz	46Gbps	

- **Wi-Fi 6**: OFDMA ( ) MU-MIMO ( ) TWT ( )
- **Wi-Fi 6E**: Wi-Fi 6 6GHz
- **Wi-Fi 7**: MLO (Multi-Link Operation) 320MHz Snapdragon 8 Elite Gen5, Dimensity 9400 Wi-Fi 7 Xiaomi 17, Huawei Mate 80, Samsung S26 Wi-Fi 7 (Wi-Fi 7)
- **2.4GHz vs 5GHz**: 2.4GHz 5GHz 6GHz Wi-Fi 6E/7

:

:

-
- VPN
- Wi-Fi
- HTTPS ( )

:

—

:

2.4GHz ISM (FHSS) 79
: Class 1 100 , Class 2 10 ( ), Class 3 1

:

- **4.0**: (BLE)
- **4.2**: IPv6
- **5.0**: (2Mbps) (300)
  ( )
- **5.1**: (AoA/AoD)
- **5.2**: LE Audio LC3 ( )
- **5.3/5.4**:

LE Audio: 2025-2026 LE Audio
(Huawei FreeBuds Pro 5, Xiaomi Buds 6 Pro, Samsung Galaxy Buds3 Pro) LE Audio LC3 LE Audio

:

- : LE Audio 5.2
- : Wi-Fi Direct
- :
- : (Beacon) (AirTag)
- : Mesh

:

(Xiaomi Huawei)

:

(38kHz)
NEC, RC5, RC6, SONY

:

- :
- : 5-8
- : iPhone

NFC:

NFC (Near Field Communication,) (10)
13.56MHz 424Kbps

:

NFC RFID :

- : NFC NFC
- : NFC Android Beam ( )
- : —

:

- : 10
- : (SE) HCE ( ) root

- / :
- **(Tokenization)**: Apple Pay, Huawei Pay

:

- / :
- : Apple Pay, Huawei Pay
- : NFC
- : NFC

: Wi-Fi

- : > /
- :
- **USB**:

:

1. [ ] ” ” ” ”
2. ( ) Wi-Fi
3.

4: — “ ”

App

: “ ”

- : Google Play ( ) (AppGallery Huawei, Xiaomi)
- **iOS**: App Store
- **HarmonyOS**: AppGallery Huawei

: : **!**

/ : “ ”

(Huawei, Xiaomi, OPPO, vivo, Honor) .

?

(“ ”) **App** : **App**

- : ” ” ” ”
- :
- : —

(Huawei/Xiaomi)

- **Huawei/HarmonyOS**: [ ] > [ ] >[ ] ” ”
- **Xiaomi**: [ ] > [ ] >[ ] [ ] > [ ] > [ ] ” ”

- **OPPO/vivo**: [ ] > [ ] >[ ]

“ ” :

- :
- : ” ” — ?
- : ” ”
- :

“ ” : !

: “ ” .

“ ” ?

“ ” — “ ” “ ” :

- ” ” ” ” ” ” ”
- ” ” ” ” ” ”
- ” ” ” ”

SMS —

:

- : (Phone Manager Huawei, Security Xiaomi)
- : SMS — ” ”
- : — ” ”

:

1.
2.
3.

: “ ”

- : ” ” ” ”
- : **!**
- : [ ] > [ ]

: **“** **”**

( ) ( ) ( )

- : ” ” ” ” ” ” ”
- :
  1. :
  2. : ” ”
  3. :

- / SMS:
- :
- / :
- :

?

APK

APK (Android Package Kit) ZIP :

- **AndroidManifest.xml**: ” ” ” ” Activity
- **classes.dex**: Java/Kotlin Dalvik dex (classes2.dex, classes3.dex)
- **resources.arsc**: ID
- **res/**: drawable (drawable-hdpi, drawable-xhdpi)
- **assets/**: AssetManager

- **lib/**: .so (C/C++) CPU (arm64-v8a, armeabi-v7a)
- **META-INF/**: MANIFEST.MF, CERT.SF, CERT.RSA APK

IPA iOS

IPA (iOS App Store Package) :

- **Payload/**: .app
- **Info.plist**: AndroidManifest.xml ID
- : .app Mach-O
- **Frameworks/**:
- : .app
- **_CodeSignature/**:

?

- : APK APK ( ) Google Play App Signing Google
- **iOS**: IPA iOS App Store
- :
  1. :
  2. :
  3. :

:

- : APK/IPA
- :
- :

:

- **App Bundle ( )**: Google Play
  ( )
- **App Thinning (iOS)**:
- :

: ?

3D

: 3

3 ( ) [ ]
“ ”

5: —

“ ”

( ) ( )

- :
- :
- :
- :

- :
- / :
- : (OLED)
- : (120Hz)
- : 15 30
- : ?

( )

: iOS

: “ ”

“ ”

/ :

- : ” ” ” ”
- :
  : • : ” ”
  • :
  • / / :
- : + / iOS ” ”

:

- : 90%
- : (SMS) ” ”
- :

:

- : + (iOS /)
- / : ” ”
- : ” ” /
- **(iPhone)**: ” ” ” ” [ ] > [ ] > [ ] > [ ]
- : Siri / Xiaoai / Xiaoyi / Xiaobo

:

- :
  : • Sogou / Baidu:
  • iFlytek:
  • Gboard (Google):
  • iOS:
- :
- : ” ” ” ”

:

- / **(PiP)**:
- / : WeChat / Douyin ” ”

- : Dock ( ) 4

” ”

10 :
(1) + (2) + (3) / /
(4) / / (5) / (6)

: “ ”

“ ” —

...

:

“ ” —

VoiceOver iOS TalkBack

iOS VoiceOver ( )

: [ ] > [ ] > [VoiceOver] ( )
:

1. : :
2. : /
3. : (” ”)
4. :
5. : ( )

: [VoiceOver] > [ ] (50% 70-80%) ,
iOS “ ” “ ”

TalkBack

: [ ] > [ ] > [TalkBack] - 3 ( )
:

1. : :
2. :
3. / :
4. : ( )

5. : /

: [TalkBack] > [ ] , , (Google Text-to-Speech)

—

- **iOS**: [ ] > [ ] > [ ] ( )
  - (1.5x 15x)
- : [ ] > [ ] > [ ] (8)

/

- **iOS**: [ ] > [ ] > [ ] ” ” ” ” ” ” ” ” ” ”
- : [ ] > [ ] > [ ] ” ” Huawei/Xiaomi ” ” ” ”

( )

- **iOS**: [ ] > [ ] > [ ] > [ ] / / /
  ” ” ” ” ( )
- : [ ] > [ ] > [ ] + ” ”

AI : AI “ ”

“ ” AI

AI

- **iOS 15+**: [ ] > [ ] > [VoiceOver] > [ ] ” ”
  AI (iPhone 12)
- **TalkBack** : Google TalkBack ” ” AI
  (TalkBack)
- : ”Seeing AI” (iOS) ”Lookout” ( ) ”Xiaoyi Bangbang” (Huawei)
  AI

“ ” AI — “ ” .

“ ” —

(Live Captions)

iOS: [ ] > [ ] > [ ] (FaceTime)

12+: “ ” [ ] > [ ] > [ ]
:

1. :
2.
3. ” ”
4. ;
5.

: +

( ) “ ”

(TTY / RTT)

iOS: [ ] > [ ] > [RTT/TTY]

: [ ] > [ ] > [ ] + [ ] “ ”
“ ”

:

- **RTT (Real-Time Text)**:
- **TTY ( )**: TTY RTT
- WeChat/QQ : ” ”

“ ”

- (iOS 14+): [ ] > [ ] > [ ] > [ ] /
- **AssistiveTouch /** : iOS [ ] > [ ] > [AssistiveTouch] [ ] > [ ]
- **Switch Control**: iOS [ ] > [Switch Control] [ ] > [ ]

- (iOS): [ ] > [ ] ” ”
  +
- : iOS [ ] > [ ] > [LED] [ ] > [ ] > [ ] .
- : [ ] > [ ] > [ ] > [ ]

iOS

30

:

1. **VoiceOver / TalkBack** WeChat
2.
3. 200%
4. ” ”

6: —

= +

:

:

1. / **PIN**: (/)
2. :
3. : (3D 2D)
4. :

:

- (123456)
- ( )
- ” ”

“ ” :

! :

- :
- :
- :
- : ( )

[ ] “ iPhone” “ ” ( )

:

(iOS 17 iOS 18)

AI “ ” Deepfake AI “ ”
“ ”

AI

(1) AI

Deepfake “ / ”
.

(2) /

AI

(3) AI

AI “ ” “ ”
.

— “ ”

AI :

- : ” ” ” ” 5 ”
AI
- : ” ” ” ”
.
- : ” ” —
.

” 24 ”

( ):

- / :
- :
- : (AI)

:

-
- OTP
- SMS

:

! “ ” 5

“ ” :

“ ” ( )

EXIF 2 “ ” **(EXIF)**
GPS

- iOS: [ ] > [ ] > [ ] ” ” ” ”
- : (Google)
- : ExifTool, Metapho EXIF

4 “ ” “ ”
“ ”

- ” ” : ?
-

IoT

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2022 ( )
= .

:

“ ”

“ ” **:**

- : / ” ” ” ”
- : (WeChat)
- / : iOS 14+ ” ” 13+
- / : ? ” / ”
- : iOS 14.5+ IDFA ” ”

:

- **(iOS)**: [ ] > [ ] > [ ]
- ( ): [ ] > [ ] > [ ]
- : ” ” ” ” ” ”

“ ” /

:

1. : OTP
2. : ” ” (iCloud /)
3. : ( )
4. : IMEI
5. :
6. ( ) :

eSIM

- **PIN** : PIN ” ”
- **eSIM**:
- : ” ” —
  ” ”

- / 123456 / qwerty
- OTP (” ”)
- SMS 00
-
- ” ” ” ”

” ”

1. ” ”
2. (iCloud.com)
3. ” ” ” ” ” ”

7:

StarFlash

:

:

:

- **Apple**:iPhone 14 SOS iPhone 17 iMessage SMS
- : Mate 50 BeiDou ( ) Mate 60 Pro Tiantong Mate 70
- **Xiaomi/OPPO/vivo**: 2024 BeiDou
- : BeiDou

:

- :
- :
- :

:

1. :

2. : ( )

3. :

4. :

:

“ ” Wi-Fi

:

: ( )

:

- :
- :
- :

:

- **Apple**: ”NameDrop” ” ” iOS 17
- **Huawei**: ChangLian
- **Xiaomi**: Xiaomi (Mi Smart Hub)

:

:

:

1. :

2. :

3. :

:

- **Apple**: iPhone 14
- **Android**:

:

- :
- : ( )
- : 100%

StarFlash (SparkLink):

StarFlash (SparkLink) Huawei Wi-Fi
: 900Mbps (50Mbps) 20 μ s (15ms)
4096 (8)

:

- : 20 (0.02ms)
- : 900Mbps lossless
- :
- + : 50%

:

- :
- :
- :
- :

:

Huawei Huawei Mate 60 Huawei

StarFlash StarFlash

:

1. : ” ”
2. : ” ” ”SOS ”
3. : Huawei StarFlash
4. :

2. (8-14)

8:

( ):

- :
- :
- : ” ”

” ”

- :

- : Android ” ” iOS Bluetooth

-
- ( )
- ” ”

:

- : ” ” ” ”
- : Android > > iOS > > 13 ” ”

- : Tencent Mobile Manager, 360 Mobile Guard Tapir (Tapir) (China Mobile/China Unicom/China Telecom) ” ” ” ” (Huawei, Xiaomi, OPPO, vivo)
- : ” ” ” ” ” : Tapir: (haoma.tk.cn)
- 360: ” ” 360 Mobile Guard
- Tencent: ” ” ”Tencent Mobile Manager”
- : China Mobile ”KTFSR” 10086 China Telecom ”Tianyi Anti-Harassment” China Unicom ”Unicom Mobile Manager”

: ” ” (China Mobile) AI

: / :

/ :

: ” ” ” ” ” ” iOS :

:

: National Anti-Fraud Center Qingzhuo /

13

VoLTE/VoNR :

- VoLTE (Voice over LTE): 4G 2G/3G
- VoNR (Voice over NR): 5G VoLTE 5G + 5G + 5G
- : iOS ”VoLTE” ” > > > ” Android ”VoLTE ” ” > > > ” VoLTE

VoLTE/VoNR 2G/3G

“ ”

iMessage/RCS

SMS

- iMessage: Apple
- RCS: Google iMessage WeChat
- iMessage RCS: iOS 18 Apple RCS iPhone Android RCS ” ” RCS

RCS :

- China Mobile: RCS ” 5G”
- China Telecom: ”China Telecom 5G Message”
- China Unicom: 5G Android ” ” ” 5G”
- : RCS WeChat WeChat RCS ” ” ( )

- : ” ” ” ” ” ” ” ” ” - - ” =
- : ”ETC ” ” ” ” ”
- : 95 1069 + ” ” 11
- :

WeChat/QQ:

WeChat QQ “ ”

:

1. WeChat (PC Mac)
2. > [ ] > [ ]
3. ” ”
4. : WeChat [ ] > [ ] > [ ] > [ ] > [ ] > [/]

:

- : WeChat [ ] > [ ] > [ ] > [ ]
- : > ”•••” > [ ] > [ ]
- / : ” ” [ ]
- : [ ] > [ ] > [ ] ” ” WiFi
- : [ ] > [ ]
- : WeChat ” ” GB 30%-50%

:

- : / ”•••” > [ ]
- : ” ”
- : ”# ”
- : ”+” > [ ]
- : > [ ] ” ” @
- / @ : 500 @

Moments :

- : ” ” ” ” ” ” (” ” ” ” ” ”)
- : [ ] > [ ] > [ ] > [Moments] > [Moments] ”3 /1 /6 / ” ”6 ”
- Moments : Moments > [Moments /] [ ]
- Moments: [ ] > [ ] > [WeChat] > [ ] > [ ]
- : [ ] > [Moments] > [Moments 10]

- WeChat : 20 (” ”)
- FaceTime: Apple iOS 15 Android/Web ” ” Apple
- Zoom: 100 40 /
- Tencent Meeting: WeChat 300 ( )
- : WeChat Apple FaceTime Zoom Tencent Meeting (Tencent Zoom)

:

- WeChat: 500Kbps-1Mbps 3-5MB 8-12MB
- FaceTime: 1Mbps, 720p 2Mbps, 1080p 4Mbps 5-15MB
- Zoom/Tencent Meeting: 720p 1.5-3Mbps, 1080p 3-5Mbps 300-800MB

- : WiFi (” ”)

:

- : ( )
- :
- : ” ”
- : WiFi 4G/5G
- : Tencent Meeting Zoom AI
  ” ”

“ ” “Tencent Meeting”

:

- : iOS > > > Android ” ”
- :

IMAP vs POP3 vs Exchange:

- POP3:
- IMAP: (/ /) —
- Exchange (ActiveSync): Microsoft Outlook, NetEase Enterprise Mail Tencent Enterprise Mail
- : IMAP Exchange IMAP POP3

:

1. iOS: [ ] > [ ] > [ ] > [ ] > ”Microsoft Exchange” ” ”
2. Android: ” ” Outlook ” ”
3. (Exchange): ”mail. .com” ”outlook.office365.com”
4. ” ” : (IMAP) 993 SSL (SMTP) 465 587 TLS
5. ” ” (Push) [ ] ” ”

“ ” “ IMAP/SMTP”

: WeChat

1. : WeChat > > [ ] > [ ]
2. : ” ”
3.

9:

“ ”

:

:

- **WPS Office**: Microsoft (docx/xlsx/pptx) PDF
” ” — WPS
- **Microsoft 365**: Office WPS
OneDrive
- **Shimo Docs / Tencent Docs**: ” ”
Shimo

: — OneDrive WPS Shimo
“ ”

:

— “ ” “ ”

- — : Notes iOS
- — : ” ” WeChat ” ”
” ” Cubox Pocket
- **—Notion Feishu**: Notion — ” ”
” ” Feishu ” ”
Notion

: “ ” — ( ) WeChat ( )
Notion ( )

OCR:

- **iPhone**: Notes iOS Notes ” ”
— ” ” ” ”
- : ” ” Microsoft Lens (OneDrive
)
- **OCR**: OCR ” ” WeChat —
Baimiao CamScanner OCR Baidu
WPS
- :
PDF

:

“ ”

- **iCloud**: :
Windows 5GB
- **Windows OneDrive**: Windows + Office 365 OneDrive Word
1TB Office
- **Baidu Netdisk**: Baidu Netdisk
Baidu Netdisk
- **Aliyun Drive / Quark Netdisk**: Aliyun Drive ( )
Quark Netdisk Quark

: iCloud ( ) Baidu Netdisk ( )
OneDrive (Office)

:

1. WeChat (+)
2. OneDrive WPS
3. ” PDF” WPS WeChat
4. WPS PDF
5. ” ” WPS
6. PDF Baidu Netdisk iCloud Notion

40

:

1. (WPS Microsoft 365)
2. PDF
3. (” Wi-Fi”)
4. Notion Feishu ” ”

10:

“ ” “ *?”* *:*

?

Alipay WeChat :

1. ” ” ”Wallet” ” ”
2.
3. SMS 6
4.

: II III

“ ”

- : + ” ”
  .
- :

“ ”

5

- / : Alipay WeChat

- : Alipay WeChat ” ” ”Wallet” : Alipay
- : ” ” Alipay WeChat ” / ”
- : : Alipay

:

:

- **Wallet** : Apple Wallet, Huawei Wallet, Xiaomi Wallet :
- **Alipay/WeChat**: ” ” :

Wallet 10-20

- : ” ” Alipay WeChat
- : Meituan Ele.me ”Pinhaofan” Meituan
- : Didi Amap ” ” ” ” ” ” ” ”

- : ” ” Alipay WeChat
- : ”Zhangshang 12333” ” ” Alipay
- : ” ” (
)

- ” ”
-
- SMS ” ETC ” ” ”
-

:

1. Alipay WeChat /
2. ” ”
3. ” / ”
4. ” ”
5.

11:

?

:

- :
- :
- : LED
- : (0.5x/1x/2x/3x)
  0.5x ( ) 1x ( ) 2x/3x ” ” —
  3x

:

- + : ” ” ( ) ” ”
- ” ”: — —
- :
- : ” ”
- : —
  2 4

- : ( ) AI —
  AI 1-2
  AI ” ”

- : 100% : ISO

:

- **Snapseed**: Google : ” ”
  ” ” ” ” ” ” ” ” Snapseed

- **Xingtu / Meitu**: : 30%

- **Lightroom** : HSL

- : iPad

?

— :

- : iPhone ” ” ” ” ” ” ”
  AI ” ” ” ”

- : iOS ” ”
  ”screenshots” GB

- : ” ” ” ” ” ”
  ” ”

- : —
  iCloud Baidu Netdisk/Aliyun Drive :

- : +
  ( ) — ” ”
- : ” ” —

: + +

1. ” ” ( )
2. Snapseed ( )
3. ” ”
4.

11 :

:

(Resolution):

- **720P (HD)**: 1280×720
- **1080P (Full HD)**: 1920×1080
- **2K (QHD)**: 2560×1440 1080P
- **4K (UHD)**: 3840×2160
- **8K**: 7680×4320

(Frame Rate):

(fps)

- **24fps**: ” ”
- **30fps**:

- **60fps**:
- **120fps/240fps**:

(Bitrate):

(Mbps)

-
-
-

: H.264, H.265 ProRes

H.264 (AVC):

- :
- :
- :
- :

H.265 (HEVC):

- : H.264 50%
- :
- :
- : 4K

Apple ProRes:

- : Apple
- :
- :
- :

ProRes RAW:

- : ProRes RAW
- **RAW** :
- :
- :
- : iPhone 15 Pro Apple ProRes (4K 60fps) iPhone 17 Pro ProRes RAW

HDR:

HDR (High Dynamic Range)

- **HDR10**: HDR
- **HDR10+**: Samsung
- **Dolby Vision**: HDR Dolby
- **HLG**:

HDR HDR

(Color Gamut):

- **sRGB**:
- **Display P3**: Apple
- **Rec.2020**: HDR

(Bit Depth):

- **8-bit**: 16.7
- **10-bit**: 1,070
- **12-bit**: 68,000

- : : 1080P/4K + 30fps + H.265
- : : 1080P/4K + 60fps
- : : 4K + ProRes ( )

- : :

:

1. :
2. :
3. : HDR, ProRes
4. : , ” ” ” ” , ,

RAW HEIF/JPEG

RAW :

RAW “ ” “ ”
JPEG HEIF
“ ” RAW

RAW HEIF JPEG:

- **JPEG**: ( ) 8-bit (2-5MB)
- **HEIF/HEIC**: Apple JPEG (1-3MB)
  10-bit Windows
- **RAW**: (20-100MB) 12-16bit
  (Lightroom, Snapseed)

RAW:

- **(White Balance)** : JPEG (
) RAW
- : JPEG 2
  RAW ” ” 3-4
- : RAW JPEG
- : ” ” RAW

RAW :

- **iPhone ProRAW**: Pro iPhone 12 Pro ” ” ”Apple ProRAW” ”RAW” 25MB/
- **Android Motion Raw / Ultra RAW**: Xiaomi/OPPO/vivo RAW ” ” ” ” Huawei RAW ” ” Samsung Galaxy S RAW ”Pro” 20-50MB
- : RAW (, ,)

“ ”

:

- : 1-3
- **HDR ( )**: 3 ( )
- **Deep Fusion (Apple) / Photonic Engine ( )**:

:

AI “ ”

- :
- :
- / :
- : AI

Deep Learning:

- **AI**: ” ” ” ” ” ” ” ”
- **AI**: Depth of Field DSLR Deep Learning
- **AI /AI** : AI ” ” Generative AI

vs :

- : ” ”
- : ” ” — ” ”
- : + Artifact ” ”
- : flagship ” + ” —

flagship 3-4

:

- **(13-24mm)**: ( ) ” ” ” ”
- **(24-28mm)**: ”1x”
- **(70-200mm)**: ( )
- **Periscope (200mm)**: flagship

:

“ ” “ ” —

- : ( )
- :
  Noise
- **flagship** : 1/1.3 1 (1) 50
  —

Pixel Binning:

“Pixel Binning” —

- **4 (Quad Bayer)**: 50 12.5 4
  4
- **9 (Nonacell)**: 100 11-12
- : — ( )
  ” ”

:

- —
- —
- ” ” ” ”
- —

Slow Motion Time-lapse:

- : 120fps 240fps
- **Time-lapse**:
- : Time-lapse 15-30 Time-lapse 10

/ :

-
-
- ( )
- ” ” iPhone (30); ” ” ” ”

Macro:

- 1-2
- Macro (2 5) ; Macro
-
-

- : HEIF/JPEG
- ( ): RAW
- :
- :
- (1)

:

1. RAW ( ) JPEG RAW
2. 3
3.
4. Panorama

12:

MP3 *Kindle*

:

:

- **NetEase Cloud Music**: ” ” ” ”
- **QQ Music**: Jay Chou Mayday QQ Music
  ” ” QQ Music
- **Apple Music**: Dolby Atmos
  NetEase
- **Spotify**: ”Discover Weekly” ” ”

: “ ” “ ”
“ Wi-Fi”

: “ ”

“ ”—

- **Xiaoyuzhou**: ” ” ” ”
  ”Huzuohuyou” ( ) ”Buheshiyi” ( )
- **Apple Podcasts**: iPhone CarPlay
- : 1.2 1.5 1.2 1.5
  ” ” ” ”

:

- **VLC**: —MKV, AVI, FLAC, ISO VLC .rt VLC
- **nPlayer**: — DTS Dolby
- **Bilibili (B)**: Bilibili
- : Wi-Fi ”TV” iPhone AirPlay

:

—

- **WeChat Reading**: ” ” ” ” ” ” ” ”
- **Kindle**: Kindle Kindle Kindle PDF Kindle
- **Apple Books**: iPhone EPUB PDF ” Books”
- : ( ) ” ” —

:

:

- / : — /
- : ( ) 60 30 120 ( ) —

- : : 70%
  ( ) ( )
- : ” ” iOS ”Digital Wellbeing”
- : (Tencent START, NetEase Cloud Gaming)
  RPG
  FPS

:

1. 3
2. Xiaoyuzhou Apple Podcasts 3
3. VLC
4. WeChat Reading
5. ( )

13:

:

Qingzhuo *Sunlogin/ToDesk* *AI*
Trae *LAN* *LocalSend* *Aegis/Microsoft*
Authenticator *TickTick*

:

“ ”

( )

- TickTick : :
/ / /Web + + ” 3 ”
” ”
” + + ”
- TickTick : : ” / ”
- :
- :
- : ” ”
- : ” ...”

Todoist : : ( ) karma

Microsoft To Do : Outlook : ” ”
Microsoft
/ : : Siri/Xiao Ai

( )

- Forest : : ” ”
- Tide : / : ( ) + +
- : 25 + 5 50/10 90/20

( )

- : : Siri/Xiao Ai
- Google Calendar : : ” ” / Gmail

( )

- Notion : wiki :
- Obsidian : : Markdown + ” ”
- flomo : : Moments #
- Apple Notes : iPhone/Mac : iCloud
- Evernote : : OCR

PDF

- PDF Expert : iPad/iPhone : PDF iOS /
- WPS : Word/Excel/PPT/PDF : / WeChat
- Adobe Acrobat : PDF ( ) : PDF/A

( )

... :

PDF/ OCR

- CamScanner : : OCR PDF
- Microsoft Lens : / Word : OneNote/Word
- Apple Notes : iPhone :
- Baimiao / CamScanner / ” ”: Huawei, Xiaomi, OPPO, vivo ” / ” Android Google Camera
- : PDF ” ”

“ ” —

- Keep : : HIIT AI
- Nike Training Club (NTC) : : Nike /
- Apple Fitness : Apple Watch : (/ /)
- AutoSleep : Apple Watch / :
- Sleep Cycle : : /
- Mi Fitness : Mi Band/ Xiaomi : Mi Band / /

- Calm : : ” ” + ( )
- Tide : / : + +
- Xiao Sleep : / : (+ +)
- Flo : : AI /
- Meiyou : : / / /
- Medication Assistant : :
- Apple Health/Google Fit : : ( ) ” ”
- Umetrip : : / /
- 12306 : : ( ) /
- Flight Master : :
- Amap : / : /

- Baidu Maps : : / AR
  /
- Google Maps : : Street View
- Google Translate : : ” ” ” ”
- Minimalist Exchange : : 160
- / China Weather : :
  ”China Weather” /
- Caiyun Weather : : ” ”
- Carrot Weather : : AI
- :
  7 ” / / ”
- Mafengwo : / :
  / /
- Xiaohongshu : / : ”
  3 XX”

“ ”

23:

- iOS ” ”: Apple ” WiFi + + ”

- Android: (” ” Xiaomi, ” ” Huawei) Tasker
- : ” ” ” + ” 23

:

:

: (1) IP DNS (2) HTTP/HTTPS IP (3) (HTML, CSS, JavaScript) (4)

:

- Microsoft Edge: Chromium Copilot AI
- Safari: Apple
- Chrome:

:

- Bing: Microsoft Edge AI
- Yandex:
- Baidu:
- Google:

:

- AI : AI Bing, Copilot Edge
- : ” ” ” ”
- : ” + ” ” WeChat”
- : Windows 11 microsoft.com Chrome google.com

:

“ ” :

- : .com —
- .cn —
- .org —
- .net —
- .edu —
- .gov —
- .xyz/.tech/.info —

/ : .jp —
.kr —
.uk —
.de —
.tw —
.hk —
: .gov .edu .xyz, .top

:

- AdGuard : : ( )

:

- Brave : : Chromium
Tor Chrome
- DuckDuckGo : ” ” : ( )
(” ”)

:

Microsoft Edge	Chromium + Copilot AI / /	AI ( )
Safari	Apple	iPhone/Mac
Chrome		
Brave		
DuckDuckGo		

” ”

1. : (22:30)
2. : > > >
3.

:

: “ ”

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- Qingzhuo : Android ”Mobile Manager”
  : ” ”
  : WeChat

- : : Android [ ] > [ ] > [ ] iOS [ ] > [ ] > [iPhone]

- WeChat/QQ : : WeChat [ ] > [ ] > [ ] > [ ]
  ” WeChat”

- : / / ” ”

- National Anti-Fraud Center : / :
  ” ”
- / Mobile Manager : : Huawei Mobile Manager, Xiaomi Security Center, OPPO Mobile Manager, vivo Mobile Manager
- : AI : China Mobile KTFSR 10086 China Telecom ”Tianyi Anti-Harassment” China Unicom ”Unicom Mobile Manager”
- AdGuard : : ( )
- ” / ” :

- :
- : National Anti-Fraud Center
- :
- : ” ”

:

“ ” “ ” “ ” 8
“ ” :

iPhone / iOS	[ ] > [ ] / [ ]	” ”
Xiaomi / Redmi	> >	
Huawei / Honor	> Mobile Manager >	
OPPO / realme / OnePlus	> >	
vivo / iQOO	> i Manager	
Samsung	> > /	( )

“ + + National Anti-Fraud Center” :

—

:

1. ” / ”
2. : + (China Mobile KTFSR 10086)
3. ”National Anti-Fraud Center”
4. ( )

“ ” “ ”

—

:

- : China Meteorological Administration / / ”China Weather” WeChat, Weibo ”China Weather”
- : Huawei Weather, Xiaomi Weather, OPPO Weather, vivo Weather, Apple ”Weather” ” / ”
- Caiyun Weather : ” ”

- ( ): :
- :
- :
- :

:

: “ ”

- :
  12379
- : +
- WeChat ” ” ” ”

- 12379 (/)
- / —

:

” ”

: 32 / 7

: “ ”

(EEW) (S)
P “
”

- : China Earthquake Early Warning Chengdu Institute of Disaster Prevention

- : ” ” (” ”)
- : ( )
/

- : ” / ” 0

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“ ” “ ” “ ” iOS

Xiaomi / Redmi	Mobile Manager / > ” ”	
Huawei / Honor	> > /	” ”
OPPO / realme / OnePlus	> (SOS) >	
vivo / iQOO	> / >	
iPhone / iOS		
Samsung	> /	” ”

- ≠ : ” ”
- :
- :
- : ” ”

: “ + / + ( )”

: 10

1. ” / ”
2. ” ” ” ” / (Android) iPhone
3.
4.

:

“ ” : AI
LAN 2FA
17

: Sunlogin / ToDesk

- Sunlogin Remote Control : Windows Mac /
(27) : ( )
- ToDesk : / : +
( )
- : / + ( )
/
- ( ): ” / - - / ” Sunlogin, ToDesk
— (27)
- :
-

AI : Trae / WorkBuddy / Dumate

- Trae : : AI IDE/ ByteDance SOLO / ( ) 41 AI
- WorkBuddy : AI : AI ” — — ”
- Dumate : / : AI
- ” ” : AI / Sunlogin/ToDesk / ” ” AI :
- :

: miHoYo Live Wallpaper (+ PC)

- Live Wallpaper : miHoYo : miHoYo PC Live2D/ /
- : GPU ( )
- : / ” ” PC

LAN: LocalSend / IPMsg / AirDrop

- LocalSend : Android/Windows/iOS/Mac/Linux : LAN Wi-Fi
- IPMsg : Windows LAN : / LAN IT
- AirDrop : iPhone, iPad, Mac : Apple Bluetooth + Wi-Fi ” ”
- : Mi Share, Huawei Share, vivo Share ” ” WeChat LocalSend/AirDrop
- : Wi-Fi Wi-Fi

: KDE Connect /

- Apple ”Universal Clipboard” : iPhone + Mac : Apple ID Handoff Mac ( )

- KDE Connect : Android + Windows/Linux/Mac
  : LAN /
- ” ” : Huawei Multi-screen Collaboration/Super Device, Xiaomi MiMo+, vivo Share, OPPO Connect, Honor Connect
- Windows + Android: ”Phone Link / Link to Windows” (Your Phone)
  Samsung
- :

2FA: Aegis / Microsoft Authenticator

- : SIM TOTP
  ( )
- Aegis Authenticator : Android :
  /
- Microsoft Authenticator : Microsoft, Office 365/ iOS/Android :
  ( ) Windows/
- Google Authenticator / 2FAS :
  (token)
- : ” / ”
- : TOTP ” +
  ” 14

:

- : TickTick / Trae/
- : PDF LocalSend/AirDrop AI
- : ToDesk/Sunflower (OTP)
  Aegis/Microsoft Authenticator
- : TickTick

- OTP ” ” ” ”
- AI IT

-

: ” ”

1. Wi-Fi LocalSend (AirDrop)
2. TickTick ” 20:00 ”
3. 2FA WeChat ( ) Aegis Microsoft Authenticator
4. Sunflower ToDesk

14:

? “ ”

?

:

1. :
2. :
3. :
4. :

AES-256

AES (Advanced Encryption Standard)
(NIST) “256” 256 —
Brute Force (1Password, Bitwarden,
KeePass) AES-256

(Zero-Knowledge Architecture)

“ ” :

1.

2. (/)

3.

4.

: —

”Zero Knowledge” ?

Bitwarden, 1Password “ ” —

: PBKDF2 Argon2

— “ ”

- **PBKDF2**: iCloud , Bitwarden, LastPass (100,000)
- **Argon2**: 2015 PBKDF2 ” ” ” ” — GPU/ASIC Bitwarden, KeePassXC

: LastPass 2022

2022 LastPass
( ) :

- **Zero Knowledge** —
-
- URL

LastPass

(16);
Argon2id ; (,)

:

- **iCloud Keychain (iCloud)**: Apple Safari

- **Google Password Manager**: Android Chrome Android
- **Huawei Password Vault (Huawei)**: HarmonyOS Huawei

:

- **Bitwarden**:
- **1Password**:
- **LastPass**:
- **KeePass**:

“ ” (/ /)
“ ” “ ”

(Entropy):

“ ”

- **”123456”**:
- **”P@ssw0rd”**:
- **”correct horse battery staple”**: 4 44 ”P@ssw0rd”

- **16** (+ + +): 100

: “ + ” “ ”

HaveIBeenPwned:

Troy Hunthaveibeenpwned.com

:

1.

2. Pwned Passwords ( )

k-Anonymity

HaveIBeenPwned ? “k-Anonymity” :
5 5
1Password Bitwarden

- **1Password Watchtower** 2FA
/ /
- **Bitwarden Vault Health**:
(HaveIBeenPwned)
- ” ” **iCloud** :iOS
- ” ” **Google** : Chrome

:

1. haveibeenpwned.com
2. iPhone:
3. Chrome:
4. 1Password/Bitwarden: Watchtower/Health Reports ” ” ” ”

Passkey:

2024-2026 —**Passkey ( )**
Apple, Google, Microsoft FIDO World Wide Web Consortium (W3C)

Passkey ?

Passkey FIDO2 / WebAuthn :

- 1 : ( ) + ( )
- ” ”
- (Face ID /) PIN
- ,

Passkey

		Passkey
(Replay Attack)	/ /	1 Face ID/

- **Passkey Apple iCloud** : iOS 16+ / macOS Ventura+ Passkey Apple ID
- **Passkey Google** : Android + Chrome Passkey Google Android 2023 Google Passkey
- **Microsoft**:Windows 11 / Edge Passkey Microsoft Microsoft, Outlook, Office Passkey Microsoft Authenticator
- **1Password / Bitwarden**: Passkey

Passkey “ ” — 2023 Apple, Google Microsoft Passkey : iCloud Keychain Apple, Google Password Manager Android 1Password Passkey

Passkey

iOS (iPhone):

1. Passkey (Google, Microsoft, Amazon, eBay, PayPal) ” Passkey / Add passkey”
2. Face ID / Touch ID Passkey iCloud Keychain
3. ” iPhone ” Face ID
4. : Passkey

Android:

1. Passkey ” Passkey”

2. Google Password Manager Google (/ /PIN)
3. Passkey Google Password Manager Android
4. ” Passkey”
5. : Android google.com

Passkey

Windows Passkey iPhone QR Code :

1. ” Passkey”
2. ” ”
3. QR Code iPhone
4. iPhone Face ID

Bluetooth Private Key Passkey iPhone

:

1. Google Passkey Passkey Face ID/
2. Microsoft Passkey
3. Passkey Taobao, JD, eBay, PayPal
4. Passkey

Passkey — “Passkey + ” : Passkey
Password Manager
Passkey Passkey “ Password Manager” “ ”

7 · (Hardware Security Key)

— / / —
SMS

?

(USB Flash Drive) Private Key
NFC :

- **YubiKey**(Yubico): FIDO2/WebAuthn, U2F, OTP, PIV, OpenPGP
  USB-C / USB-A / NFC Series 5 50-70

- **Google Titan**: Google USB-C + NFC / Bluetooth 30-40
- **OnlyKey**:
- : ePasskey Feitian

SMS?

- : Passkey
- **SIM**: SMS SIM-swap
- **Man-in-the-Middle**: FIDO2 Challenge Value
- : Private Key

- Wallet
- SSO , VPN , SSH
- , GitHub, ” ” —
-

Hardware Key 2

Google + YubiKey:

1. YubiKey (2)
2. myaccount.google.com 2
3. ” ” YubiKey USB-C NFC
4. YubiKey
5.
6. ” 2 ” Hardware Key SMS

: Google, GitHub, Microsoft, Apple ID, Facebook, Twitter, Dropbox, GitLab, AWS, Coinbase, Binance

Hardware Key “ ” : (1) 2 ; (2) ( ), ; (3) “ ”

—

YubiKey Yubico

/

Netflix, Wi-Fi , backend , ;
, , **WeChat** —

“ ” (Vault) 1Password

1Password :

- :
- : Netflix, Wi-Fi ,
- : ” ” ” ”

Master Password

“ ” Bitwarden

Bitwarden (“Send”);
“ ” Bitwarden Self-hosted

: Netflix, Wi-Fi

- **Netflix** / : ” ” 1Password
- **Wi-Fi** : iOS Wi-Fi AirDrop ; Android 12+ ” Wi-Fi ”
- , : ” ”
- **backend** : 1Password SSID Wi-Fi

1Password

1Password Families 4.99 / 5 + 4
1 / Bitwarden 10 /
6

: /

- **1Password Teams / Business**:
- **Bitwarden Teams / Enterprise**: Self-hosted SSO
- : — ” ” 2-3
- : Password Manager

(Emergency Access):

:
“ ” — /

Bitwarden:

1. Bitwarden Web Vault
2.
3. ” ” (7-14):
4. ( )
5.

1Password:

1. 1Password ” ” ” / ” — ” ”
2.
3. 1Password ” ”

: ” ”

1. ” ”
2. (Bitwarden) (1Password)
3. +
4.

- **:** (Bitwarden)
- ( )
-
- :
- :

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:

1. : (Bitwarden)
2. :
3. : ( )
4. :
5. :

3. (15-20)

15:

“ ”

?

( ) : ( ) ( ) :

- :
- :
- :

:

- 60% 40% 100%
- 80%
- ?iPhone: > > : ’ ’

?

- ?
- :
- : 20%-80% (0%) (100%)

” 80%” iPhone —

iPhone 15/16 (iOS 17) “ 20%-80%” 80%

:

> > > > 80%

: (1) — 100% (2) — 100%

(3) **80%**— 80% / 80%

GaN —

GaN ( ) GaN

:

- : GaN GaN 65W 5W
- : 95%
- : GaN 65W 100W ” ”
- : GaN PD (65W) Baseus, Anker, Nubia, Xiaomi (Type-C E-Marker)

: “ ”

“ ” —

= × (P = U × I)

:

1. **(Handshake)**:
2. :
3. :
4. :

USB PD (Power Delivery)—

- : USB-IF PD 3.0 100W (20V/5A) PD 3.1 240W (48V/5A)
- : 5V, 9V, 15V, 20V PD 3.1 28V, 36V, 48V
- : iPhone Google Pixel Switch PD
- : PPS ( )
- : Type-C

Qualcomm QC (Quick Charge)—

- **QC 3.0**: 3.6V-20V 0.2V 18W

- **QC 4.0/4+**: USB PD 27W
- **QC 5.0**: 100W+ 0-50% 5
- : Snapdragon

OPPO VOOC/OnePlus WARP/realme Dart—

- : (5V/4A)
- **VOOC 1.0**: 20W (5V/4A) 2014
- **SuperVOOC**: 50W-65W
- **SuperVOOC 2.0**: 65W 4000mAh 30
- **150W/240W**:
- : ( )

Huawei SuperCharge—

- **FCP (Fast Charge Protocol)**: 9V/2A, 18W QC 2.0
- **SCP (Super Charge Protocol)**: 22.5W 66W 100W
- : 98%
- : Huawei/Honor PD

Xiaomi Surge —

- **Surge P1**: Xiaomi 120W
- **Surge G1**: P1
- : +

—

- : USB PD
- : iPhone 8 27W (iPhone 15 Pro Max 27W+)
- :
- : USB-C to Lightning (USB-C to USB-C) PD

USB PD 3.1	240W	/	
QC 5.0	100W+		Qualcomm
VOOC/Dart	240W		OPPO
SuperCharge	100W		Huawei/Honor
	~27W	PD	iPhone/iPad

:

- **Qi**: WPC ( ) Qi 5W-15W
- **Qi2**: 2023 (MagSafe) 15W+
- : Xiaomi 50W Huawei 50W

- :
- : 60W
- : MFi
- :

-
-
-
-

:

2024-2026 “ ”

:

- : (372mAh/g)
- : 10 (4200mAh/g)

” ”

- : **15%-20%** 2024-2026
  5500-6000mAh 7-9mm

?

- **Honor**: Magic Magic6 Pro, Magic7, Magic V3
- **Xiaomi**: Xiaomi 14 Ultra, Xiaomi 15
- **vivo**: X100 Ultra, X200 Pro
- **Huawei/OPPO** : 2025-2026 ” ”

: “ ” —
4500mAh 5500mAh “ ”
80%

- ( ): OLED
- :
- **5G**: 4G
- : ” ”
- : iPhone > >

:

1. :
2. :
3. :

16:

“ ” “ ”

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- :
- :
- : ( )
- **App**:
- :

iPhone Apple ID Android Huawei/Xiaomi/OPPO/vivo Google “ ”

1. : iPhone Android ” ” ” ”
—

2. : — ? ” ”
” / ”

3. ” ” :

iCloud: “ ” Apple

iPhone : iCloud ?

- **Apple 5GB 5GB ” iCloud ” :**
- **: iCloud**
- **: 50GB 6 200GB 21 50GB**

: iCloud

: iCloud ” ” — ” ” iCloud ” ” iCloud iCloud Backup

iCloud (50GB/200GB/2TB) **iCloud+** 3 :

- **Private Relay ( ):** ”VPN” Apple Safari Apple (Apple) Wi-Fi : > > iCloud >
- **Hide My Email ( ):** /App : > > iCloud >
- : (@ .com) iCloud

: Private Relay /

iCloud (ADP) — End-to-End

2022 Apple **iCloud (Advanced Data Protection, ADP)** End-to-End (E2EE) **iCloud** — , , iCloud, 23

- **E2EE** : Apple Apple Apple
- : > > iCloud > / ( )
- : (1) (2) , iCloud E2EE (3)Apple ID ADP

: ADP

:

- : iCloud
  ” ”
- **(Backup)** : ” ”
  iCloud Backup

Android

Android Apple :

- (Huawei/Xiaomi/OPPO/vivo):
  +
- **Google**: Gmail, Google Photos, Google Google Google
  15GB Apple Gmail Drive
- **WeChat/QQ**: WeChat Google
  WeChat

Android : Google WeChat

:

“ ” Apple App iCloud 6 :
Huawei Xiaomi “ ” “ ”
: ( )

:

:

1. :
2. : ” ” ” ”
3. : iCloud
4. : ” ”

5. : Android — Google WeChat —

17:

“ ”

:

- **Apple (AirDrop +)**: iPhone Mac (AirDrop) AirDrop Mac Wi-Fi Wi-Fi ” (Universal Clipboard)” — Mac
- **Android + Windows (Windows)**: Microsoft ” Windows” Samsung -
- : ” ” Huawei HyperConnect Xiaomi — WeChat

:

:

- **iPhone** : ( ) Wi-Fi AirPlay ( )
- **Android** : Huawei ” ” Xiaomi ” ” OPPO/vivo ” ” ” ”
- : BiliBili, YouTube — Wi-Fi
- : 90% Wi-Fi 10% ” ” ”AirPlay”

/ :

“ ”

/ —

- : — WeChat
- : 24
  —
- / :
  ” ”
- : ” ” —

: “ ”

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- :
  :
- :
- : ” Siri / ”
- : ” ”

- :
- ? :
- : iPhone Apple HomeKit ( )

:

: Huawei HarmonyOS Connect Huawei

- :

- : ( )

- : AirPods iPhone Apple Huawei Free-Buds Huawei Xiaomi Xiaomi

- : DJ

UWB (Ultra-Wideband) —" " Bluetooth

Bluetooth UWB (Ultra-Wideband) 1
—
Bluetooth " " UWB " 1.2 "
:

- **AirTag** : iPhone AirTag " 7 " UWB Bluetooth " "

- : iPhone Android UWB
Relay

- : " " UWB Beacon GPS

- : iPhone 11 UWB (U1/U2) Android Xiaomi, Samsung, vivo OPPO

LE Audio Auracast — Bluetooth

Bluetooth (A2DP) **LE Audio**( ) Blue-tooth 5.2 :

- : LC3 SBC

- :

- **Auracast** : LE Audio / —

- : 2024-2026 LE Audio
LE Audio/Auracast

:

CarPlay (Apple) Android Auto (Android)

- : ( ) USB CarPlay Android Auto

- : ( ) ( ) ( ) WeChat (Siri)
- : CarPlay

:

- :
- : — ” ” ” ”
- :

:

1. : Wi-Fi
2. : ” ” ”AirPlay”
3. : : iPhone ” ” Android ” ”
4. :
5. : ” ” :

1:

- : Xiaomi 14
- :(1) (2) (3) 5G 4G (4) (5) (6)
- : 30-50%

18:

—

:

“ ” :

- : ( ) SMS
- : WeChat Taobao
- : WeChat (WeChat) (Google Authenticator)

:

“ ” —

: ( )

- **iPhone**: iTunes (Windows) Finder (Mac) ” ” ” ”
- : Huawei HiSuite, Xiaomi Assistant, Samsung Smart Switch
- —

: ( )

- **iPhone**: iCloud iCloud Backup Wi-Fi 5GB 6 50GB

- : Huawei ” ” Xiaomi ” Xiaomi” OPPO/vivo Google ( )
- ” ”

:

:

( )

- : iCloud Photos/Google Photos
- : Google Photos 15GB ” ”

( )

- iPhone iCloud iCloud.com vCard
- Google .vcf

WeChat ( )

WeChat

- **:** . WeChat / Wi-Fi : WeChat
- **:** . WeChat PC
- **:** . WeChat iCloud :

:

:

( )

- **iPhone iPhone** : ” ” Wi-Fi 20-30
- : Huawei ”Phone Clone” Xiaomi ”Mi Mover” OPPO ” ” vivo ”EasyShare” :

( )

( ) : SMS
: ( ) WeChat (WeChat)

(**iPhone**)

- **iPhone**: ”Move to iOS” iPhone ” Android” SMS WeChat WeChat
- **iPhone** : :
  WeChat

: 3-2-1

:

- **3**: ( )
- **2**: ( )
- **1**: ( )

“ ”

WeChat

WeChat ! “ ”
WeChat WeChat PC —

:

1. iCloud
2. : DCIM Pictures
3. : iPhone iCloud.com vCard .vcf
4. WeChat: WeChat
5. vcf ” ” 3-2-1

19:

“ ” ... “ ”
—

—

“ ” —

:

1. : 10% :
   15%-20%

2. : 24

3. : ? ( ) ?
   ( )

4. : CPU

( ): (60%) ( ) :

—

:

1. :

2. :

3. :

4. : ” ” ” ”!

5. : —WeChat

—

- : —
- :
- : ( ) —
- : (iPhone PD 20W) (3A) ( )

Wi-Fi

Wi-Fi Wi-Fi :

1. : 5
2. **Wi-Fi ” ”** : Wi-Fi ( ) ” ” ” ”
3. : 30 Wi-Fi
4. : Wi-Fi ? (Wi-Fi)

:

- : —
- / ( ): (TikTok) ” ” OLED
- : PWM OLED ”DC dimming” ” ”
- / :

—

:

- : 30
- : iPhone (10-15) : 10
- : ” ” (iPhone) ” ” ( ) iPhone iTunes/Finder ” ” (” ”) Fastboot Recovery + ” Recovery”
- :

:

- : (iPhone) 40%
- : —
- / : — ! ( )

:

- **” ”**: (Apple.com.cn, Huawei.com.cn) ” ”
- : ( ) (Apple ID, WeChat, Alipay) ” iPhone” —
- : ” ” ( ) ” ” ( ) ” ” ( )
- :
- : / ” ” —

:

— —

“ ”

:

1. : iPhone 10-15
2. : ( )
3. **Wi-Fi** : Wi-Fi ” ” Wi-Fi
4. / : 5
5.

20:

“ ” *Root* *BL* ...

:

:

- **Android**: > > ” ” 7 ” ”
- : > > ” HarmonyOS” 7
- **iOS**: Apple Developer

:

- **USB (USB Debugging)**: (, , ADB) ” ADB ”
- **(Show Taps)**: ( )

:

- (/ /): App 1x 0.5x ” ” ( ) 0.5x ” ”

- **GPU (Force GPU Rendering)**: 2D GPU CPU App GPU GPU

- **GPU (Profile GPU Rendering)**: 16.6ms (60fps) App App — App App App

:

- : ” ” ” 1 ” ” 2 ” Android App ( ) ” ”

- : App App

USB Debugging

OEM: “ ”

— “ OEM” (“ OEM” “ ”) BL “ ” : BL fastboot

- : ” BL” BL

- **BL Lock**: OEM ” ” BL ” ” fastboot (`fastboot flashing unlock`)

- : ” OEM” BL — `fastboot flashing lock`

- : Root ”USB Debugging”

:

Android Google Android “ OEM” “ ” BL OEM

BL Lock (Bootloader Lock):

Bootloader BL Lock

BL Lock:

- :
- :
- : BL

BL Lock :

- **Xiaomi**: (7-30)
- **Huawei**: 2018
- **OnePlus**:
- **OPPO/vivo**:
- **Samsung**: Knox Samsung Pay
- **Apple**: iOS BL Lock

ADB :

ADB Android Debug Bridge Google An-
droid USB Wi-Fi Root “USB Debugging”
“ ” — App Log

USB:

1. : ”USB Debugging”
2. : Platform Tools Google (`adb` `fastboot`)
3. ” USB debugging ” ” ”
4. Command Line (PowerShell cmd Windows) `adb devices` `List of devices attached /xxxxxxxx device`

> Windows `adb devices` “unauthorized” “ ”
> Samsung/ Xiaomi “USB ” “ ”
> Mac Linux

ADB :

- `adb install app.apk`: APK (`-r` ; `-s` SD card)
- `adb uninstall com.xxx.xxx`: App
- `adb push /sdcard/` :
- `adb pull /sdcard/` : (WeChat)
- `adb shell`: Linux Shell `ls`,`pm`,`am`
- `adb reboot bootloader`: Fastboot ( )
- `adb reboot recovery`: Recovery
- `adb shell pm list packages`: ( ) `-s` `-3`
- `adb shell pm uninstall --user 0` : Root

Root :

ADB: `pm uninstall --user 0` “ ” Root “ ”

1. `adb shell pm list packages -s` Xiaomi `com.android.browser` App Store Huawei `com.vmall.client`
2. ? `adb shell pm list packages | findstr` ”Package Viewer”
3. :`adb shell pm uninstall --user 0` `adb shell pm uninstall --user 0 com.android.browser` `Success`
4. ? 2 : (1) ( ); (2) :`adb shell cmd package install-existing`

“pm uninstall –user 0” Root (`com.android.systemui` ,`com.android.phone` ,`com.android.settings`) Google “ ”

: –user 0

Android 4.2 ID (0 10)`--user 0` “ ” APK Root “ ” Root:

Root ADB “ ”

Root: “ ” Android

Root Super User Linux (/) Android Administrator Windows: Windows (TrustedInstaller) Administrator Linux root `sudo rm -rf /*` Root Root

Root:

- :
- : (Titanium Backup, Greenify)
- :
- : Recovery ROM

Root:

- : Root
- : Root
- :
- : Root
- **OTA** :

Root ( )

SafetyNet Play Integrity: Root

Root : Root — “ ”Google Pay Netflix 2 Google: SafetyNet Play Integrity API

SafetyNet / Play Integrity :

- **SafetyNet Attestation (, 2017)**: API Google Google ” CTS ” Basic Attestation ( ) Strong Attestation ( ) Google SafetyNet 2024

- **Play Integrity API (, 2022)**: Google SafetyNet " " ( ) " / " (Google) Play Integrity
- : (Google Pay, , Alipay/WeChat Pay), (Netflix), (Tencent/NetEase), (MDM)

Play Integrity 3 :

- **MEETS_DEVICE_INTEGRITY ( )**: Android , Bootloader , Root, CTS
- **MEETS_BASIC_INTEGRITY ( )**: (Netflix)
- **MEETS_STRONG_INTEGRITY ( )**: (Titan M Google Pixel) Root
- **(NO_INTEGRITY)**: Root, BL , ROM /

Root:

Google Play Integrity API Google — Bootloader, , su , SELinux permissive “ ”

: Magisk Hide / Shamiko / Zygisk:

Root “ Root” : “ ”

- **, ::** Systemless Root — mount Root OTA Magisk BL
- **Magisk Hide ( )**: Magisk Root 2021 Magisk Google " "
- **Zygisk**: Magisk 24.0+ Riru Zygote Zygisk
- **Shamiko**: Zygisk Root (" ") Magisk Bootloader Shamiko LSPosed GitHub

- **Play Integrity Fix(osm0sis)**: Magisk Google Play Integrity (fingerprint) Google Root

Google Play Integrity
/ (,) : Root, BL
“ Root + Root”

- : , GPU
- **OEM BL Lock**: Root ( )
- **ADB**: Root — `pm uninstall --user 0`
- **eSIM**: Root
  Root (ADB , Shizuku Root)

:
1. : >
2. : ” ” 7
3. : ” ” (” ”)
4. : ” ” ” ” ” 0.5x
5. :

20 : Termux — Linux Android

“ ” *Linux* *Termux* *root*
Android *Linux*

Termux

Termux Linux Android root
Linux shell (30MB) APT Python,
Node.js, Git, SSH

Termux

Linux

Termux **Bash**Shell Shell Zsh fish
:`pkg install zsh` `pkg install fish` Termux Vim, Nano, Emacs Command Line 2000 “pkg install” GCC, OpenSSH, SQLMap

APT (Debian Ubuntu) Linux
Python, Ruby, Node.js, Go, Rust, Swift

- **SSH**: SSH SSH
- : OpenSSL
- : Nginx Apache

- :
- Termux: Styling
- :

Termux :

Termux:API	: Android API
Termux: Boot	:
Termux: Float	:
Termux: Styling	:
Termux: Tasker	: Termux Tasker
Termux: Widget	:

Termux sharedUserId (F-Droid GitHub)

1. : Termux (F-Droid GitHub) root
2. : ”Installing bootstrap packages”
3. : `termux-setup-storage`:
4. : `pkg update && pkg upgrade`:

- : `termux-change-repo` ( )
- : Ctrl Esc
- : Zsh + Oh My Zsh

Python C GCC Git
SSH

SSH

Linux

Linux Command Line Shell
Script, Linux (`chmod 755` rwxr-xr-x,`chown user: group` ,`ls -l` 9 3 owner/group/other) “ ”
Unix/Linux (owner / group / other × / /) ACL
Windows NTFS

Nmap, SQLMap, Hydra, Metasploit

Termux v0.118.2 Android $\geq$ 7 Termux Android 12

Termux Linux Android
Termux “Linux ”

: Linux

1. : Termux F-Droid
2. `pkg update && pkg upgrade`:
3. Python:`pkg install python`
4. `python3 -c "print('Hello from Termux!')"`:

4. (21-26)

21:

... ?

— —

VPN:

VPN (Virtual Private Network)

- :

- **VPN**: VPN VPN
IP VPN IP

VPN ? : ; , ( );
, **(ISP)** : **VPN !**
VPN VPN VPN
(NordVPN, ExpressVPN)
VPN VPN

—

Wi-Fi ? : Wi-Fi Star-
bucks_Free — HTTP
TLS —
: WeChat Alipay HTTPS
( )
HTTP (Man-in-the-Middle)
Wi-Fi Wi-Fi
Wi-Fi (ChinaNet CMCC)
? :

- **HTTPS** : https:// s Secure http:// (s) ! HTTPS
- : iPhone Wi-Fi
- : ··· 4G/5G
- **Wi-Fi** : Wi-Fi Wi-Fi

HTTPS ? — SSL/TLS

HTTPS ? HTTPS SSL/TLS handshake :

1. : ( )
2. : CA
3. : CA ? ? ?
4.

: ( )

: HTTPS HTTPS —
URL

DNS: ?

DNS (Domain Name System) : baidu.com DNS IP (110.242.68.66) DNS ! ISP Wi-Fi

DNS: DNS baidu.com URL

DNS :

- **DNS over HTTPS (DoH)**: DNS HTTPS DNS
- **DNS over TLS (DoT)**: TLS DNS DoT

Cloudflare 1.1.1.1 Google 8.8.8.8 DNS DNS Private DNS one.one.one.one dns.google iPhone

?

:

: — Cookies

:

- , IP
- (HTTPS)
- /
- ; : VPN+Tor

:

:

- **:** **Cookies** — **Cookies** — **GB**
- **:** **Do Not Track**
- **: Firefox Focus, DuckDuckGo, Brave Chrome Safari**
- **: DuckDuckGo Baidu/Google Google**
- **: ?**

:

:

Wi-Fi

1. **: iPhone -Wi-Fi Wi-Fi** DNS
2. **: Private DNS dns.google iPhone DoH 1.1.1.1** (2FA)

3. **：WeChat Alipay Apple ID Google 2FA SMS 2FA TOTP（Google Authenticator, Microsoft Authenticator）**

4. **：：SMS** —

5. **：Cookies**

6. **：USB （Juice Jacking）**

：

DNS

1. **：：> > Private DNS > one.one.one.one dns.google**
   iPhone: Safari 1.1.1.1 DNS profile 1.1.1.1 Cloudflare Wi-Fi

2. **：Wi-Fi**

3. **：> /（）**

4. **：Apple ID Google 2FA**
   ：AI AI
   （22）

22: AI

“ ” *(AI)*
“ ”

AI: “ ”

AI AI :
“ ” “ ” “ ”

: “ ”

1. ( ):
2. ( ):
3. : ” ”

AI vs. AI

AI AI

AI :

2025-2026 AI (DeepSeek R1, ChatGPT-5, ERNIE Bot 5.0, Qwen 3
) AI (NPU
80 TOPS) AI

AI ?

AI (Large Language Model: LLM) AI
“ ”

AI :

- : DeepSeek, ChatGPT, Doubao, Kimi

- : AI
- : AI
- :
- : AI

: AI

AI (7B) :

- : AI
- :
- : AI
- : AI

AI :

- **Apple Intelligence** : AI iOS 18 Siri

- **Pangu** : HarmonyOS Xiao Yi AI

- **HyperAI Xiaomi**: AI AI AI AI
- **Galaxy AI** : AI (Circle to Search)

“ ” AI AI “ ” :

1: —

: ? “ ” iPhone
Galaxy AI

: 3 (ASR) (NMT) (TTS)

1

2: AI — “ ”

: ? “ ” “ ”

: (Inpainting) “ ”
AI

3: AI — “ ”

: ? “ ” AI
AI “ ”

: (Diffusion Model) AI (Generative AI)
“ ” AI

4: —

: AI PDF
10 AI 10

: (Long Context LLM) “ ” Apple Intelligence, Xiao Yi , WPS AI, Tencent Yuanbao

5: AI — “ ”

: AI “ ” “ ” AI

: (CV) (Embedding) “ ”

: “ ” AI

AI “ ” “ - -
- ”

?

AI Baidu — AI “ ”
“ ” AI

(Role): AI

“ ” AI

:

- ” **Xiaomi 10** .” ( )
- ” .” ( )
- ” .” ( )
- ” .” ( )
- ” .” ( )

? AI AI “ ”
“ ”

(Background): AI

“ ” AI

:

- ” **Xiaomi 14** .” ( )
- ” **60** .” ( )
- ” **CET-6 2026** .” ( )
- ” **30,000** .” ( )
- ” .” ( )

? AI “ ” “ ”
AI “ ”

(Task): AI

(...)

:

- ” **5** .” ( )
- ” .” ( )
- ” **4 3,000** .” ( )
- ” **5G** .” ( )
- ” **3** .” ( )

? “ ” AI **AI** “ ”
“ ” “ ” “ ” AI

(Constraints): AI

“ ” **AI** :

:

- ” **200** .” (+)
- ” ”.” ( )
- ” .” ( )
- ” **3** .” ( )
- ” **:** .” ( )
- ” .” ( )

? AI “ ” — =

: 5

AI:

1. : ” ” ” 800 3 ”
2. : AI ” ” ” ”
3. : JSON?
4. :
5. **AI** : AI

/ / AI	(/) + ” ” (200) ” ’ ’” + ” ... ”

1: ( )

:

( ) 15 6 60
( ) **7** ( ) : (1) 15 (2) (+ +) (3)
(4) ( )

: 4

2: (CET-6)

:

CET-6 ( ) **CET-4 480** **CET-6 500** **3**
( ) **12** ( ) : (1) (2)
(3) 3 ( ) (4) ( )

: “ ” “ ” “ ”

3: ( )

:

( ) 1
2( ) ( ) : (1) “ / /
” (2) (“ ”) (3)
600 (4) markdown ( )

: 4

: AI “ ”

AI:

- : AI ” ”
- **(Chain-of-Thought)**: AI ” ” ” ”
- **AI** : A B C —
- **(System Prompt)**: AI

AI:

AI “ ” :

- ” ”: AI (Hallucination)

- : AI (2024)
- : AI

- : AI AI
- : AI

AI

- AI (” ”)
- AI ( )
- (Prompt) : AI
- AI
- ( ) AI

: AI

1. **1:** : : ” 7 3 4 ”
2. **2:** **AI**: ” ” ” AI” AI ” ”
3. **3:** : AI ” ” CRISPE

2:

- : iPhone 13 Xiaomi 15
- : (1) iCloud/WeChat (2) Apple ID + ” (Find My)” (3) ” Xiaomi (Mi Mover)” (4) (5) WeChat ” ” (6) (/ /WeChat) (7)
- :

23:

“ ” “ AI” :
“ ” “ ”
— “ ”

?

: —

: WeChat
“ 7 ” ?
“ ....” if-else
:

1. : ? ” 7 ” ” Wi-Fi ” ” ” ” 20%”
2. : ? WeChat ...
3. ( ): A X Y ” ”

iOS “Shortcuts ( )” — Apple

“Shortcuts” Apple
:

- ” **(Shortcut)”**: Siri ” Siri ”
- ” **(Automation)”**: ” ”
” ” ” ”
- ” **(Action)”**: action action action
action action

:

1: “ ”

(Siri “ ”) :
: action “ ” “ ” “ ” “ ” “ ”
NetEase Cloud Music

2: “ ”

“ ” “ ” “ 30%” action “ ” “
30%” “ ” 30%

3:

WeChat/Alipay Numbers
“ SMS/ ” “ ” Shortcuts (iOS 17+ Wallet)
action “ Numbers”
: “ ”

Android —Tasker

Android :

:

Android “ ” :
:
( )

: Tasker — Android

Tasker Android
Tasker “ ” :

- **(Profile)**: ” ” Wi-Fi
... Tasker
- **(Task)**: ” ” action action : (
Wi-Fi) SMS HTTP request Shell

Tasker:

- **Wi-Fi** : ” ” ( ) ” Wi-Fi” GPS Wi-Fi
- : ” ’ ’” ” WeChat ( )”
- : ” 22:00-06:00” ” 20% ( )” —
- : ” / ” ” ”

Tasker :**MacroDroid** Android (MIUI Xiaomi, HarmonyOS Huawei, ColorOS OPPO) “ ” “ ” Tasker

“ X Y ” :

- :
- : WeChat
- : SMS ” ”
- : ( )
- : ( ) ” ”

“ + action” Shortcuts Tasker “ ” —

:

:

- : ” / ” iOS geofencing GPS Wi-Fi
- **Shortcuts** : iOS ” ” (iOS) Wi-Fi
- **Tasker (Accessibility)**: ( ) Accessibility Android

- : ”2-3 action” 20 —

:

5 : 3 ? ...

5

: 3

:

1. **1:** : iOS: Shortcuts ” ” ” ” ” ” Siri ” ” Android: Tasker ” ” ” ” ” ” 7:00
2. **2:** : iOS: ” ” 25% ” ” + ” 25%” Android: Tasker Profile 0-25% ” ” + ” ”
3. **3:** : iOS: ” ” ” ” + ” 80%” Android: Tasker Profile ” ” + ” ”

3: ” ”

- : 7:00 + +
- : iOS: ” ”+” ”+” App”; Android: MacroDroid (7:00) (/ URL/TTS)
- :

23 : GitHub —

“ ” *GitHub*

GitHub

GitHub ?

GitHub Git 2008 2018

: **Git** **GitHub** —
“ ” “ ” —

GitHub
“ ” GitHub

Git GitHub (!)

:

	Git	**GitHub**
	( )	( )
		( )
	(	Git
	)	
	Linus Tor-	
	valds (Linux)	
	SVN, Mercurial	GitLab, Gitee, Bitbucket

:**Git** “ ” **GitHub** “ ” Git

GitHub Git Git GitHub (
) GitHub Git (GitHub Git)

GitHub

GitHub :

- **(Repository Repo)**:
  ” ”
- **(Commit)**: ” ” Git/GitHub
  ” ”
- **(Branch)**: (merge)
- **(Clone)**:
- **(Push) / (Pull)**: Push GitHub Pull GitHub
- **Pull Request (PR)**: PR
  GitHub
- **(Issue)**: Issue
- **Fork**: PR

GitHub :

- :
- :
- : Issue
- : (Projects) Trello
- : **GitHub Actions** ”CI/CD
  (/)”
- : README Wiki Pages
- : ” ”
- : Star ( ) Watch ( ) Follow ( )

: 5

1. github.com "Sign up"
2. (`github.com/yourname`)
3.
4. (Free)

1. "+" "New repository"
2. (Repository name) `my-first-project`
3. (Description)
4. (Public) (Private)
5. "Add a README file" —README " "
6. "Create repository" GitHub !

: README

README.md Markdown README : README

:

GitHub : :

1. :
2. **git add**: " " Git " "
3. **git commit**: " " " "

4. **git push**: GitHub
5. **git pull**:

: " "

: Pull Request

GitHub :

1. : (main master) `feature/login`
2. :
3. **GitHub**:
4. **Pull Request**: PR GitHub PR
5. **(Code Review)**: PR
6. **(Merge)**: ”Merge” !

— PR

GitHub

ACM GitHub —
GitHub

GitHub (GitHub Pages) Markdown
+ Git +

GitHub Fork PR Star

CI/CD

GitHub Actions “ + ”

git pull

- : ”update” commit message ” ”
- :
- :
- **.gitignore** : Git ( )
- : GitHub Trending README
- **GitHub Student Pack**: (Copilot)

- API GitHub —
- ( ) `git filter-branch`
- (environment variables)

GitHub

GitHub : (1) (GitClone, gh-proxy) (2) Hosts (3) **Gitee (Mayun)** GitHub Gitee

: GitHub Pages

GitHub Pages Jekyll, Hexo, Hugo Markdown HTTPS

: GitHub

GitHub “ ” :

“ ”

“ + ” —

GitHub : “ ”

: GitHub

1. GitHub ( )
2. `hello-github` ”Add a README file”
3. README.md ( ) Markdown
4. GitHub (`python-tutorial`) Star
5. ( ) Git README ” ”

24: Root BL

20

Bootloader

Bootloader Bootloader
“ ”

BL Lock:

- : BL Lock
- :
- :

BL (Locked/Unlocked) Public Key (dev-
info Qualcomm, eFuse RPMB) Bootloader

Private Key Bootloader Public Key
(boot.img, recovery.img) :

1.
2. Public Key
3.

BL (Kernel)

vs : BL

BL 5 :

	( )	
		: ROM Root : ROM , , (,) OTA : BL :
	: OTA BL : ( ) : BL (,)	: (Google Pay,) BL DRM

BL (Xiaomi)

1. **4.** :
2. **ROM**: ,
3. **Recovery**: TWRP, , ROM
4. **ROM**: LineageOS, Pixel Experience ,

Root

- , : ( )
-
- , ,

XDA Developers

- : **Android** **ROM** Coolapk
- :
- : **Xiaomi, OnePlus** 25:

Samsung Knox: BL

Android BL Samsung “ ” : **Samsung Knox** **(Samsung Pay);** **;**
:

- : eFuse BL
- : Samsung Pay (/), Knox, MDM ,

- : ( )
- : Knox

Knox Samsung Root
Root

:

- : ( )
-
-
-

:

- **XDA Developers**: Android ROM
- **Coolapk**:
- : Xiaomi, OnePlus

(Q& A)

Q1: Bootloader ?

A: Flash Flash (Xiaomi`flash_all_lock.bat`) (Knox Samsung)

Q2: Root ?

A: Bootloader Root :

1. Flash Recovery (TWRP)
2. Flash Magisk Root Recovery boot image Root

25:

“ ” “ ” —

“ ” *?*

“ ” :

: “ ” “ ”

— “ ”

1. : (iCloud, Huawei/Xiaomi/Samsung, Google) ” ” —

2. ” ” : ” ” —

3. ” ” ( ): (iOS 8+, Android 6+) (FBE) (2015)

: “ ”

	> > >	Huawei ” ” Pure ” SD ” ” ” Xiaomi ” ”
Xiaomi / Redmi	> >	
OPPO / OnePlus	> > >	ColorOS ” ”
vivo	> > >	” ” SD card vivo ” ”
	> > >	Samsung ” ” ” ”
( )	> > iPhone >	Apple ID iCloud Activation Lock iOS 15+ ” iCloud ”

3

- :
- **SD card** SD card
- **USB/OTG**:

: “ ” **?**

“ ” “ ”

? —

: 49

2021 49 :

:

-
- ” ”
- ( )
- ” ”

WeChat Alipay:

2 :

- **WeChat**: Tencent WeChat Tencent **WeChat** **Qian-qiantong** Tenpay
- **Alipay**: Alipay, Yu’ebao, Licaitong ” ” Alipay Ant Fortune
- :

- ” ”—
- ” ”—
-

“ ” Apple:

iOS 15.2 Apple “ ” + iCloud (, ,) :

1. iPhone [ ] > [ ] > [ ] > [ ]
2. ” ” Face ID
3. ( )
4. SMS,
5.

Apple Apple 7

“ ” Google:

Google “ (Inactive Account Manager)”

1. **myaccount.google.com/inactive** Google
2. : 3 / 6 / 12 / 18
3. (10) (Gmail, , Google Drive)
4. :
5. :
6.

:

2 : “ ” ( ) “ ” ( )

- **Weibo**: ” ” ” ”
- **Douyin**: ” - - - - ”
- **Bilibili**: (UP)
- **Xiaohongshu**:

-
-
-

:

- **(Bitcoin, Ethereum)**: ” ” ” ” : / (+)
- **Exchange**: Exchange (Binance, OKX) Exchange

- : (Alibaba Cloud, GoDaddy)
- (, , ,): ( )
  50
- :

“ ”

Chainalysis **Bitcoin 3**

——

“ ”

“ ”

:

( )

1: (,)

- : WeChat (xxx), Weibo, Douyin, Bilibili, Xiaohongshu
- : Alipay, WeChat Pay,
- : iCloud, Huawei Cloud, Xiaomi Cloud, Google, Microsoft
- : QQ Mail, NetEase Mail, Gmail
- : , ,

2:

- : (Bitwarden, 1Password)
- / : 2 (+)
- : , , ,

3:

- ” ” ” ”
- iCloud / Google ” ” ” ”
- Private Key
-

4:

-
-
- : _____ ___ ___ , : _____ ___ ___

” ”

: “ +
+ ”

3 :

Trade-in (Huawei, Apple, Xiaomi) (Aihuishou, Zhuanzhuan, Huisoubao) (,) ( )			

- / : Apple, Huawei, Xiaomi ”Trade-in”
- : SF Express ” ”
- : **SIM/SD Card**
- : , , , ,

, , ,

: ” ”

1. : , ,
2. : iCloud / Huawei / Xiaomi / Google Account ” ”
3. SIM SD card OTG
4. ” ”
5. ( )
6. : Trade-in , ,
7. ” ”

1. ” ” Apple ” ” Google
2. ” ”
3. 2

26:

“ ” *2024-2026* :
AI , *Flagship* , “ ”
“ ”
“ ” “ ”

AI: ?

2024 Android “ AI” Flagship Apple Apple Intelligence iOS 18 “ AI” ? 3 :

- **Large Model** : 7B (Qwen, Llama, DeepSeek) , , : AI On-device
- **NPU (TOPS)**: NPU ( ) TOPS ( ) Flagship 2024 (Snapdragon 8 Gen3, Dimensity 9300) NPU 40-70 TOPS 2025-2026 100 TOPS
- **AI Agent**: — ” Zhang San” , , WeChat AI ” AI”

AI : ?

AI :

- **Apple Intelligence (iOS 18+)**: — , , , AI On-device Private Cloud Compute 2025
- **Google Gemini Nano**: Pixel 8/9 ” ”, Magic Compose, Gemini Nano On-device AI Android

- **Pangu Huawei**: XiaoYi HarmonyOS On-device, AI, AI ( ), HarmonyOS NEXT 5.0 AI
- **MiLM Xiaomi**: HyperOS AI Portrait, AI, AI Painter, Super Resolution Xiaomi ” ”
- **OPPO (Andes) vivo (BlueLM)**: ColorOS 15 OriginOS 5 AI , AI, AI, (” ”)

: AI On-device AI AI

: “ ” Flagship

2022 Huawei Mate 60 iPhone 14 Flagship 2025 2,000-3,000

- **Beidou Short Message** : Beidou Short Message Beidou (SMS 1,000 +) Redmi, realme, iQOO
- **Tiantong Satellite Call**: Huawei Mate 60/70 Honor Magic6 Tiantong-1 Satellite Call / (10-200)
- **iPhone Satellite Messages (SOS)**: iPhone 14 Satellite SOS 2
- : —

: ?

“ ” “ ” 3 : , ,

- : 500,000 (- 100 13) IPX8 UTG + ” ”
- : (WeChat, Taobao, Douyin, Bilibili, Office, WPS) 30%

- : 90°-120° ” ” ,

- : 2 3 HyperOS, ColorOS, OriginOS iOS (iPadOS) iPhone

- : ( ) 5,000-7,000 ( ) 4,000-5,000 2025 3,000 mass market

:

30

6G Wi-Fi 7:

6G :

- **6G**:3GPP 6G 2027 2030 6G 1 Tbps (5G 50) 0.1ms ” ” (Communication + Sensing)

- **Wi-Fi 7(802.11be)**: 2024 46 Gbps (Wi-Fi 6 4) 320MHz, 4K-QAM MLO ( ) Wi-Fi 7 2-5 Gbps Wi-Fi 6E 2-3 Wi-Fi 7 Wi-Fi 7

- **Bluetooth 5.4 LE Audio**: 2024-2026 Bluetooth 5.4 LE Audio ( ) 2 Auracast

- :

2024 -

- : 372 mAh/g 4200 mAh/g (10) 300% - - 15%-30%

- : - 2024-2025 (Xiaomi 14 Ultra, Honor Magic 6 Ultimate, vivo X100 Pro) 5,000-6,000 mAh 7-8 . 800-1,000 mAh

- **100W**: - 100W/120W 17 2025 150W-240W 80% 10

- : 2024 ” - + 100W”

:

- AI 3 : , NPU TOPS, AI Agent
- : BeiDou Tiantong
- : , 70%, /
- 6G , Wi-Fi 7 , Bluetooth LE Audio
- - + 100W:

: ” ”

3 “ / / ” AI, , , Wi-Fi 7, “ ” :

- : 5,000mAh, 80W, NFC, IP68
- : AI , Wi-Fi 7, BeiDou
- : ( ), ( ), 240W

- AI 3 : , NPU TOPS, AI Agent
- AI : Apple , Google , Huawei/Xiaomi/OPPO/vivo
- : BeiDou Tiantong
- : , 70%,
- 6G 2027, Wi-Fi 7 , Bluetooth LE Audio
- - 5,000-6,000mAh

5. (27-32)

27: —

“ ”

WeChat *2* “ ”

: “ ” “ ”

:

—

3 : (1) — (2) —
“Wi-Fi ” (3) ——

: “ ” ——

“ ” (“ ” “ ” “ ”) :

- ——
- —— , , , , WeChat App
- —— (, ,)
- —— , ,

( ):

iPhone	( )	> > ” > ” ” ” ” ”
Huawei/Honor		> > >
Xiaomi/Redmi		> > >
OPPO/OnePlus		> >
vivo/iQOO		> , >
Samsung		> >

“ ” “ ” “ ” “ ”

—— “ + ” 5

“ ” 5 :

1. : > > ” ” (” ”)

2. : ( ) > ( )
(3×5)

3. ” ”: ” ”

4. : > App , WeChat

5. : > > , ,

: ” ”

1. ( )

2. (> >)

3. Bluetooth, NFC,

4. 4 : , , ( ), /

——

6 ( ):

- **(1)** ” ”—— ” / ”
  : ” ” .
- **(2)** ” ”——SMS ” ”
  : **SMS** . 12333
  ( )
- **(3)** ” ”—— ” ” ” ” WeChat SMS
  : **5%** ” ” **99.9%**
- **(4)** ” / ”——
  : ” ”
- **(5)** ” / ”—— ” ”
  :
- **(6)** ” ”—— App (Sunlogin, TeamViewer)
  : **!**

3 90%:

1. **1:** —— Android ” ” (App App) iPhone ” ” (> >)
2. **2:** —— SMS ”KTFSR” 10086 China Mobile ; China Telecom China Unicom
3. **3: App** —— App ”National Anti-Fraud Center” ( ) App Store App

” ” ( )

- **!**
- **SMS !**
- **!**
- **!**

- !
-

—

“ ” —

:

- **Huawei/Honor**: ”Changlian” ” ” Changlian ” ” ” ”
- **Xiaomi/Redmi**: App Xiaomi ” ” +
- **OPPO/vivo**: ” ” ” ” ” ”
- ( ): ”Xiangrikui Yuancheng Kongzhi” ( )
- **iPhone**: FaceTime ” ” FaceTime iOS ” ”

“ ” WeChat —

WeChat App :

- **WeChat** :WeChat > > > > WeChat
- ” ”:WeChat > > > > ” ” ( )
- : — ” - ” ” - ” WeChat
- ” ” : (Android) (iPhone) ” ”
- ” ”:WeChat > > > > ” ”” ”” ”” ”” ”

? “ ”

:

1. ” **iPhone** ” ” ” :iPhone > [ ] > > iPhone ; Android ” ”
2. : Apple ID / Huawei / Xiaomi
3. :” ” iCloud Android ” ”

—

iPhone Android MFi (Made for iPhone)

- :iPhone > > Bluetooth
- : ” > / ” ” ” ” ”
- **Android**: ” > ” ” ”

—

- **SOS** : / 5 ( ) —iPhone: ” > SOS ”, Android: ” > > ”
- :Apple Watch
- :App ” ” iPhone Android ” ” App ” ”

- /
- 3 : , ,
- 6 3 90%
- ” ” WeChat + ” ” 2
- SOS 5

28: —

“ ”

3 :

“ ” :

3

1. : ? ? ? ? ? WeChat ?
2. : ” ” ” ”
3. :

— 5

5 :

(SoC)	2026: Snapdragon 8 Elite Gen5 / Dimensity 9500 = ; Snapdragon 8s Gen4 / Dimensity 8400 = - ; Snapdragon 7+ Gen4 / Dimensity 7350 = 3 Snapdragon 6	SoC
	OLED LCD; 120Hz 60Hz ; 1080P PWM (1920Hz)	OLED + 120Hz + PWM =
	;	+ >
	(3x) ;	
	4500mAh , 5000mAh : 60W , 100W	5000mAh + 60W =
	Apple 5-6 ; Samsung 4 + 5 ; Google Pixel 7 ;	
	(2-3)	

(2026)

2026

1000 : WeChat

, , , WeChat

- Redmi Note , Honor X , realme Snapdragon 6

Dimensity 6000 , LCD ,
2 : (5000mAh), (6GB+128GB, 4GB) Redmi

1000-2000 : 3-4

, ,
Redmi Turbo , iQOO Z , Honor X , OnePlus Ace
Snapdragon 7+ Gen3 Dimensity 7300 8GB+256GB + Honor of Kings 3-4

2000-3500 :

, ,
Xiaomi (Xiaomi 17), vivo S , OPPO Reno , Honor , Redmi K
“ ” — 2000 :
(OIS), OLED , (80W),
12GB+256GB 12GB+512GB 4-5

3500-5000 :

Xiaomi , vivo X , OPPO Find X , iPhone (iPhone 16)
:
“ ”

5000 :

, , ,
iPhone 17 Pro/Pro Max, Samsung S26 Ultra, Huawei Pura 80 Ultra, Xiaomi 17 Ultra, vivo X300 Pro
vivo Huawei, iPhone, S Pen Samsung, Xiaomi ,

— ?

“ ” :

	(1), , (3x),	vivo X , Huawei Pura/Mate , Xiaomi Ultra
	, , , + 5500mAh , , LTPO	iQOO , ROG Phone, Red Magic, Redmi K Honor Magic Series, One-Plus, Redmi Note
	, , , ,	Huawei Enjoy, Honor Play, OPPO A Series, Redmi Note
/	, , ,	Redmi Note/Turbo, realme GT, iQOO Neo
Apple	Mac/iPad/AirPods iPhone	iPhone 17 ( ) Pro ( )
	eSIM, , Google	iPhone, Samsung S Series, Pixel, OnePlus

—

1. **1:** — 2 200 ” ” (+) + +

2. **2:** **100** — , , 50 100

3. **3:** —” ” , , ( ) ” ”

4. **4: ” ” —** — :
, , 2026 ” ” ” ” ” ”

5. **5: ”AI”** — 2026 ” AI” AI : , AI ( ), AI, ,

?

- **618 (** **):** 800-1500
- **Double 11 (** **):** 618

- : ( ) 200-500
- :
- : —

— ?

- / **Tmall / JD** :
- **Pinduoduo ” ”**: ” ”
- : : ( ) 2-3 ” ”
- **(Xianyu / Zhuanzhuan)**: : , IMEI ,

:

1. 2 (: ,)
2. (3500)
3. JD Tmall 3-5
4. Bilibili ” XX 1 ” ( ) 2-3
5.

- 3 :
- 5 : , , , / ,
- 1500-3500
-
- 5 : , , , , AI
- 618 Double 11 2

29: —

: 5 ? 2?
“ ”

“ ” —

“ ” (iPhone)
“ ” (Android)

- **iPhone**: > > , ,
- **Android**: > Huawei ” ” Xiaomi
  ” ”

80% Douyin 4 28 —
“ ”

“ ” —

“ ” :

1. **iPhone**: > > > (Douyin) 30
2. **Android**: > >
3. : ” ”

“ ” —

“ ”
15-20 “ ”

- : 4 7 SMS

- : ” ” ” ” ” ” (3)
- : ” ” iPhone — ” ” ” ”

— 20-20-20

( )

:

20-20-20

20 , **20** **(** **6** **)** **20**

- : > > (Night Shift)
- : ” ”
- : 30

“ ”

5-6

:

- :
- **30** : 3 3 5 ” ” — ” ”
- **3** : 4 3

—

“ ” “ ”

:

- **1** : ” / ”
- : —

- ” ”:” ” iPhone Android
- : ” ”

— “ ”

“ ” “ ” :

- : Official Accounts, UPzhu ”
?”
- **App** : WeChat, , SMS App
—
- ” ”: ( )

: 1

1. : /
2. : App (30-60)
3. : (22:00-07:00)
4. : 1
5. : 5
6. : ” ” —
7. :

-
- App
- 20-20-20
- 5-6 +
- 1
- ” ” ” ”

30:

7 *55* *100* “ ”

3

3 —

“ ” 3 :

AirPlay	iPhone/iPad/Mac	Apple Apple TV AirPlay
Miracast	Android	Wi-Fi Direct Android
DLNA		” ” (/ /)

:

- **iPhone**: > > /Apple TV AirPlay Apple TV AirPlay
- **Android**: > ” ”/” ”/” ” > Wi-Fi
- : Chromecast Xiaomi/ Huawei HDMI

—

:

- **iPhone**: Lightning/USB-C HDMI (100-200) HDMI /

- **Android**: USB-C HDMI (DP Alt Mode flagship) USB-C to HDMI
- : HDMI / HDMI

—

Android “ ” — :

- **Samsung DeX**: Samsung flagship Bluetooth
- **Huawei**: Huawei Mate/Pura App
- **Ready For**:Moto flagship DeX
- : ( ) PPT Excel

—

:

- **Apple CarPlay**: iPhone USB iPhone — (Amap/Baidu), , , , Siri
- **Android Auto**: Android CarPlay Google Maps, Spotify Google
- **Huawei HiCar**: Huawei Huawei
- **Baidu CarLife**: iOS Android CarLife

—

:

- : 10 Wi-Fi

- **Plan B**: HDMI
- : WeChat
  ” ” ” ” ( )
- ” ” ” ”: PPT PPT WPS

:

1. : Wi-Fi
2.
3. (iPhone , Android)
4. DLNA push (” ” /)
5.

- : AirPlay (Apple), Miracast (Android), DLNA ( )
- HDMI
- Samsung DeX/ Huawei
- : 4 CarPlay/Android Auto/HiCar/CarLife
- : , Plan B,

31:

2026 “ ”

— *Premiere*

: ?

“ ” 3 :

1. ?——

2. ?—— ( ) ( ) ( ) ( )

3. ?——Douyin/Kuaishou ( ), Xiaohongshu ( ), Bilibili ( ), ( )

—5

1. **1080P** **30fps**: 4K 60fps

2. :

3. : 2 ”AE/AF Lock”

4. ?—— Douyin/Kuaishou/ /Xiaohongshu (9:16), Bilibili/YouTube (16:9)

5. : > > Rule of Thirds

5 — “ ”

“ ” — 5 :

- **(Push)**: ” ”

- **(Pull)**:
- **(Pan)**:
- **(Track)**:
- **(Follow)**: Vlog

: “ ”
(DJI OM 500-800)

— Jianying

Jianying (CapCut)
:

1. : Jianying > >
2. : :
3. : ” ” ”Cross Dissolve” ”Fade to Black” 3D
4. :Jianying ” ” 90%
5. : Jianying BGM 1/3
6. : 1080P 30fps ” ”

—70%

“ ” :

- : ”3 ”
- : ”clickbait” —
- :
- **Jianying**: Jianying

3

- **3** : 3 3 ” ” —

- :
- : ” ” 20 2-3

: 30

1. (” ”)
2. 3-5 5-10
3. Jianying BGM
4.
5.

- :
- 5 : 1080P/30fps, , , ,
- 5 : , , , , —
- Jianying: BGM 6
- 3 , ,

32:

“ ”

0: / -

13 “ ”

:

1. : ( )
2. :
3. :
4. : (” XX / ”) SMS/WeChat
5. : 7 SOS / ( ) Bluetooth
6. : - Bluetooth

/ / :

- :
- : ( ) 119/110
- :
- : ” WeChat/SMS ”

4

1.

2. ” ” 2-3 (iOS/Android)

3.

4. ” ” ( )

1: / - 30

30 “ ” :

30

1. (1):

2. (2-5): : iPhone: iCloud.com > Find My > >

3. Android : Huawei cloud.huawei.com / Xiaomi i.mi.com / OPPO cloud.oppo.com

4. SIM (5-10): (China Mobile 10086/China Unicom 10010/China Telecom 10000) SIM SIM

5. (10-20): Alipay: 95188 Alipay > > > >

6. WeChat: 110 WeChat WeChat >

7. : Internet Banking

8. (20-25): Moments WeChat

9. (25-30): 110

2: - 72

:

1. : ” ”

2. : ! !

3. 96110 : (24)

4. 110 :

5. / : chargeback

6. : Apple ID/ WeChat Alipay

3: / -

:

- 1: (Safe Mode) - Android - -
(”Safe Mode”)
- 2: - [ ] > [ ]
- 3: - Tencent Mobile Manager 360 Mobile Guard
- 4: ( ) - :

- iPhone sandbox iOS iPhone
99%

4: -

1. :

2. : ” ”

3. : 7

4. :

5. : /

6. : WeChat/Alipay

5: -

: (1) “ / ” (2) “
” :

- 99.9% :

- : ” ” ” ”
- :
- :
- :

6: AI /AI -

2024 AI AI “ ”
“ ”

1: - 4 AI

- : 15-20 AI
- :
- : ” ”
- : ” ”

2:

- : - AI ” ”
- : ” ?”
- :

3:

1. : ” ” ” ”
2. :
3. 96110: ”AI / ”
4. 110 : AI
5. : Moments

7: SIM swap ( ) - “ ”

SIM swap ( )
SMS
Alipay WeChat

? 3 SIM swap

- ” ”/” ”
- SMS ” ” ” ”
- SMS

-

- PIN : [ ] > [ ] > [PIN] PIN
- : China Mobile/China Unicom/China Telecom ” ” ” ” SMS
- SMS : Apple ID (Google Authenticator, Microsoft Authenticator) SMS
- : ” XX ”

-

1. : China Mobile 10086 / China Unicom 10010 / China Telecom 10000
2. : Alipay WeChat Apple ID
3. : Alipay 95188 , WeChat , Internet Banking
4. :
5. : SIM swap (, ,)
6. :

- 5

1. +
2. ” ”
3. PIN ([ ] > [ ] > [PIN])
4.
5. ( )

: ” ”

1. : (China Mobile 10086/China Unicom 10010/China Telecom 10000) 96110 110
2. Apple ID/
3.
4. : ” ” PIN

- 30 :
- 72 : 96110
- : ( )
- :
- 99.9%
- AI / : / / /
- SIM swap: ” ” PIN
- 5 : +2FA PIN

6. (33-34)

33: —

“ ” *3* : , “ ”

AR/VR Spatial Computing: “ ”

AR/VR Spatial Computing:

- **Vision Pro**: Vision Pro Apple ”Spatial Computing” (29,999) 2026-2027 Vision Spatial Computing ” ” ” ” iPhone Vision
- **AR**: ” ” Meta Ray-Ban Rokid, XREAL, INMO AR AR 2025-2026 , HUD AI Assistant Waveguide ” ” 3-5
- : Samsung Galaxy Ring Oura Ring ” ” ” ” + ”

Brain-Computer Interface:

:

- **Neuralink**:Neuralink Elon Musk 2024 Noland Arbaugh BCI (Brain-Computer Interface) ” ”
- **BCI** : (Kernel, OpenBCI) ” ”
- :Brain-Computer Interface 10-20 (” ”)

“ ”

“ ” — “ ” :

- : ” ”
- : ” ”
- **2 :** : Cloud

: ?

: “ ” “ ” ?

- **AI**: AI ” ” ? AI Agent ?
- : Cloud? GDPR ” ” ” ”
- : AI, , ” ” ” ” — 20 ” ”
- ” ” ?
- ” ” ?

—

10 :

1. :
2. ” ”
3. ” ”
4. ” ” ” ”
5.

: ” ”

“ ” (AR AI) :

- : 1: ” ”
- : 2:

- : 3:

- Vision Pro AR
- Brain-Computer Interface 10-20
- ” ”
- AI, , 10
-

34: —

“ ” “ ”

3

“ ” 3 :

- **Cast** : ” ” ” ”
  : Apple CarPlay, Google Android Auto, Huawei HiCar, Baidu CarLife App

- : ” ”
  : HarmonyOS (AITO/Luxeed), BYD DiLink, NIO OS, Li Auto

- **CarPlay 2.0** :CarPlay Apple 2025 ” Cast ”
  Ferrari, Aston Martin, Porsche
  ” ”

CarPlay / Android Auto / HiCar

Cast 3 :

Apple CarPlay	iPhone		/ / /	
Android Auto	Android (Service Framework)	/	CarPlay	VPN
Huawei HiCar	Huawei Android		HarmonyOS	
Baidu CarLife	iOS Android			

HarmonyOS:

CarPlay “ ” HarmonyOS “ ” Huawei Wenjie/Zhijie :

- : App
- : ” ”
- : Huawei

“ + ” :

- **NIO OS NIO**: NOMI NIO Phone ” + ” NIO
- **Li Auto**: ”Task Master” ” ”
- **Xiaomi HyperOS**: Xiaomi HyperOS ” ”
- **DiLink BYD**: App App Store

— ?

?

- **Motorola Ready For /** : Motorola, Samsung Huawei ” ” ( )
- **vs** : + : (1) (2) (3)
- : CarPlay/HiCar Aftermarket

“ ” “ ”

- **CarPlay** : (1) Bluetooth (2) (3) Bluetooth (4) (5) CarPlay
- : (1) Wi-Fi (2) App ” ” ( ) Wi-Fi (3)
- : (1) ” ” iOS ” Bluetooth ” (2)” ” Android App (Tasker) Bluetooth (3) Huawei ” ”

: (1) CarPlay/HiCar (2) (3) App Store App (4) OTA —

:

1. (CarPlay / HiCar / Android Auto / CarLife)
2. Bluetooth
3. ” ” Bluetooth
4. App (, Li Auto, NIO App)

- : (CarPlay), (HarmonyOS), CarPlay 2.0
- CarPlay ; HiCar ; Android Auto ; CarLife
- HarmonyOS
- NOMI NIO, Li Auto, - - Xiaomi, BYD
-
- : CarPlay Bluetooth

(35)

35: —

“ ” “ ”

?

:

- **3-6** : 15
- **7-9** : ” ”
- **10-12** : -
- **13** : ” ”

” ”

: 1500-2500

“ ”

“ ” :

- **Xiaoai**: ” ” App
- **Huawei**: Huawei
- **Xiaomi**:

: **4G** (2G/3G)

:

iOS:

- **App**: > > App App (30)
- : " " (22:00-7:00) App

- : App App
- **Family Sharing**: iPhone

Android: Google Family Link /

- **Google Family Link**: Google Android
- **Xiaomi**: > App
- **Huawei**: > > App
- **OPPO/vivo**: " " App " "

	iOS	**Android Family Link /**
App App	Family Sharing	App ( ) Family Link / Family Link

" "

(" ")

" " :

- **Douyin/Kuaishou**: 40 (22:00-6:00)

- **WeChat**: WeChat Pay ( )

- : 20:00-21:00 . 1

“ ”

“ ”

90% 10% “ ”
“ ” :

- :

- :
WeChat

- : ” ” ” ”
” ”

- : QR Code App Store
(” ”)

: ” ”

:

- : ” 19:30-20:30 .”
- :
- : App
- : ” 1 ”
- : ” ”

:

- : 3-6 1 6-12 1 12
2

- : ” 20-20-20” — 20 20 (6) 20 1
- : 30
- : ” ” ” ”

” ”

(, , ,) :

- : ” ” ” ” (Apple, WeChat)
- : 14 App ” ” ” ”
- : /

” ”

“ ” — , , : ? “ ”

- 3-6 , 13
- ” ”:
- iOS Android Family Link/
-
- : , ,
- —
- ” ”

36: -

“ ” “ ”
“ ” “ ”

:

“ ”

:

- Matter: 2022 CSA (Connectivity Standards Alliance) Apple, Google, Amazon, Samsung 2023-2026
  Apple HomeKit, Google Home, Amazon Alexa
  Mi Home
- Thread: Matter ( )
  Thread Border Router (HomePod mini, Apple TV 4K, Xiaomi)
- Wi-Fi:
- Bluetooth (BLE): -
- Zigbee: Matter+Thread

2026 “Matter over Thread” Wi-Fi Bluetooth Matter

“ ” “ ” Matter

:

Apple HomeKit Mi Home (Xiaomi) Huawei AI Life (HarmonyOS)	( ) iPhone/Siri/Shortcuts Xiao AI / / HarmonyOS	HomePod Apple TV /	Apple Huawei ”Super Device”

: iPhone HomeKit Huawei AI Life Android Mi Home Matter “ ”

“ ” “ ”

:

- : HomeKit/Mi Home/AI Life -
- : ” ” : : + + + + +
- : ( ) + + + +
- : 23:00 + + 26 +

NFC : NFC ( ) iPhone Shortcuts Android Wallet

: ” ”

“ ” : “ ”( ) “ ”“ 26 ” “ ”

:

- : Xiao AI (Mi Home) / HomePod mini (Apple Thread hub) / Alexa ( ) :
- : Yeelight (Mi Home/Matter) / Philips Hue (Hue Bridge) : (E27/B22)
- : Xiaomi Smart Plug ( ) / TP-Link Kasa (Matter) : ( )
- : Loock/Xiaomi (Mi Home) / Kaadas ( ) : Matter

( ) ( ) ( )
“ ”

:

1. :
” ”
2. : ” ”
3. : IoT Wi-Fi ( )
4. : ” ” ( ) 24

- ” ”
-
- ”white label”

- Matter+Thread 2023-2026
- : Apple , Mi Home , Huawei
- ” / / ”
- :
- -

7. AI (37-47)

37: - “ ” “ AI ”

2025

?

Vibe Coding:

2025 Andrej Karpathy - - :
“ Vibe Coding (vibes) AI
”

:

AI Karpathy :
AI “ ” “ ”

: Vibe ?

Vibe “ ” Vibe Coding “ AI ” AI
“ ”

“ ” :

Vibe Coding 2026
“ ”

AI

2026: Agentic Engineering

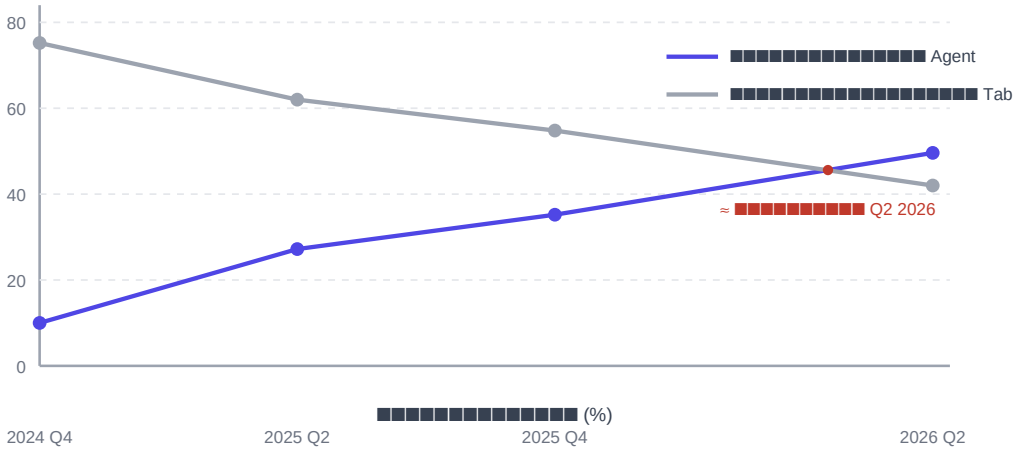
2026 AI Vibe Coding :
AI “ ”
Agent

Karpathy 2026 Agentic Engineering Vibe Coding “ ”
Agentic Engineering : AI AI
(Agent)

:

1. : ” ” ” ”
2. : ” ” ” ”
3. : ” ” ” ”

: Cursor 3 2026 Agent Tab completion
Claude Code AI 2026
2.5 :



1: AI (2/2026 Agent Tab completion)

: Cursor Claude Code GPT-5
Claude 4.5 AI

: Coder Conductor

Coder - API AI
“ ” (Conductor):
“ ” “ ”
AI

:

	Coder	**AI (Conductor)**
	API /	AI

Coder AI

: AI
AI “ ”
“ ” “ ”

AI “ ” “ ”
2026 2025 - Trae, Qoder , Cursor IDE
Cursor 3, Claude Code

- :
AI

“ (methodology)” “ ”
: ( ) “ ” “ ” ( )
“ ” “ ”

: ” ”
: (1) AI ( )? (2) “Coder”
“ ” ? (3) ? 47

- 2025 Karpathy AI ”Vibe Coding”:
- 2026 Agentic Engineering: AI
- Agent Cursor 3 Tab completion ; Claude Code 2.5
- Conductor ( ):
- -

38: AI

“ ”

: Andrej Karpathy CLAUDE.md 70
AI 4 Star GitHub 70,000 AI programming
- -

70 ?

Star 70,000 ? GitHub 70 :
AI “ ” 70 “ ”

: Think Before Coding ( )

AI “ ” ? : AI
AI “ ” - “AI ”
“ ” AI “ ” :

1. ? ( )
2. ? ( )
3. ? ( )

	( )	( )
	( )	AI 3 (responsive) HTML+CSS+JS

: ≠

“ ” “ PRD ” 30 30
(40)

: Simplicity First ( )

AI “ ” “ ” AI “ ”
- AI
“ ” “ (minimum viable)” :
Karpathy : “ ”
“ ” :

- AI ;
- dependency ;
- 50 AI 500 ;
- AI ” ”

	( )	( )
	MySQL + ORM OAuth2.0 + JWT + React + Redux + Router Docker + CI/CD + K8s	JSON SQLite HTML + JS

: Surgical Changes ( )

AI AI
“ ” :
“ ” : AI “ calcu-
late_discount” “ ” “ ” AI
Karpathy diff ( ) AI “ ” :
- “ ”
AI “ ”
( )

( )

```
      calculate\_discount   cart.py: 1.                2.          10\% 3.
                  4.
```

: Goal-Driven Execution ( )

: AI “ ” “ ” ? Karpathy AI
“ ” “ CSV ” AI

“ ” AI
“ ” “ ” _

: ≠

AI : ? ? ?
- “ ” “ IndexError ”

( )	” Python Data-Parser parse filename csv ...”	AI
( )	” : CSV JSON ”	AI

: Iterate and Verify ( )

Karpathy : AI AI
“ - - ” “ - ”
“ ” “ ” AI :

- : ? - ?
- : ?
- : ? hard-code ? ?

AI

2: AI

“ ” “ ” “ IndexError
15 ” AI

Harness Engineering: Agent = Model + Harness

2026 Harness Engineering AI programming :
Agent = × Harness
AI Harness “ ”

: GPT-4 Claude ?
Harness Harness :
Harness

- IQ
Harness Engineering “ ”

Harness : 39 ( ) 40 (PRD)
43 (Prompt) 45 ( ) Harness

: ” ”

( ) 3 : (1) ? (2)
? (3) ( )? “Think Before
Coding” -

- CLAUDE.md 70 Karpathy (Star 70K) ” ”
- Think Before Coding: - ? ? ?
- Simplicity First: AI ” ” -
- Surgical Changes: ” ” AI
- Goal-Driven Execution: ” ” ” ” AI

- Iterate and Verify: (/ /) +
- Harness Engineering: = × Harness AI ” ”

39: - AI

: AI AI “ ” “ ” AI 2026 - (Spec-Driven Development)

: “ ” AI

AI : “ ” AI “ React” “ Node.js” “ PostgreSQL” “ ” -

“ ”- AI :

Cursor	`.cursorrules`
Claude Code	`CLAUDE.md`
GitHub Copilot	`copilot-instructions.md`
	`AGENTS.md`

: AI “ ” : ( ) ( ) ( ) ( ) ( ) CLAUDE.md Karpathy 70 -

```
#     :
##

##
- Frontend: React + TypeScript + Tailwind CSS
- Backend: Node.js + Express + TypeScript
-     : SQLite (   ) / PostgreSQL (     )
-    : Vitest
##
-
-     2
-       camelCase
```

```
-      PascalCase
-            JSDoc
##
/src
  /frontend  -
  /backend   -
  /shared    -
  /tests     -
##
-
-
- API        { success, data, error }
```

: ” ”

- “ AI ”
“ ” “ AI ”

(SDD)

Spec-Driven Development (SDD) 2026
: TDD ( ) “ ” SDD
AI

SDD :

1. (.md);
2. ( );
3. AI ;
4. ;
5. AI

“ (single source of truth)” -

?

AI

/
/

3: (SDD)

SDD : AI
“ ” -

(context window)

“ ” ( ) AI
AI “ ” -
“ ” :

:		CLAUDE.md
:		@
:		
:		

Context Engineering

Context Engineering :

1.

2. @ AI @file @folder
AI

3. ” ” context win-
dow AI ” ”

4. AI ” ” ” ” bug @ log AI
” error ” AI

5. AI

: 70

( ) CLAUDE.md: (2-3) ( )
(3-5) (tree) (2-3) 70 —

- AI " ":
- (CLAUDE.md) " " AI
- = + + + + 70
- SDD : " "
- : @
- : , @, , ,

40: Requirement Engineering — AI

Context Engineering “ *AI* ” *Requirement Engineering* “ *AI* ” *AI* —

— “ ” “ ” “ to-do list”

AI Requirement Engineering

AI

: PRD “ 5 ” :

		:
Who	?	
What	?	
Why	?	
How	?	front-end React back-end
Constraint	?	2 de- ploy

5

“ ”

“ ” :

“ ” :

1. ( );
2. ( );
3. ( );
4. ( )

: ” ” ” ”

—

( ) AI “AI ” — AI

PRD

(full-stack front-end, back-end) 5
(PRD) : PRD AI —
AI

PRD : ( ) ( ) (tech stack) ( ) ( ) ( )

```
# PRD
##

##
1.      :          (   /   )
2.      :
3.      :
4.       :
5.      :     CSV
##
- Frontend: React + TypeScript + Tailwind CSS
-     : localStorage (       )
-   :   Chart.js
-   : Vite
##
Record {
  id: string
```

```
  amount: number
  type: 'income' | 'expense'
  category: string
  note: string
  date: string  // YYYY-MM-DD
}
##
Step 1:          (CRUD)
Step 2:
Step 3:
Step 4:
Step 5:
```

PRD : type
PRD AI AI

:

Requirement Engineering : “
learning tracker ”
5

- Who:
- What:
- Why:
- How:
- Constraint: 1

1. Step 1: ” ” ( );
2. Step 2: ” ” ( );
3. Step 3: ” ” (contribution GitHub);
4. Step 4: ” ” ( )

PRD — PRD .md
AI
AI :

```
                —
                                ( )
    localStorage
     React + TypeScript
```

: ( ) ; (React, TypeScript, localStorage); AI “ ” “ ” AI AI Requirement Engineering

: 5 ” ”

5 : Who / What / Why / How / Constraint
3-5 “ ” : “ ” “to-do list”

- 5 : Who / What / Why / How / Constraint —
- : ” ”
- PRD :
- = + + ” ” ” ”
- : 5 PRD AI

41: AI 2026

: “ ” 2026 —
“ ” “ ”

AI 2026 :

AI IDE Agent Builder	Cursor 3, Trae, Windsurf, CodeBuddy Claude Code, Codex CLI, Aider v0, Bolt, Replit Agent	AI AI	 CI/CD

AI

“ ” “ ” :

- ? Builder AI IDE Agent
- ? Builder AI IDE refactor Agent
- deploy ? Builder AI IDE workflow Agent

AI IDE

AI IDE : AI AI
2026 Cursor 3, Trae, Windsurf CodeBuddy

Cursor 3

Cursor AI IDE Cursor 3 2026 AI:
Agent Tab — “AI ” “AI ”
Cursor 3 : (AI), ( )
(AI)

Trae

Trae AI IDE ByteDance SOLO (Builder)
“ ” — AI Trae

Windsurf (Codeium) “ ” CodeBuddy
Tencent Cloud VS Code deploy Tencent Cloud
WeChat Mini Program Tongyi Lingma Alibaba Comate Baidu IDE

Agent

Agent 2026 : AI —
dependency Claude Code

Claude Code

Claude Code AI Anthropic 2.5 2026
IDE :
—

Claude Code “ ” :
CLAUDE.md AI refac-
tor Agent IDE

Codex CLI Aider

Codex CLI OpenAI /import — AI
Aider AI (GPT, Claude, DeepSeek)
: AI

Builder

Builder AI

- v0 (Vercel): UI front-end
- Bolt: full-stack front-end, back-end
- Replit Agent: IDE Replit deploy

“ ” — :
IDE “ 0 1”: Builder IDE

AI :

2024 2026: 2024 GitHub Copilot 2025 Cursor
2026 Claude Code “ ” :

: ( )
— Cursor Claude
Code refactor v0

: ” ”

(deploy) : Bolt v0
; Cursor Trae ; Claude Code :
?

- : AI IDE ( ), Agent ( ), Builder ( )
- : ? ? deploy?
- Agent Cursor 3 Tab ; Claude Code 2.5
- Builder ” 0 1” IDE
- :

42: API

“ AI ” : “ AI ? ?” 2026
— GPT, Claude, Gemini DeepSeek, Qwen, GLM, Kimi, Doubao
: API
Token Plan ( )

:

2026 : OpenAI (GPT), Anthropic (Claude), Google (Gemini) —

OpenAI GPT-5.6:

OpenAI GPT GPT-5.6 2026 : GPT-5.6 Sol reasoning GPT-5.6 Terra GPT-5.6 Luna context window 1.05 Token input structured output

		Input (/ Token)	**Output (/ Token)**	**Context Window**
GPT-5.6 Sol	(reasoning)	US$5.00 (36)	US$30.00 (217)	1.05 Token
GPT-5.6 Terra	( )	US$2.50 (18)	US$15.00 (109)	1.05 Token
GPT-5.6 Luna	( )	US$1.00 (7)	US$6.00 (44)	1.05 Token

OpenAI GPT-5.6 (2026)

GPT “ ” —reasoning, , , OpenAI toolchain “ ” GPT “ ” : “ ” API

OpenAI 2026 GPT-5.6 Luna
GPT reasoning API

Anthropic Claude:

GPT Claude “ ” Claude Anthropic SWE-bench (benchmark)
hallucination ( ) Claude 1% —
Claude : context window 1 Token “ ”
refactor Claude
Code AI 2.5 2026 —
Claude

		Input (/ Token)	**Output (/ Token)**	**Context Window**
Claude Fable 5		US$10.00 (72)	US$50.00 (362)	1 Token
Claude Opus 4.8		US$5.00 (36)	US$25.00 (181)	1 Token
Claude Sonnet 5		US$2.00 (14)	US$10.00 (72)	1 Token

Anthropic Claude (2026, Sonnet 5)

Claude — output Fable 5 Token 50
( ) Sonnet 5

Google Gemini:

Gemini Google 3.1 Pro 2026 Gemini native —
OCR Google
(Search, ,) Gemini input 2 Token
context window
Gemini “ reasoning ”
Gemini Claude

	GPT-5.6	Claude Fable 5	Gemini 3.1 Pro
(SWE-bench)	94.8 74.9%	95.0 64.3% (Pro)	92.1
Context Window	1.05 Token	1 Token	2 Token
hallucination		(1%)	
API			Google

(2026)

:

GPT, Claude, Gemini Azure proxy, API

:

2026 “ ” “ ” “ ” 2023 “ ”
2026 logical reasoning, 5%
deploy

DeepSeek V4:

DeepSeek 2025 DeepSeek-V3 “ GPT ”
“ AI ” — 2026 DeepSeek
V4-Pro : MoE 1.6 49B 1

DeepSeek V4 Agent — DeepSWE benchmark 12.8 62.7 (+390%)
NL2Repo 38.5 61.5 API Responses API Anthropic API
Codex Claude Code API DeepSeek API Claude
Claude Code —

		Input (/ Token)	Output (/ Token)	Context Window
DeepSeek V4 Pro	(reasoning)	3 (US$0.43)	6 (US$0.87)	1 Token
DeepSeek V4 Flash	( )	1 (US$0.14)	2 (US$0.28)	1 Token

DeepSeek V4 (2026,)

: input 1 Token + output 1 Token DeepSeek V4 Flash 0.42
GPT-5.6 Sol 35 Claude Fable 5 60 80-140

: DeepSeek

DeepSeek 6 2026 “ API ”
30 14 DeepSeek V4
(/) —

Qwen3.7 Max: Alibaba Cloud

Qwen Alibaba Cloud Alibaba Cloud deploy Qwen3.7
Max 2026 : dense 235B native 256K 1M Qwen
: (7B 235B) Alibaba Cloud

	Output (/ Token)	
qwen-turbo	0.6	
qwen-long	2.0	10M
qwen3.7-flash	7.2	
qwen3.7-plus	12.0	(+)
qwen3.7-max	36.0	

Qwen (2026)

Qwen — qwen-turbo (output 0.6 / Token) qwen3.7-max
(output 36 / Token) reasoning Alibaba Cloud
Qwen

GLM-5.1 Zhipu: SWE-bench

GLM Zhipu AI 2026 GLM-5.1 SWE-bench: MoE
dense 744B 1 deploy private
— GLM-5.1 input 6 output 24 Token

Kimi K3:

Kimi Moonshot AI Kimi K3 2026 2.8
Kimi retrieval augmentation —
“ ” Kimi (output 12-20 /
Token)

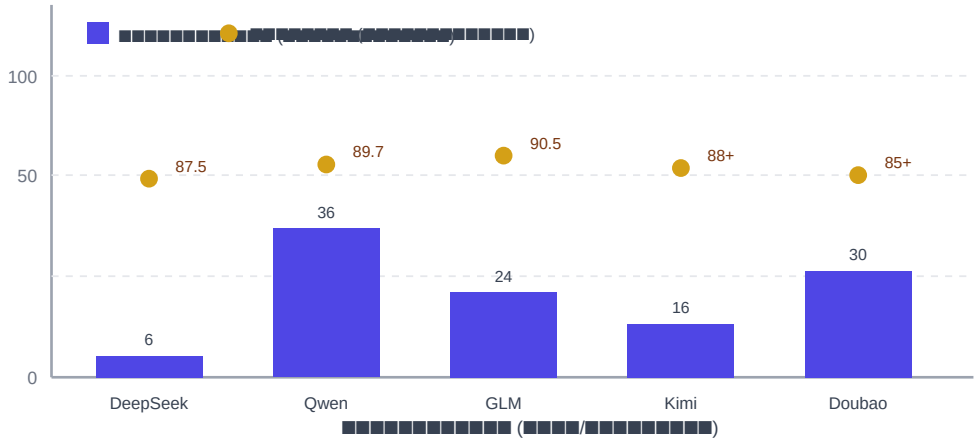
Doubao ByteDance:

Doubao ByteDance 2026 Doubao Seed 2.1 Pro Volcano Engine
256K Doubao : Token
Volcano Engine AI Doubao

	DeepSeek V4	**Qwen3.7 Max**	**GLM-5.1**	**Kimi K3**	**Doubao 2.1 Pro**
	87.5	89.7	90.5 1 SWE	88+	85+
Context Window	1	256K-1M	1		256K
Output (/)	6 MIT	36	24	12-20	30
deploy private					

(2026)

: “ ” GPT/Claude logical reasoning
GPT-5 reasoning Claude
— “ ” deploy



4: ( )

? “ ” —

?

|

?

Claude DeepSeek

|

(>500K) ?

Gemini Kimi

|

?

DeepSeek Qwen-turbo

|

GPT-5.6 Claude

5:

:

— : DeepSeek Qwen ( )
Claude (reasoning hallucination) Gemini ( )
“ ”

API:

AI “ AI” — — API API (Application Programming Interface) API

Token

API “Token” Token “
AI” Token : 0.75 (4) 0.5-1 1000 Token
1500-2000 Token

Token (Tokenizer) Token Token
— “ ” 3 Token GPT 2 Token DeepSeek API
Token API response

Token

1 1.5 Token DeepSeek V4 Flash input 1 + output 1 0.42 (3); Claude Fable 5 60 (434)

API Key

API API Key — API API Key API :

	API		
DeepSeek	platform.deepseek.com	API Key	
Qwen	bailian.aliyun.com (Bailian)	Alibaba Cloud Key	
GLM Zhipu	open.bigmodel.cn	API Key	
Kimi	platform.moonshot.cn	API Key	
Doubao	volcengine.com (Volcano Engine)	Ark API Key	Token
OpenAI GPT	platform.openai.com	API Key	( )
Anthropic Claude	console.anthropic.com	API Key	

API Key (2026)

: API Key =

API Key — Key : API Key (environment variable); push Key Git repo ; Key Key

API

API Key API Python — DeepSeek

```
#    SDK: pip install openai
# DeepSeek      OpenAI SDK
from openai import OpenAI
import os

# 1.    client (   API Key          )
client = OpenAI(
    api_key=os.environ.get("DEEPSEEK_API_KEY"),
    base_url="https://api.deepseek.com"
)
```

```
# 2.
response = client.chat.completions.create(
    model="deepseek-chat",  #
    messages=[
        {"role": "system", "content": "         Python"},
        {"role": "user", "content": "      quick sort    "}
    ]
)

# 3.
print(response.choices[0].message.content)

# 4.      Token
print(f"Token   : {response.usage.prompt_tokens}")
print(f"Token   : {response.usage.completion_tokens}")
print(f"Token   : {response.usage.total_tokens}")
```

: client (API Key) (system , user input) request OpenAI — api_key base_url OpenAI SDK:

— :

```
# DeepSeek
client = OpenAI(
    api_key=os.environ.get("DEEPSEEK_API_KEY"),
    base_url="https://api.deepseek.com"
)
# model="deepseek-chat"

# Qwen (     Bailian)
client = OpenAI(
    api_key=os.environ.get("DASHSCOPE_API_KEY"),
    base_url="https://dashscope.aliyuncs.com/compatible-mode/v1"
)
# model="qwen-plus"

# Zhipu GLM
client = OpenAI(
    api_key=os.environ.get("GLM_API_KEY"),
    base_url="https://open.bigmodel.cn/api/paas/v4"
)
# model="glm-4"

# OpenAI GPT
client = OpenAI(
    api_key=os.environ.get("OPENAI_API_KEY"),
```

```
    # base_url
)
# model="gpt-5.6-terra"
```

streaming response

AI — streaming response AI

```
#   :
stream = client.chat.completions.create(
    model="deepseek-chat",
    messages=[
        {"role": "user", "content": "      bubble sort    Python"}
    ],
    stream=True  #
)

#
for chunk in stream:
    if chunk.choices[0].delta.content is not None:
        print(chunk.choices[0].delta.content, end="", flush=True)
#
```

Token Plan

: (Pay-as-you-go) (Token Plan / Subscription)

vs

	(Pay-as-you-go)	(Subscription)
		20%-50%
	API	OpenAI Plus/Pro, Claude Pro, Cursor Pro

— (50)

ChatGPT Free		GPT-5.6 Luna	
ChatGPT Plus	US$20/	GPT-5.6 Sol, , ,	
ChatGPT Pro	US$200/		
Claude Free		Claude	
Claude Pro	US$20/	Claude	
Cursor Free		AI	
Cursor Pro	US$20/	AI , request 500 /	
Cursor Business	US$40/ /	, ,	

AI (2026)

DeepSeek — Qwen resource pack Bailian (Token) Doubao Volcano Engine : DeepSeek ( ) + Doubao ( )

API

API API Key

OpenRouter (openrouter.ai)	Key		
SiliconFlow (siliconflow.cn)	API	20%-30%	
Azure OpenAI (Microsoft)	OpenAI API Azure Microsoft	Batch API	OpenAI async
Baidu Qianfan (qianfan.baidubce.com)	AI Baidu Wenxin (ERNIE)		Baidu

API (2026)

: “Shared Key”

“API Key ” Taobao Xianyu—Key
API Key

“ ” :

1. cache hit ” ”— (system prompt)
cache hit DeepSeek 1/60 system prompt
cache hit

2. — DeepSeek V4 Flash (
2 / Token) Claude (181 / Token) 80%

3. API Token max_tokens
AI ” ”

4. Batch API ( ) Batch API
50% OpenAI Azure

5. Token API Token
Prompt (” ” ” 3 ”) Token 50%

Cache hit	98% 50%-80% 30%-60%		
Batch API	50%		max_tokens batch processing
Prompt	20%-50%		Prompt

(2026)

: API

— “ AI”: API
API : API

```
# ai_summarizer.py -        AI
import os
from openai import OpenAI
```

```
# 1.    (  API Key              hardcode    )
client = OpenAI(
    api_key=os.environ.get("DEEPSEEK_API_KEY"),
    base_url="https://api.deepseek.com"
)

def summarize(text: str, max_length: int = 200) -> dict:
    """

    Args:
        text:
        max_length:          (    Token)

    Returns:
        dict:            Token
    """
    try:
        response = client.chat.completions.create(
            model="deepseek-chat",
            messages=[
                {
                    "role": "system",
                    "content": "           "
                               "              "
                               "     3    "
                },
                {
                    "role": "user",
                    "content": f"       :\n\n{text}"
                }
            ],
            max_tokens=max_length,  #
            temperature=0.3  #     =
        )

        return {
            "success": True,
            "summary": response.choices[0].message.content,
            "input_tokens": response.usage.prompt_tokens,
            "output_tokens": response.usage.completion_tokens,
            "total_tokens": response.usage.total_tokens
        }

    except Exception as e:
        return {
            "success": False,
            "error": str(e)
```

```
        }

# 2.
if __name__ == "__main__":
    long_text = """
    (           ...)
    """

    result = summarize(long_text)

    if result["success"]:
        print(f"  : {result['summary']}")
        print(f"  Token: {result['total_tokens']}")
        #       : DeepSeek V4 Flash
        cost = result['total_tokens'] * 0.000004  #    0.4    /  Token
        print(f"       : {cost:.4f}    ")
    else:
        print(f"      : {result['error']}")
```

API : API Key ; type hint ; max_tokens ; temperature (0.3) ; ; Token AI —

: ” ” ” ”

API ” ” —

: API

(DeepSeek Zhipu) API Key (!) “ API ” AI “ AI” “ AI”

- : GPT-5.6 ( ), Claude (+ 1%), Gemini (+ 2 Token)
- : DeepSeek V4 ( ), Qwen3.7 ( ), GLM-5.1 (1 SWE), Kimi K3 ( ), Doubao ( )
- 4 : ? ? ? ?
- Token : 1 ≈ 1.5-2 Token; 1 ≈ 1.5 Token
- API Key = : Git
- OpenAI SDK : api_key base_url
- 5 : cache hit, , , Batch API,

43: — AI

AI : “ ” “ ” — *AI*
“ ” :

“ ” *AI*

Prompt

AI Prompt

	AI AI	” React Type- Script” ” ” ” / ”	

Prompt

Prompt (CLAUDE.md) —
“ ” “ ” “ ” — AI

:

“ + + ” “ 25 console.log ( )
( ) handleError ( )”
“ ”

: AI

AI AI

:

1. AI ” ” error log, stack trace @ AI
” error ” AI

2. ” ” ” ” ” ( ) ( )”
— ” ”

3. ” ”— AI

4. ” network request 200 ”— AI

```
```

```
                    :        /home        :                  error      : 1.
              admin     123456 2.    "     " 3.             : - network request
       200 -      onClick      -          LoginButton          : @src/components/
   LoginButton.tsx @src/pages/Login.tsx
```

: vs

AI

“ ” : AI

; : AI

“ ” : AI

AI ; AI : refac-
tor

	(5-10 /)	(1-3 /) refactor

refactor “ ” AI “ ”

“AI ”

: “ ” “ localStorage
” — AI

: “ ” AI “ ”

: “ ”

: AI “ ” “ ” — AI

: ” ” AI

( ) AI “ ” : ,
vs , , AI — “ ”

- Prompt : ( ), (+ +), (+ +)
- : , vs , ,
- : ; :
- : , , ,

44: —

“ ” “ ”

: ( )

: 3-4
“ ”

: Who— What— Why—
How— HTML+CSS+JS Constraint— 30

Think Before Coding AI :

```
                    :
1. HTML      (  CSS    JS)
2.
3.          4  :
   -
   -
   -
   -        QR Code
4.
5.
6.
7.
   HTML
```

AI : ? 4 ? ?
: ? ? : ? CSS style

? ?

( ) — “ ”

: ( )

PRD

: 40

PRD 3 40

Simplicity First— React Redux state management Node.js

SQLite PostgreSQL ( )

5 : Step 1 API (Express + SQLite + CRUD); Step 2

(+); Step 3 - (API +); Step 4 ; Step 5

“ ”

context engineering— AI

```
#     :
##

##
- Frontend: React + TypeScript + Vite
- Backend: Node.js + Express + TypeScript
-     : SQLite (   better-sqlite3)
-
##     API
-       : /api
-       : { success: boolean, data: any, error: string }
-    RESTful
##
Record { id, amount, type, category, note, date }
##
- TypeScript
-     2
-         JSDoc
-
```

:

```
      API
    :
1.      Express    3001
2.     SQLite      records
3.      records:
   id (TEXT PRIMARY KEY)
   amount (REAL)
   type (TEXT: 'income'|'expense')
   category (TEXT)
   note (TEXT)
   date (TEXT: YYYY-MM-DD)
4. API CRUD:
   GET /api/records -
   POST /api/records -
   PUT /api/records/:id -
   DELETE /api/records/:id -
5. Middleware
```

: ( ) ; (AI) AI ;
Goal-Driven Execution— “ ” “ ”

AI : curl Postman Step 2
( ) “ - - ”

: AI ( )

: API — “ AI” AI :
AI API, prompt engineering

: “ AI AI” prompt engineering prompt
API —

:

- API:
- :
- Token: token
- Prompt: prompt

AI “ ” — ? ? token
? “ ” AI

AI

1.		Think Before Coding
2.		Simplicity First
3. PRD		Goal-Driven
4.	CLAUDE.md	Harness Engineering
5.	- -	Surgical Changes
6.		Iterate & Verify
7.		Iterate & Verify
8.		Iterate & Verify

AI

: ” ”

“ ” 44 AI : 5 AI ( )
30 — AI “ ”
“ ”

- ( ): — 30
- ( ): —PRD + + 5 ” ”
- ( ): AI— AI AI prompt engineering
- 8 : PRD
- ” - - ”

45: — AI

AI : “ ” — “ ” “ ”
AI *AI*—
“ ” “ ”

: AI

AI : AI —
“ ”

AI : AI
AI : —

```
     unit test      calculate_discount
         :
1.    :
   -    100      20% -> 80
   -    200      50% -> 100
2.     :
   -    0      20% -> 0
   -    100         -> 100 (    )
   -    100      0% -> 100 (    )
3.     :
   -      ->    throw error
   -       > 1 ->    throw error
   -         ->    throw error
      Vitest
```

: ( ) AI “ ” AI
“ ”

AI —
AI :

	SQL injection error N+1	AI AI catch console.log AI $O(n^2)$ AI AI

AI

— AI : “ ”
AI

: Git AI

Git AI AI
—
Git AI “ ” AI

“ ” AI ( ) merge ;

1. AI ( ) AI git checkout .
2. AI ”feat: ”
3. : git checkout -b refactor/auth-system merge
4. diff AI git diff

AI

:

“ AI ?” — AI — —
“ ” : “ ”

：

“ AI ?” — AI “ ”：

： Simplicity First AI “ ”

：

“AI ?” — AI “ ”

“ ” ： AI “ ”

：

“ AI ?” — AI

：

— “ ”； —AI “ ”； — “ ”； —

“ ” ： , , ,

：

( ) ： (1) AI 3 ； (2)

(/ / / /); (3) git

- AI ” ” ≠ ” ”—
- AI ： / / ” ”
- ：
- Git ” ”: diff
- ： ( ), ( ), ( ), ( )

46: —

37 44 AI ? “ *AI*” “ *AI*” “ *AI* ” “ *AI* ”

MCP

MCP (Model Context Protocol) 2025 2026
AI MCP AI “ ” — API

MCP AI MCP : MCP
AI MCP AI “ ” “ ”

MCP : AI MCP
AI “AI ” Agentic Engineering

Skills

AI “ ” “Skills” —
Skill AI
“ ” —

:

Cursor Rules Custom Instructions, Claude Code
, Replit Runme : “ ” — AI

Agent

2026 Agent : AI AI AI
—

Agent “ ” “ ” : Agent ;
Agent :

: ” Agent”

Agent — “ ” Agent :
AI ; (CLAUDE.md) AI “ ” ; AI
“ ” “ ” Agent

AI :

: Andrej Karpathy CLAUDE.md, Anthropic
OpenAI, —

: “ ” “ ” (API
)

: — :
?

: —
“ ”

: ” ”

(.md) AI — “ ”

- MCP AI : AI API
- /Skills ” ”:
- Agent = + ; ” ”
- : , , ,

47:

“ ” :

Think Before Coding		: ? ? ?
Simplicity First		
Surgical Changes		” ” AI
Goal-Driven Execution	” ” ” ”	AI
Iterate & Verify	- -	:

AI : “ ” “ ”

?

Builder (v0/Bolt/Replit)

AI IDE (Cursor/Trae)

Agent (Claude Code)

Builder Trae SOLO

6: AI

FAQ

1: AI ?

AI 2026 AI :
Builder (v0, Bolt) ; — “ ”
AI

2: ?

—JavaScript/TypeScript; AI—Python; —Java Go
: AI “ ” AI “ ” —

3: AI ?

2026 AI
: (1) AI ; (2)
; (3) AI

4: AI ?

AI “ ” “ ”
AI “ ” “ ” —
“ ” “ ”

5: AI ?

: — ? - ? : — ?
: — ? ? ?
—

2026 :

	CLAUDE.md Andrej Karpathy Anthropic (Claude Code) Skills Vibe Coding YouTube Cursor awesome-claude-code GitHub r/cursor Reddit MDN Web Docs (developer.mozilla.org) Python (docs.python.org)	70 Agent AI IDE Claude Code Cursor Python

:

: 2026
AI — refactor
—

AI ; AI
AI ; AI AI ;
AI
“ ” —
;
AI
— AI

1. Andrej Karpathy. (2025). Vibe Coding. Social media post introducing the concept of vibe-driven programming with AI.
2. Andrej Karpathy. (2025). CLAUDE.md - Project guidelines for AI-assisted development. GitHub repository, 70+K stars.
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5. OpenAI. (2026). Codex CLI: Bringing AI to the terminal. Technical documentation.
6. Model Context Protocol (MCP) Specification. (2025-2026). Open protocol for AI-tool integration.
7. Vibe Coding Community. (2026). Community-curated skills and best practices for AI-assisted programming.
8. Spec-Driven Development (SDD) methodology. (2026). Emerging practices in specification-first AI programming.

8. (48-58)

48: Android—

— “ ” “ ” : *Android* *?*
? *?*

Android: “ ” “ ”

Android 2003— Andy Rubin Android :
“ ” Google 2005 2007 Open Handset Alliance (OHA)
2008 Android HTC Dream (G1)
: 2026 Android
Android ?

: Android

Android (AOSP, Android Open Source Project) —
Xiaomi MIUI (HyperOS), Huawei HarmonyOS, Samsung One
UI : “ Android” “ ”

: 1.5 Cupcake —Donut, Eclair, Froyo, Gingerbread... Android 10 2019 2026 Android 17
“ ” Android 8 (API 26) Android 10 (API 29)—

	/	
Android 8 (API 26)	Oreo • 2017	—
Android 10 (API 29)	2019	(scoped storage)
Android 12 (API 31)	2021	Material You
Android 13 (API 33)	2022	” ”
Android 14-15 (API 34-35)	2023-2024	16KB
Android 16-17 (API 36-37)	2025-2026	AI (Gemini)

Android (2026)

: Java Kotlin?

Android Java “ ” —Android ; Kotlin “ ” Google 2017 2019 “Kotlin ” (Kotlin-first) 2026 Kotlin

: Kotlin Java : Kotlin Java 30%-40% (Kotlin); “null pointer” ; AI (AI ?) Kotlin Java

Java ( )

```
public class MainActivity extends Activity \{
@Override
protected void onCreate(Bundle b) \{
super.onCreate(b);
setContentView(R.layout.activity);
\}
\}
```

Kotlin ( )

```
class MainActivity : ComponentActivity() \{
override fun onCreate(b: Bundle?) \{
super.onCreate(b)
setContent \{ App() \}
\}
\}
```

Kotlin 50 : “ ” Android
Kotlin

?

“ ” :

(UI+)

(APK/AAB)
(+)
(/)

1: Android

:

- APK: Android (.apk) .exe Windows
- AAB: Android App Bundle (.aab) ” ” Google Play

SDK (Software Development Kit) “ ” Android —
API Activity, Button SDK

Android: “ ”

Android 2026 —
:

	(2026)	
	Kotlin 2.x Jetpack Compose + Material 3 MVVM + ViewModel Room ( ) + DataStore ( ) Retrofit + OkHttp + Kotlin coroutines Gradle + Kotlin DSL Android Studio	( ) IDE ” ” Google

Android 2026

— “ ”

: Android Studio

Android vs iOS: “ ”

Android Apple — “ ”
“ ” :

	Android	iOS
	Windows / Mac / Linux Android Studio ( ) APK / ( )	Mac (Xcode Mac) Xcode ( ) + 99 1-7 / TestFlight App Store

Android iOS (2026)

: Mac —Android Mac 8 9

: ”Android”

[ ] [ ] (“ ”) “ Android” —
“ ”

- Android 2008 2026
- Android 17; Android 8-10
- Kotlin : 30%+ NullPointerException
- APK AAB ; SDK
- 2026: Kotlin + Compose + MVVM + Room + Retrofit + Gradle + Android Studio

49: —Android Studio

“ ” : *Android Studio* *Android*
—

Android Studio

Android Studio IDE Google :
4 :

1. developer.android.com/studio (2026 Narwhal Windows/Mac/Linux);
2. ” ” — Android SDK ;
3. SDK (GB);
4. — ” ”

:

Android Studio “ ” : 16GB (8GB) 20GB
CPU (VT-x/AMD-V BIOS) “ ”
(4)

New Project Empty Activity ( ) :

Name	My First App	
Package name	com.example.myfirstapp	
		” ”
Minimum SDK	API 26(Android 8.0)	95%+

Finish Android Studio Gradle (
) — “Hello Android!”

:

(package name) “ ” com.example.xxx
(com. .) com.example —

:

“ ” :

1) (Emulator)

Android Android Studio Device Manager Create Device
(Pixel) (Google APIs) Run
“ ”

2) ( )

— Android :

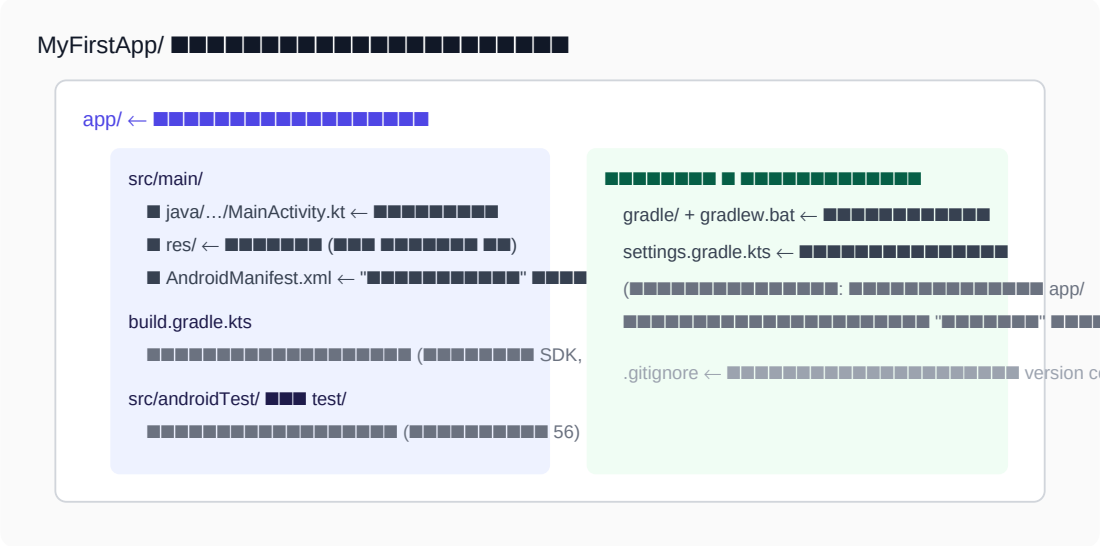
1. [ ] [ ] ” ” 7 ” ”;
2. [ ] USB debugging;
3. ” USB debugging”;
4. Android Studio Run

: USB debugging

USB debugging “ ” USB debugging

:

90% :



2: (: app/)

- MainActivity.kt: ” ” ”Hello Android!”
- AndroidManifest.xml: ” ” — ( )
- res/: Android ( )
- build.gradle.kts: — (Compose)

: “Hello”

MainActivity.kt Compose Text(“Hello Android!”) Text(“ ! ”) Ctrl+R (Mac Cmd+R)

```
setContent {
    MyFirstAppTheme {
        Surface(modifier = Modifier.fillMaxSize()) {
            // v
            Text("      !        ")
        }
    }
}
```

“ ” :

— “ ”

error : “ ”

error :

error		
Gradle sync failed / SDK /	repository SDK	Gradle repository (maven Aliyun/Tencent Cloud) ”gradle mirror” Settings SDK Manager VT-x/AMD-V BIOS; ;

error

: error

error — “android studio” “ ” “Gradle sync failed: Could not resolve com.android.tools.build”

: ” ”

1) Android Studio ; 2) ; 3) Hello ; 4) : Android ( )

- Android Studio = IDE ; 16GB + 20GB +
- : , (com.example), SDK (API 26)
- : ( ) (USB debugging)
- : app/ —MainActivity.kt , Manifest , res/
- = Run

50: Kotlin — Android

—Kotlin “ ” : 20% 80% : > >

:

“ + ” Kotlin :

```
val name = "   "        // val:          (          )
var age = 18             // var:       (       )

age = 19                 // <i class="fas fa-check-circle mr-1"></i> var
// name = "  "          // <i class="fas fa-times-circle mr-1"></i> val
    error

val score: Int = 100     //
val pi = 3.14            //
```

:

Int		42
Double		3.14
String		” ”
Boolean	/	true / false
List< T>		listOf(1, 2, 3)

Kotlin

: val

Kotlin: val var val var IDE
“ ” —

: “ ”

— ( ) Kotlin fun:

```
fun greet(name: String): String {
    return "   $name!"     //       $
}

println(greet("   "))        //    :        !

//             expression
fun add(a: Int, b: Int) = a + b

println(add(3, 5))            //    : 8
```

: ( ); ; $
Kotlin — “ + ” Java

: “ ”

: (if/when) (for/while)

```
//       : if / else
val age = 17
if (age >= 18) {
    println("  ")
} else {
    println("   ")
}

//       : when (    switch    )
val score = 85
val level = when {
    score >= 90 -> "   "
    score >= 60 -> "  "
    else -> "      !"
}

//   :   for
val fruits = listOf("    ", "   ", " ")
for (fruit in fruits) {
    println("    $fruit")
}
```

when Kotlin Android ( )

Null Safety: “ ” Kotlin

Java NullPointerException — “ ”
Android Kotlin : null

```
var name: String = "   "   //    :     null
// name = null                // <i class="fas fa-times-circle mr-1"></i>


var nickname: String? = null //         :       null
nickname = "    "

//   nullable type          :
println(nickname?.length)    // safe call:   null   null
val len = nickname?.length ?: 0   // Elvis operator:   null
println(len)
```

: ? ?

String? “ null ” a ?: b “ a null b” (Elvis operator) “error null” null —

: “ ”

“ ” “ ” Android — — Kotlin data class ( ) “ ” Java :

```
//    :
class Dog(val name: String) {
    fun bark() = println("$name     !")
}

val dog = Dog("   ")      //
dog.bark()                 //        !

// data class:       ( equals/copy    )
data class User(val id: Int, val name: String, val email: String)

val me = User(1, "   ", "zhangsan@example.com")
val twin = me.copy(name = "     ")   //
```

data class : data class Record(val amount: Double, val category: String, val date: String)

lambda high-order function: “ ” Kotlin

Android “ ” { ... } lambda ( ) onClick = { count++ }
Button lambda — “ ”

```
//    :
fun double(x: Int) = x * 2
val result = listOf(1, 2, 3).map(::double)

//    lambda:
val result2 = listOf(1, 2, 3).map { it * 2 }   // it =

//    +      (           )
val bills = listOf(12.5, 88.0, 5.0, 200.0)
val bigBills = bills.filter { it > 50 }.sorted()
// [88.0, 200.0]—"    50        "
```

“ ” : filter ( ), map ( ), sorted ( ), sum ( ) 90%
Kotlin :

:

	`val x = 1`
	`var x = 1`
	fun ( ): { ... }
	if ( ) {...} else {...}
	when { -> }
	for (x in) { ... }
	” $name”

null	`x?.length ?: 0`
data class	`data class X(val a: Int, val b: String)`

Kotlin ( )

: Kotlin 10

play.kotlinlang.org (Kotlin)
10 “ ” 51

- val var ;
- = ; $
- if/when for ; when ” ” Android
- null safety Kotlin: null null ? / ?:
- data class
- 20% 80%:

51: — Jetpack Compose

Compose Android XML — ; 2021 Google Jetpack Compose: Kotlin “ ” 2026 Compose “ ”

Compose:

Compose : = ( ) “ ” (Composable) @Composable

```
@Composable
fun Greeting(name: String) {
    Text(text = "   $name!")     //    "   "
}

//       :
@Composable
fun HomeScreen() {
    Column {                          //
        Greeting(name = "    ")
        Greeting(name = "  ")
        Button(onClick = { /*        */ }) {
            Text("   ")
        }
    }
}
```

“ ” @Composable — “ ” Compose

:

Compose “ ” :

Text		text ( )
Button		onClick ( )
TextField		value, onValueChange ( )
Image		painter ( )

Compose

```
@Composable
fun BasicWidgets() {
    Column {
        Text(text = "     ")
        Button(onClick = { println("    ") }) {
            Text(" ")
        }
        var input by remember { mutableStateOf("") }
        TextField(
            value = input,
            onValueChange = { input = it },
            label = { Text("     ") }
        )
    }
}
```

remember { mutableStateOf(“ ”) } — “ ” Compose 5

: Column, Row, Box

? “ ” :

- Column: ( );
- Row: ( );
- Box: ( )

```
Column {
    Text("    ")
    Row {
        Text("  ")
        Text("  ")
    }
    Box {
```

```
        Text("   ")
        Text("         ")
    }
}
```

? Modifier ( ) — :

```
Text(
    text = "      ",
    modifier = Modifier
        .padding(16.dp)          //
        .fillMaxWidth()          //
        .background(Color.Blue)  //
        .clickable { /*    */ }//
)
//   :              (padding                )
```

: Modifier

padding(16.dp).background(Blue) background(Blue).padding(16.dp) —
“ ” ( ) “ ” ( ) “ ”

: “ ”

+1 (State) Compose:

```
@Composable
fun Counter() {
    // remember + mutableStateOf:
    var count by remember { mutableStateOf(0) }

    Column {
        Text(text = "     $count   ")
        Button(onClick = { count++ }) {   //
            Text("    +1")                //
        }
    }
}
```

Compose : “ ” : remember
(recompose) mutableStateOf (Compose “ ”)

(count++)

Compose

UI (recomposition)

3: “ ” Compose

:

MyFirstAppTheme { ... } Material 3
:

1. / (ColorScheme);
2. isSystemInDarkTheme() ;
3. (MaterialTheme.colorScheme.primary)

: Material 3

Android 12 “Material You” Compose (dynamicColor) “ ” Android

: “ ” Compose

Compose :

- @Preview: annotation @Preview Composable Android Studio —
;
- (Live Edit): Android Studio 2026 ” ” /
;
- API : animateFloatAsState, AnimatedVisibility —” ”

```
@Composable
fun AnimatedBox() {
    var big by remember { mutableStateOf(false) }
    //
    val size by animateDpAsState(if (big) 120.dp else 60.dp)

    Box(Modifier
        .size(size)
        .background(MaterialTheme.colorScheme.primary)
        .clickable { big = !big }
    )
}
```

:

“ ” — “ ” :

: ” ”

: : when
“ ”

- Compose: = @Composable
- : Text / Button / TextField / Image
- : Column , Row , Box
- Modifier : padding, fillMaxWidth, background, clickable —
- : remember + mutableStateOf
- Material 3 +

52: Activity

Compose “ ” “ ” *Android* *Activity* “ ” *Android* “ ”

: “ ” Android

Android :

Activity		—
Service	” ”	—
BroadcastReceiver	(broadcast)	— ” ” ” ”
ContentProvider		—

Android

95% Activity Service BroadcastReceiver ContentProvider ( ) : “ ” AndroidManifest.xml

Activity: “ ” Android

: ... Activity :

onCreate (:)

onStart ( )

onResume ( )

onPause ( )

onStop ( )

onDestroy ( )

onRestart / onStart / onResume:

4: Activity (:)

Callback		
onCreate	Activity ( )	
onStart		
onResume		
onPause	/	
onStop		
onDestroy		

callback

: “ ” Activity

: : Activity (onDestroy onCreate) : rememberSaveable (remember Compose) 6 7

Intent: “ ”

Intent ( ) “ ” “ ” :

```
//     : Intent    —
val intent = Intent(this, DetailActivity::class.java)
intent.putExtra("name", "    ")     //       "    "    (  )
startActivity(intent)

//     : Intent    —
val web = Intent(Intent.ACTION_VIEW, Uri.parse("https://developer.android.com"))
startActivity(web)   //           /
```

: (explicit) (implicit) ( ) Compose Composable LocalContext.current Context startActivity

Compose Navigation Compose (androidx.navigation):

```
val navController = rememberNavController()

NavHost(navController, startDestination = "home") {
    composable("home") { HomeScreen(navController) }
    composable("detail/{name}") { entry ->
        val name = entry.arguments?.getString("name")
        DetailScreen(name ?: "    ")
    }
}
```

```
// :
navController.navigate("detail/ ")     //
navController.popBackStack()            //
```

: ?

(route) (AppNavGraph.kt) “ ”
“ ” “ ”

: Fragment ( )

Fragment — “ ” Activity
Compose Composable + Navigation Fragment

: Fragment “ Activity”
2023 — “ ” Android

: ” ”

51: 1) remember rememberSaveable ; 2) “ ” Navigation ; 3) log (Log.d) onCreate/onPause Logcat

- : Activity ( ), Service ( ), BroadcastReceiver ( ), ContentProvider ( )
- callback : onCreate onStart onResume onPause onStop onDestroy
- Activity : rememberSaveable
- Intent ” ”:
- Navigation Compose: NavHost + composable

53:

“ ” “ ” *Android: DataStore ( ), Room ( ), Retrofit ( ) “ ” — (coroutine)*

:

DataStore	(,)	-	54
Room	(,)	SQLite	54
Retrofit	( )	HTTP	55

: “ DataStore Room Retrofit” SharedPreferences DataStore SQL SQLite Room —

DataStore:

“ ” “ ” DataStore (Preferences):

```
val Context.dataStore by preferencesDataStore(name = "settings")

//   :
suspend fun saveTheme(isDark: Boolean) {
    context.dataStore.edit { prefs ->
        prefs[THEME_KEY] = isDark
    }
}

//   :       (Flow           )
val themeFlow: Flow<Boolean> = context.dataStore.data
```

```
    .map { it[THEME_KEY] ?: false }
```

Flow — “ ” (reactive)
Android 54

Room: “ ”

— Room SQLite “ ”
Kotlin :

```
// 1. Entity:       (     =    data class)
@Entity(tableName = "records")
data class Record(
    @PrimaryKey(autoGenerate = true) val id: Long = 0,
    val amount: Double,
    val category: String,
    val note: String,
    val date: String
)

// 2. Dao:            (        )
@Dao
interface RecordDao {
    @Query("SELECT * FROM records ORDER BY date DESC")
    fun getAll(): Flow<List<Record>>

    @Insert
    suspend fun insert(record: Record)

    @Delete
    suspend fun delete(record: Record)

    @Query("SELECT SUM(amount) FROM records WHERE category = :c")
    suspend fun sumByCategory(c: String): Double
}

// 3. Database:
@Database(entities = [Record::class], version = 1)
abstract class AppDatabase : RoomDatabase() {
    abstract fun recordDao(): RecordDao
}
```

“ ” — Flow

Retrofit:

... Retrofit “ ”

:

```
// 1.         :      HTTP
interface WeatherApi {
    @GET("v1/weather")
    suspend fun getWeather(@Query("city") city: String): WeatherResponse
}

// 2.
val api = Retrofit.Builder()
    .baseUrl("https://api.example.com/")
    .addConverterFactory(GsonConverterFactory.create())
    .build()
    .create(WeatherApi::class.java)

// 3.             (  coroutine)
val result = api.getWeather("  ")
```

:

1) — (NetworkOnMainThreadException) (viewModelScope.launch) 2) HTTP : Android 9 HTTPS http:// usesCleartextTraffic Manifest — HTTPS

JSON: “ ”

JSON — “ ” Retrofit “ ” Kotlin 2026:

Gson / Moshi kotlinx.serialization	reflection	(2026)

JSON

```
//       @Serializable   data class
@Serializable
data class WeatherResponse(
    val weather: List<Weather>,
    val main: Main,
    val name: String
)
```

```
// Retrofit   Converter (              Retrofit 2.11+):
// .addConverterFactory(json.asConverterFactory("application/json".toMediaType()))
```

: JSON “ ” — JSON

: “ ” Android

: “ ” Android

suspend launch (Coroutine) Kotlin — Android

: “ ” ( )

```
//   coroutine   ViewModel
viewModelScope.launch {
    val records = dao.getAll().first()  // suspend:
    //        ......
}

//        :
// suspend   —— "          "
// launch    —— "    !"
// viewModelScope —— "    "
```

“ ” 90%

: ” ”

API (hitokoto) Retrofit “ ” “ ” ( )
“ ” Room DataStore/Room/Retrofit/ — 7 8

- : DataStore Room Retrofit
- DataStore Flow:
- Room: @Entity ( ) + @Dao ( ) + @Database ( )
- Retrofit :
- : ; HTTPS
- = ; viewModelScope

54: 1 — ( )

“ ” :
“ ” —

:

“ ” :

Who ( ) What ( ) Why ( ) How ( ) Constraint ( )	 + Kotlin + Compose + Room + MVVM ; Android 8;

“ ”: 1) Room ( ); 2) ( ); 3) ( );
4) ; 5) “ ”

: “ ”

— :

```
@Entity(tableName = "records")
data class Record(
    @PrimaryKey(autoGenerate = true) val id: Long = 0,
    val type: String,        // "income"     | "expense"
    val amount: Double,      //      ( )
    val category: String,    //    :            ……
    val note: String = "",   //    (     )
```

```
    val date: String          // "2026-08-15"
)
```

“ ” Dao — :

```
@Dao
interface RecordDao {
    @Query("SELECT * FROM records ORDER BY date DESC, id DESC")
    fun getAll(): Flow<List<Record>>

    @Insert
    suspend fun insert(record: Record)

    @Delete
    suspend fun delete(record: Record)

    @Query("SELECT COALESCE(SUM(amount),0) FROM records WHERE type='income' AND
    date LIKE :month || '%'")
    suspend fun monthIncome(month: String): Double    // month       "2026-08"

    @Query("SELECT COALESCE(SUM(amount),0) FROM records WHERE type='expense' AND
    date LIKE :month || '%'")
    suspend fun monthExpense(month: String): Double
}
```

: Double?

BigDecimal “Int ” (0.1+0.2≠0.3) Double —

MVVM:

? MVVM : View ( ) , ViewModel ( ) , Model (Room)

View (UI Compose)

ViewModel ( )

Model (Room)

5: MVVM —

```
class RecordViewModel(private val dao: RecordDao) : ViewModel() {

    //    Flow      UI
    val records = dao.getAll()
        .stateIn(viewModelScope, SharingStarted.WhileSubscribed(), emptyList())
```

```
    //        :       suspend
    fun addRecord(type: String, amount: Double, category: String, note: String,
    date: String) {
        viewModelScope.launch {
            dao.insert(Record(type = type, amount = amount,
                category = category, note = note, date = date))
        }
    }

    fun deleteRecord(record: Record) {
        viewModelScope.launch { dao.delete(record) }
    }
}
```

(Compose) : val vm: RecordViewModel = viewModel() val list by vm.records.collectAsState() — MVVM + Flow + Compose

: +

LazyColumn ( ) — :

```
@Composable
fun RecordListScreen(vm: RecordViewModel) {
    val records by vm.records.collectAsState()

    Column {
        //        (  -   =   )
        SummaryCard(records)

        //
        LazyColumn {
            items(records, key = { it.id }) { record ->
                RecordRow(record) { vm.deleteRecord(record) }
            }
        }

        //     "    "
        FloatingActionButton(onClick = { showAddDialog = true }) {
            Text("+")
        }
    }
}

@Composable
fun RecordRow(record: Record, onDelete: () -> Unit) {
```

```
    Row(Modifier.fillMaxWidth().padding(12.dp)) {
        Text(record.category, fontWeight = FontWeight.Bold)
        Spacer(Modifier.weight(1f))
        Text(if (record.type == "income") "+${record.amount}" else "-${record.
    amount}",
             color = if (record.type == "income") Color(0xFF27AE60) else Color(0
    xFFC0392B))
    }
}
```

AlertDialog ( ) : + / +
vm.addRecord(...)

:

: “abc” (toDoubleOrNull())
“ ” :

: “ ”

:

```
//    Dao      :
@Query("UPDATE records SET amount=:amount, category=:category, note=:note WHERE id
    =:id")
suspend fun update(id: Long, amount: Double, category: String, note: String)

@Query("DELETE FROM records WHERE id=:id")
suspend fun deleteById(id: Long)

//    :            "  / " (Compose   DropdownMenu   AlertDialog)
//           —"   "
```

: ( ) ( )
— “ ”

: +

: +

“ ” : ( ); 0
( ); ( )
“ ” :

1. : (2026-08 v) ;

2. : Canvas Compose ” ” —
” ”

:

“ ViewModel ” error AI (
7 “ AI” ? “ ”)
—

- ” ”
- : Record + Dao (SQL) + Database
- MVVM: View , ViewModel , Model
- Flow + stateIn + collectAsState:
- LazyColumn ; AlertDialog
- (/ /) + (,)

55: 2 — ( )

“ ” “ ” — : API JSON “ / / ” API “ ” —

API

OpenWeatherMap QWeather — OpenWeatherMap:

1. API keys Key (apikey);
2. Current Weather ( ) : api.openweathermap.org/data/2.5/weather?q=beijing&appid= &units=metric;
3. JSON — ” ” API

:

API Key “ ” GitHub : local.properties (gitignore) buildConfig Gradle “best practice API Key Android”

: Retrofit + JSON

“ ” JSON API :

```
// JSON: {"weather":[{"description":"   "}],"main":{"temp":26.5,"humidity":40},"
    name":"  "}
data class WeatherResponse(
    val weather: List<Weather>,
    val main: Main,
    val name: String
)
data class Weather(val description: String)
data class Main(val temp: Double, val humidity: Int)
```

```
interface WeatherApi {
    @GET("data/2.5/weather")
    suspend fun getWeather(
        @Query("q") city: String,
        @Query("appid") key: String,
        @Query("units") metric: String = "metric",
        @Query("lang") lang: String = "zh_cn"
    ): WeatherResponse
}
```

Gson JSON data class —

: / /

“ ” ... sealed class ( )

:

```
sealed class UiState {
    object Loading : UiState()                    //
    data class Success(val data: WeatherResponse) : UiState()  //
    data class Error(val message: String) : UiState()          //
}

//   ViewModel:
fun search(city: String) {
    viewModelScope.launch {
        _state.value = UiState.Loading
        _state.value = try {
            UiState.Success(api.getWeather(city))
        } catch (e: Exception) {
            UiState.Error("            ?")
        }
    }
}

//    :
when (val s = state) {
    is UiState.Loading -> CircularProgressIndicator()   //
    is UiState.Success -> WeatherContent(s.data)        //
    is UiState.Error -> Text(s.message)                 //
}
```

Loading/Success/Error “ ”

:

( ):

```
@Composable
fun WeatherContent(data: WeatherResponse) {
    Column(
        Modifier.fillMaxSize().padding(24.dp),
        horizontalAlignment = Alignment.CenterHorizontally
    ) {
        Text(data.name, fontSize = 28.sp, fontWeight = FontWeight.Bold)
        Text(
            "${data.main.temp}°C",
            fontSize = 72.sp,
            fontWeight = FontWeight.Light
        )
        Text(
            data.weather.firstOrNull()?.description ?: "    ",
            fontSize = 20.sp,
            color = Color.Gray
        )
        Spacer(Modifier.height(16.dp))
        Row {
            Text("     ${data.main.humidity}%")
            Spacer(Modifier.width(24.dp))
            Text("      ${data.main.temp}°C")
        }
    }
}
```

: firstOrNull() ( ) ?: “ ” — null safety
72.sp “ ” (Light)

- : TextField (action ” ” Enter);
- : (26.5°) ( ) ;
- : DataStore — 6 ;
- : Compose PullToRefreshBox (Material 3) ” ”

:

emoji () map when(description) —
emoji

:

1) ; 2) error (!); 3) ; 4)
; 5)

- API JSON —” ”
- API Key local.properties repository
- Retrofit + data class: JSON
- Loading/Success/Error
- — error

56: — Google Play

? “ ” “ ” :

Google Play —

: Logcat, Breakpoint, UI

Logcat Layout Inspector	log /	Log.d(”TAG”, ” =$x”) TAG De- bug Tools Layout Inspector

: error?

Logcat stack trace (Stacktrace) (com.example.yourapp) “ ” “ Exception” (null pointer? ?) “at ... : ” crash 80%

:

— “ ” “ ” Android

3 :

	( )	app/src/test/ app/src/androidTest/ ——	

```
//      "              "
fun parseAmount(input: String): Double? {
    val v = input.trim().toDoubleOrNull()
    return if (v != null && v > 0) v else null
}

//      (JUnit4)
@Test
fun emptyInputReturnsNull() {
    assertEquals(null, parseAmount(""))
}
@Test
fun negativeReturnsNull() {
    assertEquals(null, parseAmount("-5"))
}
@Test
fun normalValueParses() {
    assertEquals(12.5, parseAmount("12.5"))
}
```

- : test Run

:

Android — “ ” Android Studio
debug key release key :

1. Build Generate Signed Bundle / APK;
2. keystore ( ) — ;
3. release + AAB .

: keystore =

keystore
— keystore (
+) Android

Google Play

AAB

(~)

6: Google Play

- : 25 Google ;
- : (512×512), (2), , ( );
- : — - “ ” ;
- : .

: Google Play (Huawei, Xiaomi, Yingyongbao) —AAB/APK

:

“ ” 3 :

- : 512×512 AI ;
- : 2 4-6 (“ ” “ ”);
- : “ + ” “ ” — .

:

10 : ? ? ? —

crash

:

4

1. LazyColumn: Column —lazy loading An-
   droid;
2. : Coil 5MB ;
3. main thread: coroutine ;
4. Profiler : Android Studio CPU —
   .

crash

Firebase Crashlytics (Google):
stack trace “ ” crash

: “ ” -

(1) 3 parseAmount ; (2) Logcat crash ; (3)
APK release (“ ”); (4) -

- : Logcat log, , Layout Inspector
- crash: stack
- 3 : ( ) / /
- : keystore “ ”
- : AAB

57: —

“ ” “ ” :

— *2026* *AI* *Android*

:

, , , … “ ” Android (Android 6)
: (Manifest) ( ) Compose rememberPermissionState
Accompanist :

```
//      Manifest   :
// <uses-permission android:name="android.permission.CAMERA" />

@Composable
fun CameraButton() {
    val permission = rememberPermissionState(
        permission = android.Manifest.permission.CAMERA
    )
    Button(onClick = {
        when {
            permission.status.isGranted -> openCamera()
            permission.status.shouldShowRationale -> showExplainDialog()
            else -> permission.launchPermissionRequest()
        }
    }) { Text("   ") }
}
```

: ?

“ ” launchPermissionRequest (Intent
) “ ” —

:

“ ” 3 :

:

```
// Android 13+      POST_NOTIFICATIONS    (     )

// 1.           (Android 8+   )
val channel = NotificationChannel(
    "reminder", "  ", NotificationManager.IMPORTANCE_DEFAULT
)
val manager = getSystemService(NotificationManager::class.java)
manager.createNotificationChannel(channel)

// 2.
val notification = NotificationCompat.Builder(this, "reminder")
    .setSmallIcon(R.drawable.ic_notification)
    .setContentTitle("     ")
    .setContentText("         ")
    .setPriority(NotificationCompat.PRIORITY_DEFAULT)
    .build()
if (ActivityCompat.checkSelfPermission(this,
        Manifest.permission.POST_NOTIFICATIONS) == PackageManager.
    PERMISSION_GRANTED
) {
    manager.notify(1, notification)
}
```

(“ 20:00”) WorkManager — “ ”

: dependency injection

:

- object : new new API — Hilt (dependency injection) “ ” ;
- : — (app / core / feature-xxx) .

:

Hilt - + dependency — “ ” AI

2026: AI Android

2026 AI Android :

	AI	
	Gemini Android Studio / AI error AI AI 3 “ / / ” stack trace + AI	

4 AI Android

— AI — 11 7 “ AI” (37-47)
: Android AI 7

Compose

Compose “recompose ” :

- : @Immutable data class derivedStateOf re-compose ;
- key: items(list, key = { it.id }) — key ;
- Compose Layout Inspector: Composable recompose recompose .

Compose compiler (Metrics) “ ” —

: “ ” -

(1) “ 20:00” (WorkManager + notification channel + An-droid 13); (2) API inject Hilt; (3) AI “ ” 3 “ ”

- : Manifest + +
- 3 : ; channel
- WorkManager

- Hilt object, —
- AI 4 : ; 7 (37-47)

58:

“ ” :

	var x by remember { mutableStateOf( ) }
	rememberSaveable { mutableStateOf( ) }
	`Column { }/Row { }`
	LazyColumn { items( ) { ... } }
	navController.navigate(”route”)
	`viewModelScope.launch { ...}`
	`val list by flow.collectAsState()`
	context.dataStore.edit { it[KEY] = }
	Room: @Entity / @Dao / @Database
	Retrofit + suspend
	NotificationChannel + NotificationCompat.Builder
	`Log.d("TAG", "message")`

Android (2026)

(FAQ)

Q1: Android ?

: 1-2 Kotlin; 3-4 Compose ; 5-6 - ; 7-8 1-2

Q2: Java ?

Kotlin Java “ ” Kotlin
Java —

Q3: ?
- ( ) —

Q4: Android ?
2026: “ ” “ Kotlin + + ”

Q5: error ?
: (1) error StackOverflow/ ; (2) ; (3)
“ + + error + ” AI

	developer.android.com Kotlin (kotlinlang.org) Google Codelabs Stack Overflow (android) Juejin / CSDN Android Google Developers 7 “ AI” (37-47)	Android Compose/Kotlin + error ( ) AI Android

Android (2026)

Gradle

(AI CI):

`./gradlew assembleDebug`	APK (Windows ./)
`./gradlew assembleRelease`	( )
`./gradlew test`	
`./gradlew lint`	static:
`./gradlew clean`	(“ ”)
`./gradlew :app: dependencies`	dependency (conflict)

Gradle

: clean

“ ” ./gradlew clean

Android Studio — File Invalidate Caches

: N

: N

“ Activity ” Kotlin Compose

—

: AI “ ” :

” ” —

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4. Google Play. (2026). Play Console Help: Publishing and App Signing.
5. AndroidX Release Notes. (2026). Room / DataStore / Navigation / WorkManager releases.
6. Square. (2026). Retrofit Documentation. github.com/square/retrofit.

9. iOS (59-69)

59: iOS — Apple

— “ *iPhone*” “ *iPhone*” *: iOS ?*
“ ” *Apple ? ?*

iOS:

2007 iPhone — 2008 Apple
App Store iPhone SDK iPhone “ ” : App Store

: iPhone OS iOS 2010; iOS 17 (2023) 2025 Apple
—iOS 26 2025 2026 iOS 27 :
iOS — iOS Android

App Store	2008	
Swift	2014	iOS
SwiftUI	2019	UI declarative
iOS 26 ( )	2025	; Apple Intelligence
iOS 27	2026	( )

iOS (2026)

: “ ” “ ” Apple

Apple “ ” (Walled Garden) — : Mac
Xcode App Store ? :

- : iPhone — 5 95% (An-
droid);

- : API “ ”
;
- : App Store ;
- : .

:

iOS : Mac (Xcode macOS) 99 / (
) — 60

: Objective-C Swift?

iOS “ ” Objective-C — 2008 2014; Swift Apple
2014 Swift 6 (2024 concurrency) 2026 Swift

: Swift Objective-C — legacy code
Swift: (header C), Optional “crash null pointer”
, SwiftUI Swift Playgrounds iPad
Objective-C ( )

```
NSString *name = @"   "; [name stringByAppendingString:@"  "]; //       +      +
```

Swift ( )

```
var name = "   " name += "   " //
```

?

iOS Apple “ ” :

(UI+)
(Swift + SwiftUI)

(+)
Apple
App Store

1: iOS (“ ”)

:

- (Signing): iOS “ ” — “ ” Xcode Apple ID 67
- (Simulator): iPhone Mac - CPU Mac Android
- TestFlight: “ ” Apple — 10000 TestFlight iOS

iOS: “ ”

	(2026)	
	Swift 6 SwiftUI + Apple (HIG) MVVM + @Observable SwiftData ( ) + UserDefaults ( ) URLSession + async/await + Codable Xcode Build System / xcodebuild Xcode (macOS)	( ) IDE Apple

iOS 2026

: Android — Apple “ ”

:

: “ Apple”

[ ] [ ] [ ] iPhone : iOS, , “ ” App Store — “ ” “ ”

- 2008 App Store “ ” ; 2026 iOS 27
- :
- : Mac; 99 / ;
- Swift ; Objective-C legacy code
- Android : Apple; TestFlight
- 2026: Swift 6 + SwiftUI + MVVM + SwiftData + URLSession + Xcode

60: —Xcode

“ ” : *Xcode* *iOS*
—

Mac

: iOS macOS Mac : (Mac mini Mac-Book Air), Mac , 8 “ Android” (Windows)

		( )
	Apple Silicon (M1) 8GB 40GB+ (Xcode) macOS	M2 16GB 256GB

Mac (2026)

: Mac

Swift Playgrounds iPad Swift (Mac); iPhone
61 Playgrounds Mac “ ”

Xcode

Xcode IDE Apple :

1. App Store Mac ”Xcode” — GB ;
2. ( );
3. Xcode Settings Locations Command Line Tools —
.

: Xcode

Xcode macOS Mac App Store Apple Xcode
( ) — iOS “ ”

Xcode Create New Project iOS App :

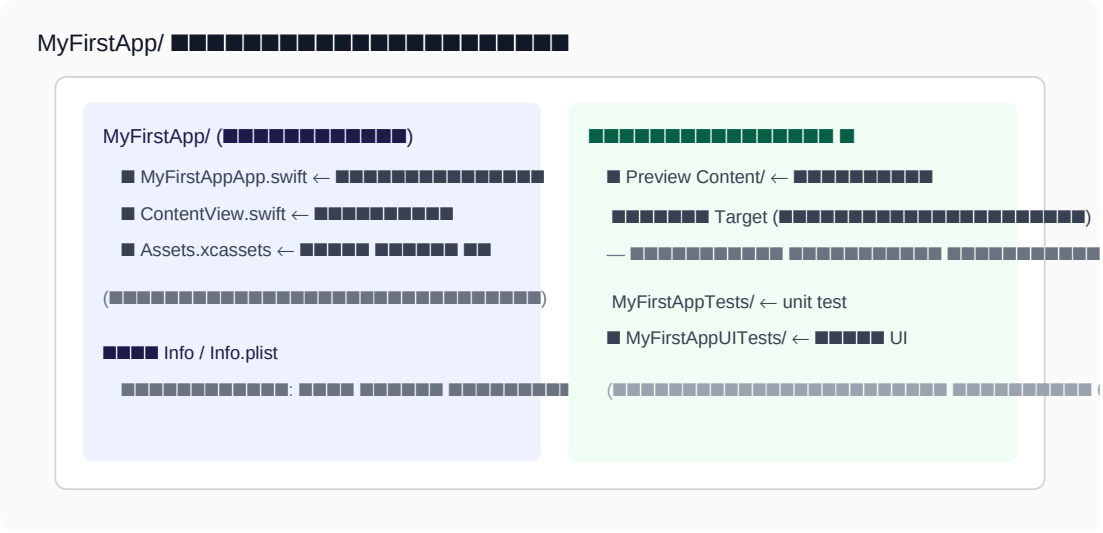
Product Name	MyFirstApp	
Team	Apple ID ( )	
Organization Identifier	com.example	Product Name
Interface	SwiftUI	
Language	Swift	

5

Create Xcode (iPhone 17) > —
“Hello, world!” iOS

:

:



2: Xcode

- MyFirstAppApp.swift: SwiftUI — “ ” ;
- ContentView.swift: ;
- Assets.xcassets: — ;
- Target: iOS .

: iPhone

Xcode: “ ” 5

“ ” :

/	CmdR OptionCmd CtrlI CmdO Cmd/	(!) -

Xcode ( )

: (iPhone iPad) (
Snippets) —

: iPhone

Apple ID :

1. iPhone (Wi-Fi);
2. Xcode Apple ID ;
3. iPhone [ ] [ ] [VPN] ;
4. > .

: “ 7 ”

Apple ID 7 ( ) Apple ;
(99 /) 67

+ error

+ error

ContentView.swift Text(“Hello, world!”) CmdR —
“ ”

error		
error Signing / Xcode /	Team App Store	Target Signing ; Device Erase All Content; ; .xip Apple

error

: “ ”

(1) Xcode ; (2) ; (3) Hello ; (4) Target “ ”

- : Mac + Xcode; Mac Swift Playgrounds iPad
- Xcode App Store;
- 5 : , Team, , SwiftUI, Swift
- : MyFirstApp/; Target
- Apple ID 7 ;
- = CmdR

61: Swift — Apple

—Swift “ ” : 20% 80% : > >

:

“ + ” Swift let var:

```
let name = "   "        // let:    (          )
var age = 18             // var:    (     )

age = 19                 // <i class="fas fa-check-circle mr-1"></i> var
// name = "  "          // <i class="fas fa-times-circle mr-1"></i> let
    error

let score: Int = 100     //
let pi = 3.14            //

print("   \(name)")   //     :          +
```

:

Int		42
Double		3.14
String		” ”
Bool	/	true / false
[Int] ( )		[1, 2, 3]

Swift

: let

Swift: let var let var Xcode
“ ” —

Optional: “ ” Swift

Swift : “ ” Swift — “crash null pointer”

```
var name: String = "   "   //    :     nil
// name = nil                // <i class="fas fa-times-circle mr-1"></i>


var nickname: String? = nil  //         :       nil (optional)
nickname = "    "

//   optional    "    "   :
if let n = nickname {        // if let:
    print("   \(n)")
} else {
    print("     ")
}

let len = nickname?.count ?? 0   // ??       (nil coalescing)
print(len)
```

: ! “ ”

nickname! (force unwrap) “ nil”— :
if let / ?? ! Xcode !

closure: “ ”

closure “ ” :

```
//    :
func greet(name: String) -> String {
    return "   \(name)!"
}
print(greet(name: "   "))     //

//    :
let numbers = [1, 2, 3, 4]
let doubled = numbers.map { $0 * 2 }   // $0 =
// [2, 4, 6, 8]
```

```
let big = numbers.filter { $0 > 2 }    //
// [3, 4]
```

SwiftUI closure : Button(“ ”) { ... } closure map, filter, sorted — 90%

struct protocol: “ ” Swift

Swift : struct class inheritance struct “value type” — :

```
// struct:
struct Record {
    var id = UUID()
    var amount: Double
    var category: String
    var date: Date

    // struct  "memberwise initializer"    :
}
let r = Record(amount: 25.5, category: "   ", date: Date())

// protocol:     "           "
protocol Identifiable {
    var id: UUID { get }
}
struct Record: Identifiable { ... }   //    protocol =
```

SwiftUI struct (View protocol) 62 — Swift struct

/	let x = 1/var x = 1 ” \(name)” func (:) -> { ... } if { ... } else { ... } for x in { ... } if let x = { ... } x ??
struct	struct X { var a: Int; var b: String }
Closure	{ $0 * 2 }

: Swift Playgrounds 15

Mac Xcode Playground; Mac Swift Playgrounds iPad
error Xcode 15 ’ ’ 62

- let var ; (:)
- Optional Swift: if let / ?? !
- ; Closure SwiftUI
- struct Protocol; SwiftUI struct
- 20% 80%:

62: — SwiftUI

SwiftUI *(declarative)* *Apple* *2019* —
’ ’ *2026* *SwiftUI*
’ ’

SwiftUI:

’ ’: ’ (20,30) ’ SwiftUI ’ ’:
’ “ ” ,

```
import SwiftUI

struct ContentView: View {
    var body: some View {      // body:     "          "
        VStack {               //
            Text("      !")
                .font(.largeTitle)      //      :
            Button("   ") {
                print("    ")
            }
            .buttonStyle(.borderedProminent)  //
        }
        .padding()
    }
}
```

: = struct View, body , .modifier()
SwiftUI

:

Text Button TextField Image		Text(" ") Button(" ") { } TextField(" ", text: $) `Image(systemName: "star.fill")`

4 SwiftUI

Image(systemName:) —Apple SF Symbols ( ) “star.fill” iOS:

: VStack, HStack, ZStack

- VStack: ( )
- HStack: ( )
- ZStack: ( )

```
VStack(alignment: .leading, spacing: 12) {
    HStack {
        Text(" ")
        Spacer()                  //     :
        Text("-25.5   ").foregroundColor(.red)
    }
    Divider()                     //
    Text("  :     ")
}
.padding(16)                      //
.background(Color(.systemGray6)) //
.cornerRadius(12)                 //

//          : padding       background
```

: modifier

padding(16).background( ) background( ).padding(16) — ’ ’ ( ) ’ ’ ( ) ’ ’

: ’ ’

SwiftUI :

```
struct CounterView: View {
    @State private var count = 0   // @State:

    var body: some View {
        VStack {
            Text("     \(count)   ")
            Button("    +1") {
                count += 1          //
            }
        }
    }
}
```

’ ’ — $ ’ ’ :

```
struct InputView: View {
    @State private var text = ""

    var body: some View {
        TextField("     ", text: $text)  // $
            .textFieldStyle(.roundedBorder)
        Text("    : \(text)")   //
    }
}
```

(count += 1)

SwiftUI

UI (body)

3: ’ ’ SwiftUI

: ’ ’ SwiftUI

- #Preview: #Preview { ContentView() } —
- : (Color(.systemGray6), .primary)
- : withAnimation { count += 1 } .animation(.default, value: count) —

```
struct AnimatedHeartView: View {
    @State private var liked = false

    var body: some View {
        Button {
            withAnimation(.spring(duration: 0.4)) {
                liked.toggle()
            }
        } label: {
            Image(systemName: liked ? "heart.fill" : "heart")
                .font(.system(size: 48))
                .foregroundColor(liked ? .red : .gray)
                .scaleEffect(liked ? 1.3 : 1.0)
        }
    }
}
```

:

’ ’ —

’ ’

: ’ ’

modifier :

	Modifier
(/)	.padding(16) .cornerRadius(12) .background(Color(.systemGray6)) .font(.title.bold())/.font(. system(size: 20)) .foregroundColor(.red) .frame(maxWidth: .infinity) .frame(width: 100, height: 44) VStack(alignment: .leading) .opacity(0.6) .shadow(radius: 5) .onTapGesture { ...} .keyboardType(.decimalPad)

modifier SwiftUI

— : padding, , ,

: :

(+), @State (index + 1) % 3 “ ”

- SwiftUI : ’ ’
- = struct View; body ; modifier
- : Text / Button / TextField / Image; SF Symbols
- : VStack , HStack , ZStack ; Spacer
- @State ; $
- #Preview ; ; withAnimation

63:

’ ’: ( ) ( )
, ,

, ,

SwiftUI @State ’ ’:

@State @Binding @StateObject / @Observable @EnvironmentObject / .environment		(ViewModel)	, , ,

SwiftUI (2026 @Observable)

```
//       iOS 17+:     @Observable (          )
import Observation

@Observable
class CounterModel {                //
    var count = 0
    func increment() { count += 1 }
}

struct CounterView: View {
    @State private var model = CounterModel()   //

    var body: some View {
        VStack {
            Text("     \(model.count)   ")
            Button("   ") { model.increment() }
        }
```

```
    }
}
```

: @State @Observable @StateObject/@ObservableObject

NavigationStack: ' '

iOS 16 NavigationStack — :

```
struct HomeView: View {
    var body: some View {
        NavigationStack {
            List {
                NavigationLink("      ") {
                    DetailView(name: "      ")   //
                }
                NavigationLink("     ") {
                    DetailView(name: "     ")
                }
            }
            .navigationTitle("      ")   //
        }
    }
}

struct DetailView: View {
    let name: String
    var body: some View {
        Text("         \(name)")
            .navigationTitle(name)
            .navigationBarTitleDisplayMode(.inline)
    }
}
```

NavigationLink ' ' navigationTitle (' ') @Environment(\.dismiss) — 90%

: sheet TabView

' ' :

```
// 1.     /        (      )
struct HomeView: View {
    @State private var showAdd = false
    var body: some View {
```

```
        Button("     ") { showAdd = true }
            .sheet(isPresented: $showAdd) {
                AddRecordView()     //
            }
    }
}

// 2.         ("  "    )
struct MainTabView: View {
    var body: some View {
        TabView {
            RecordListView()
                .tabItem { Label("    ", systemImage: "book") }
            StatsView()
                .tabItem { Label("  ", systemImage: "chart.pie") }
            SettingsView()
                .tabItem { Label("   ", systemImage: "gear") }
        }
    }
}
```

: ?

: NavigationStack ( ), sheet (, ,), TabView ( )

List: ’ iOS’

’ Apple’ iOS List — :

```
struct RecordListView: View {
    @State private var records = ["     25.5", "     6.0", "     +8000"]

    var body: some View {
        List {
            ForEach(records, id: \.self) { r in
                Text(r)
            }
            .onDelete { indexSet in          //
                records.remove(atOffsets: indexSet)
            }
        }
    }
}
```

+ NavigationLink + sheet + TabView — ’

+ ’

: ’ ’

:

Activity Android iOS ’ , , , ’ SwiftUI scenePhase:

```
@main
struct MyApp: App {
    @Environment(\.scenePhase) private var phase

    var body: some Scene {
        WindowGroup {
            ContentView()
        }
        .onChange(of: phase) { _, newPhase in
            switch newPhase {
            case .active:    print("            ")
            case .inactive:  print("       (   /      )")
            case .background: print("          ")
            @unknown default: break
            }
        }
    }
}
```

: iOS — ( )

68 ’ ’

: TabView (/ /) List

(NavigationStack) + sheet — 65

- : @State ( ), @Binding ( ), @Observable ( ), .environment ( )
- @Observable ; @ObservableObject
- NavigationStack ; @Environment(\.dismiss)
- sheet TabView
- List ’ iOS’:

64:

, , *, , iOS: UserDefaults (*
), SwiftData ( ), URLSession ( ) Codable JSON async/await

:

UserDefaults	(,)	-	65, 66
SwiftData	(,)	@Model	65
URLSession	( )		66

’ UserDefaults SwiftData URLSession’ Core Data
(SwiftData) SQLite —

UserDefaults:

```
//   :
UserDefaults.standard.set(true, forKey: "isDarkMode")
UserDefaults.standard.set("  ", forKey: "favoriteCity")

//   :       (       )
let isDark = UserDefaults.standard.bool(forKey: "isDarkMode")
let city = UserDefaults.standard.string(forKey: "favoriteCity") ?? "  "

//   :
UserDefaults.standard.removeObject(forKey: "isDarkMode")
```

— SwiftData UserDefaults
( )

SwiftData: ’ ’

SwiftData iOS 17 ’ ’ — @Model
:

```
import SwiftData

// 1. @Model:         "   "
@Model
class Record {
    var amount: Double
    var category: String
    var note: String
    var date: Date

    init(amount: Double, category: String, note: String, date: Date) {
        self.amount = amount
        self.category = category
        self.note = note
        self.date = date
    }
}

// 2.
@main
struct MyApp: App {
    var body: some Scene {
        WindowGroup {
            ContentView()
        }
        .modelContainer(for: Record.self)   //
    }
}

// 3.      @Query   , @Environment
struct RecordListView: View {
    @Query(sort: \Record.date, order: .reverse) var records: [Record]
    @Environment(\.modelContext) private var context

    var body: some View {
        List(records) { r in
            Text("\(r.category) \(r.amount)   ")
        }
    }

    func add(amount: Double, category: String) {
        let r = Record(amount: amount, category: category, note: "", date: Date())
```

```
        context.insert(r)     //
        try? context.save()   //
    }
}
```

’@Model + modelContainer + @Query’ —
@Query

Codable: ’ ’ JSON

JSON Codable Swift ’JSON struct’ —
:

```
// JSON: {"name":"  ","main":{"temp":26.5},"weather":[{"description":"   "}]}
struct WeatherResponse: Codable {
    let name: String
    let main: Main
    let weather: [Weather]
}
struct Main: Codable { let temp: Double }
struct Weather: Codable { let description: String }

//       JSON  struct
let decoder = JSONDecoder()
let data = jsonString.data(using: .utf8)!
let result = try decoder.decode(WeatherResponse.self, from: data)
print(result.weather[0].description)   // "   "
```

: JSON ’ ’ (“26.5”
Double) — JSON

URLSession + async/await:

iOS URLSession async/await ( ) Swift
’ ’ ’ ’:

```
struct WeatherService {
    //     async: async     "     " throws     "     "
    func fetchWeather(city: String) async throws -> WeatherResponse {
        let url = URL(string:
            "https://api.example.com/weather?city=\(city)")!
        let (data, _) = try await URLSession.shared.data(from: url)
        return try JSONDecoder().decode(WeatherResponse.self, from: data)
    }
}
```

```
//     (  .task):
.task {
    do {
        let weather = try await service.fetchWeather(city: "  ")
        //      ……
    } catch {
        print("      : \(error)")
    }
}
```

:

(1) async/await — Swift try await do/catch (2) ATS HTTP : iOS HTTPS http:// Info.plist (NSAppTransportSecurity) HTTPS —

: ’ ’

’ ’: error

, , 404, 500... :

```
func friendlyMessage(_ error: Error) -> String {
    if let urlError = error as? URLError {
        switch urlError.code {
        case .notConnectedToInternet: return "              ?"
        case .timedOut:               return "               ?"
        case .cannotFindHost:         return "        "
        default:                      return "    : \(urlError.code)"
        }
    }
    if let d = error as? DecodingError {
        return "         (     API    )"
    }
    return "      : \(error.localizedDescription)"
}
```

, , , , (error) error.localizedDescription —

API (hitokoto) URLSession + async/await , , , , , , SwiftData UserDefaults/SwiftData/Codable/URLSession — 7 8

- : UserDefaults SwiftData URLSession
- SwiftData: @Model , modelContainer , @Query + con-

text.insert

- Codable’ ’JSON :
- URLSession + async/await: ; do/catch
- : ; HTTPS (ATS)

65: 1 — ( )

, , :
, , —

:

Who ( )	
What ( )	
Why ( )	+
How ( )	Swift + SwiftUI + SwiftData + MVVM
Constraint ( )	; iOS 17 (SwiftData)

, ': (1) SwiftData ( ) (2) ( ) (3) ( ) (4) (5) , ,

: , ,

— :

```
import SwiftData

@Model
class Record {
    var type: String      // "income"    | "expense"
    var amount: Double    //      ( )
    var category: String  //    :            ……
    var note: String      //    (     )
    var date: Date        //

    init(type: String, amount: Double, category: String, note: String = "", date:
    Date = Date()) {
```

```
        self.type = type
        self.amount = amount
        self.category = category
        self.note = note
        self.date = date
    }
}
```

: Double?

Decimal ( ) (0.1+0.2≠0.3) Double
Decimal 'Int ' —

MVVM:

? MVVM : View ( ) ,
ViewModel ( ) , Model (SwiftData)

View (UI SwiftUI)

ViewModel (@Observable)

Model (SwiftData)

4: MVVM —

```
import Observation

@Observable
class RecordViewModel {
    var records: [Record] = []
    private let context: ModelContext

    init(context: ModelContext) { self.context = context }

    func load() {
        //          #Predicate    SwiftData
        let descriptor = FetchDescriptor<Record>(
            sortBy: [SortDescriptor(\.date, order: .reverse)]
        )
        records = (try? context.fetch(descriptor)) ?? []
    }

    func add(type: String, amount: Double, category: String, note: String) {
        let r = Record(type: type, amount: amount, category: category, note: note)
        context.insert(r)
        try? context.save()
        load()                          //
```

```
    }

    func delete(_ record: Record) {
        context.delete(record)
        try? context.save()
        load()
    }
}
```

@State private var vm = RecordViewModel(...) vm.records

—

: +

```
struct RecordListView: View {
    @State private var vm: RecordViewModel
    @State private var showAdd = false

    var body: some View {
        NavigationStack {
            List {
                Section("        ") {          //
                    SummaryRow(records: vm.records)
                }
                Section("   ") {
                    ForEach(vm.records) { r in
                        RecordRow(record: r)
                    }
                    .onDelete { idx in
                        idx.forEach { vm.delete(vm.records[$0]) }
                    }
                }
            }
            .navigationTitle("      ")
            .toolbar {
                Button { showAdd = true } label: {
                    Image(systemName: "plus")
                }
            }
            .sheet(isPresented: $showAdd) {
                AddRecordView(vm: vm)          //
            }
        }
    }
}
```

: .keyboardType(.decimalPad) ( ), /
(Picker), , ’ ’ vm.add(...) dismiss()

:

: “abc” (Double(text) nil)
’ ’ :

- : .onDelete List + ( )
- : ( )
- : filter + reduce ’ / / ’ :

```
func monthSummary(_ records: [Record]) -> (income: Double, expense: Double) {
    let cal = Calendar.current
    let monthRecords = records.filter {
        cal.isDate($0.date, equalTo: Date(), toGranularity: .month)
    }
    let income = monthRecords
        .filter { $0.type == "income" }
        .reduce(0) { $0 + $1.amount }
    let expense = monthRecords
        .filter { $0.type == "expense" }
        .reduce(0) { $0 + $1.amount }
    return (income, expense)
}
```

: +

’ ’: ( ); 0
( ); hardcode ( )
’ ’ :

1. : ’ ’ Swift Charts / — Apple

2. iCloud: iCloud SwiftData — iPhone/iPad ’ ’

:

’ ViewModel ’ error AI
(7 ’ AI’ ? ’ ’)
—

- , ,
- : @Model + modelContainer + FetchDescriptor
- MVVM: View , ViewModel @Observable , SwiftData
- : List + Section + toolbar + sheet
- (/ /) + (Swift Charts, iCloud)

66: 2 — ( )

’ ’ ’ ’— : API JSON
’ / / ’ API ’ ’ — , ,

API

OpenWeatherMap QWeather — OpenWeatherMap :

1. API keys Key (apikey)
2. Current Weather : api.openweathermap.org/data/2.5/weather?q=beijing&appid= &units=metric
3. JSON — ’ ’ API

:

API Key ’ ’ GitHub iOS: Config.xcconfig (gitignore) Info.plist ’iOS xcconfig ’

: URLSession + Codable

JSON API :

```
// JSON: {"weather":[{"description":"   "}],"main":{"temp":26.5,"humidity":40},"
    name":"  "}
struct WeatherResponse: Codable {
    let weather: [Weather]
    let main: Main
    let name: String
}
struct Weather: Codable { let description: String }
struct Main: Codable { let temp: Double; let humidity: Int }
```

```
struct WeatherService {
    let key: String   //     xcconfig

    func fetchWeather(city: String) async throws -> WeatherResponse {
        var comps = URLComponents(string: "https://api.openweathermap.org/data/2.5/
    weather")!
        comps.queryItems = [
            URLQueryItem(name: "q", value: city),
            URLQueryItem(name: "appid", value: key),
            URLQueryItem(name: "units", value: "metric"),
            URLQueryItem(name: "lang", value: "zh_cn")
        ]
        let (data, _) = try await URLSession.shared.data(from: comps.url!)
        return try JSONDecoder().decode(WeatherResponse.self, from: data)
    }
}
```

: / /

, , ... enum + switch

:

```
enum LoadState {
    case loading                  //
    case success(WeatherResponse) //
    case error(String)            //
}

@Observable
class WeatherViewModel {
    var state: LoadState = .loading
    private let service = WeatherService(key: Config.apiKey)

    func search(city: String) async {
        state = .loading
        do {
            let data = try await service.fetchWeather(city: city)
            state = .success(data)
        } catch {
            state = .error("            ?")
        }
    }
}

//    :
switch vm.state {
```

```
case .loading: ProgressView("    …")
case .success(let d): WeatherContent(data: d)
case .error(let msg): Text(msg).foregroundColor(.secondary)
}
```

loading/success/error ’ ’

:

```
struct WeatherContent: View {
    let data: WeatherResponse

    var body: some View {
        VStack(spacing: 12) {
            Text(data.name)
                .font(.title.bold())
            Text("\(Int(data.main.temp))°C")
                .font(.system(size: 76, weight: .thin))
            Text(data.weather.first?.description ?? "   ")
                .font(.title3)
                .foregroundColor(.secondary)
            HStack(spacing: 24) {
                Label("    \(data.main.humidity)%", systemImage: "humidity")
                Label("     \(Int(data.main.temp))°C", systemImage: "thermometer")
            }
        }
        .padding()
    }
}
```

: first? ( ), ?? “ ” ; weight 76 ; Label ’ + ’ HStack

- : modifier .searchable vm.search
- : UserDefaults
- : .refreshable { await vm.search(...) }
- : Image(systemName:) (sun.max / cloud.rain / snowflake)

:

OpenWeatherMap “ ” “beijing”
’ / ’ /

:

: ,

, , , , :

1. : String(data: data, encoding: .utf8) JSON —

2. : Proxyman Mac Charles

3. : Features Network Link Conditioner ( )
3G, — ’ ’

: ATS ?

error “App Transport Security” http://
HTTPS (NSAllowsLocalNetworking Info.plist)

(1) ; (2) error (Mac !); (3)
; (4) ; (5)

- API JSON —’ ’
- API Key xcconfig
- Codable + URLSession async/await: JSON
- loading/success/error
- , , —

67: — App Store

? ’ ’ ’ *’:* *App Store* —

: print, ,

print +		print(” =\(x)”) Xcode
SwiftUI +		#Preview + View Hierarchy

: ?

Swift “Fatal error” Xcode Xcode Window Organizer Crashes ( ) ’ error + swiftui’ —

:

— ’ ’ ’ ’ iOS XCTest :

		Target	
Unit Test UI Test	(,)	MyFirstAppTests MyFirstAppUITests ——	

iOS (Test Target)

```
import XCTest

//      "              "
func parseAmount(_ input: String) -> Double? {
    guard let v = Double(input.trimmingCharacters(in: .whitespaces)),
          v > 0 else { return nil }
    return v
}

final class AmountTests: XCTestCase {
    func testEmptyInputReturnsNil() {
        XCTAssertNil(parseAmount(""))
    }
    func testNegativeReturnsNil() {
        XCTAssertNil(parseAmount("-5"))
    }
    func testNormalValueParses() {
        XCTAssertEqual(parseAmount("12.5"), 12.5)
    }
}
```

: CmdU Unit Test

: ’ ’

iOS — ’ ’ :

- : Apple ID + Xcode (Personal Team) 7
- : (99 /)

: Xcode Product Archive ( ) Organizer Distribute App App Store Connect Generic iOS Device ( )

: =

— ’ ’ (.p12)
Apple ’ ’ (Automatically manage signing Xcode)

App Store

Apple (1-7)

5: App Store

- : developer.apple.com 99 /
- : (1024×1024 —Apple), (iPhone), ,
- : — , ,
- : 1-7 , , — Apple
- TestFlight: (10000)

:

, , :

- : 1024×1024 AI

- : 2 (iPhone)
  (' ' , ' ')
- : ' + ' ' ' —

: ' '

:

Apple Xcode Organizer ( )
SDK (Firebase Crashlytics)

1. LazyVStack / List: VStack —
2. : AsyncImage ( ) 5MB
3. : async/await
4. Instruments : Xcode (Time Profiler,) —

(1) Unit Test parseAmount (CmdU); (2) ;
(3) Product Archive TestFlight ; (4)

- : print, , SwiftUI
- XCTest: Unit Test (CmdU) + UI Test +
- : (7);
- : Generic iOS Device + Archive + Distribute;
- : ; TestFlight

68: —

’ ’ ’ ’.

— *2026 AI iOS*

:

, , , ... ’ ’ iOS ’ + +

’:

```
// 1.            Info.plist (          !)
// Privacy - Location When In Use Usage Description: "               "

// 2.
import CoreLocation

@Observable
class LocationModel {
    private let manager = CLLocationManager()

    func request() {
        switch manager.authorizationStatus {
        case .notDetermined: manager.requestWhenInUseAuthorization()
        case .denied, .restricted: /*          */
        default: /*          */
        }
    }
}

// 3.              :      +
// UIApplication.shared.open(URL(string: UIApplication.openSettingsURLString)!)
```

: iOS ' '

iOS — : ' ' ( )
(' ')

:

' ' iOS UserNotifications :

:

```
import UserNotifications

// 1.        (       )
UNUserNotificationCenter.current()
    .requestAuthorization(options: [.alert, .sound, .badge]) { granted, _ in
        print(granted ? "    " : "    ")
    }

// 2.     "   20:00      "
func scheduleDailyReminder() {
    let content = UNMutableNotificationContent()
    content.title = "     "
    content.body = "         "
    content.sound = .default

    var comps = DateComponents()
    comps.hour = 20
    let trigger = UNCalendarNotificationTrigger(dateMatching: comps, repeats: true)

    let request = UNNotificationRequest(
        identifier: "daily-reminder",
        content: content,
        trigger: trigger
    )
    UNUserNotificationCenter.current().add(request)
}
```

: iOS ' ' —

: MVVM

65 MVVM :

- : ViewModel (protocol) — ' '

- : ViewModel ( );
- : '@Observable + ' VIPER Clean Architecture —

: ' ' SwiftUI

iOS ' ' : (WidgetKit)
(' ') (App Intents) Siri
—

2026: iOS AI

2026 AI Apple :

	AI	
	Xcode + AI error AI AI XCTest ' / / ' + AI	

iOS AI

— AI — 11 7 ' AI' (
37-47) : Swift AI 7

: ' '

Swift: @MainActor Task

Swift 6 :

- : SwiftUI @MainActor callback
(await MainActor.run { ... } ViewModel
@MainActor)
- Task: modifier .task { } —
SwiftUI Task body

```
struct WeatherScreen: View {
    @State private var vm = WeatherViewModel()

    var body: some View {
```

```
        Group {
            switch vm.state {
            case .loading: ProgressView()
            case .success(let d): WeatherContent(data: d)
            case .error(let m): Text(m)
            }
        }
        .task {                          //
            await vm.search(city: "   ")
        }
    }
}
```

’ data race’ Swift 6 — error

(1) ’ 20:00’ (+); (2) ViewModel Unit Test ; (3) ’ ’ , ,

- : Info.plist ( ), ,
- iOS : ’ ’
- : , , ; UserNotifications
- MVVM : ( ), ,
- : WidgetKit + App Intents
- AI : , , , ; 7 (37-47)

69:

’ ’:

/	@State private var x =
	`@Observable class VM { var x = 0 }`
	`VStack { }/HStack { }`
	`Spacer()`
	List { ForEach( ) { ... } }
	`NavigationStack + NavigationLink`
	.sheet(isPresented: $show) { }
	`@Environment(\.dismiss) var dismiss`

JSON	`try await URLSession.shared.`
	`data(from: url)`
	`struct X: Codable+JSONDecoder().`
	`decode`
	UserDefaults.standard.set(, forKey: ” ”)
	`@Model+modelContainer+@Query`
	(.primary / Color(.systemGray6))
	
	CmdU

iOS (2026)

FAQ

Q1: Mac iOS ?

Swift Playgrounds iPad/iPhone 61 Xcode macOS Mac Mac Mac mini

Q2: iOS ?

: 0 (Xcode + Apple ID) : 99 / (3,500)
Mac — App Store —

Q3: SwiftUI UIKit ?

SwiftUI 2026 SwiftUI UIKit
— SwiftUI UIKit ( )

Q4: iOS ?

2026: ’ ’ ’ Swift + SwiftUI + ’
iOS

Q5: ?

— Apple ( )
App Store Connect 3
(/)

Xcode

/	CmdR / CmdU
	OptionCmd
	CmdO
	CmdF
	CtrlI
/	Cmd/
	CmdK
	Cmd
( )	Refactor Rename

Xcode

:

“ ” CmdK
Xcode (Mac) “ DerivedData” — iOS

	developer.apple.com SwiftUI Tutorials Apple Swift (swift.org) Hacking with Swift Stack Overflow (swiftui) Apple Developer (WWDC) Apple Human Interface Guidelines (HIG) 7 ” AI” (37-47)	Apple Swif- tUI/Swift Paul Hudson ” Apple” AI iOS

iOS (2026)

: N

“ SwiftUI ” Xcode Swift SwiftUI
—

: AI “ ” :

” ” iPhone —
iOS

1. Apple Inc. (2026). SwiftUI Framework Documentation. developer.apple.com.
2. Apple Inc. (2026). Swift Programming Language Guide. swift.org.
3. Apple Inc. (2026). SwiftData Framework Documentation. developer.apple.com.
4. Apple Inc. (2026). App Store Review Guidelines. developer.apple.com/app-store/review.
5. Apple Inc. (2026). Human Interface Guidelines (HIG). developer.apple.com/design.
6. Apple Inc. (2026). Xcode 27 Release Notes and App Distribution Guide.

10.

A:

:

- :
- : IMEI/
- :

:

- : Wi-Fi
- : Apple ID, Huawei, Xiaomi
- : ” ”
- **eSIM**: eSIM (iPhone) eSIM eSIM eSIM
- : iPhone ” ”

:

- : App Store
- : ” ?”

B:

- :
- : WeChat [ ] - [ ] - [ ] - [ ]
- : iPhone [ ] - [ ] - [ ]
- : !

C: “ ”

	.....	
SoC /		
RAM ( )	” ”	
ROM ( )	” ”	128G, 256G
OLED / LCD		OLED LCD LCD
NFC		” ”
OTA		” ” [ ]
Root	Android	
BL Lock	Bootloader	
NPU		AI CPU
AI	AI	AI
Qi2		MagSafe
eSIM		
LTPO		(1-120Hz)

D: ?

:

- **Apple**: apple.com/cn iCloud.com
- **Android**: ” ”
- : Apple iTunes Huawei Huawei Phone Assistant Xiaomi Mi Mover

:

- : ”[ ] ” Bilibili
- :
- :

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1. ?
2. ” ” ?
3. ?
4. WeChat ?
5. :

E:

“ ” !

1.6

- ” ” ” ” (+ 72 EXIF)
- ” / ” ” ”
- AI (/ /)
- ” (doxxing)” (social engineering)
- (VoiceOver/TalkBack AI RTT/TTY)
- ” - - - ” 3
- 3 : 1: Android (zRAM/swap)
- 20: Termux — Linux Android (/ /)
- 23: GitHub — (Git GitHub Pull Request)

1.7

1. 4

- 29: — 20-20-20 ” ” 7

- 30: — AirPlay / Miracast / DLNA Samsung DeX CarPlay / Android Auto / HiCar / CarLife
- 31: — 5 ( ) 6 CapCut ” ”
- 32: —SOP 30 72 6 5

2. 2

- 27: — 6 3 (Changlian/Sunflower) Care WeChat SOS / /
- 28: — 3 ” 5 ” 5 7 5

3.

- : / / 1 0.5x-2.5x
- :
- : R ESC
- :

4. (4 5) (1.6 1.7) (5 6)

5.

- : Snapdragon 8 Elite Gen5 / Dimensity 9500 (2026)
- : iPhone 17 Pro, Samsung S26 Ultra, Xiaomi 17 Ultra, Huawei Pura 80 Ultra, vivo X300 Pro
- AI: DeepSeek R1, ChatGPT-5 NPU 80 TOPS
- : 4,000-5,000
- : 2026
- : Wi-Fi 7 / 5G-A / HarmonyOS / LE Audio

6. : (Hero)

- : SVG

- : 8 (WeChat)
- : localStorage
- : ” / / / ”

1.8

7 “ AI”(37-47 11): Agentic Engineering 5 AI Context Engineering (SDD) Requirements Engineering AI 2026 API Conversation Engineering 0 1 (MCP/ Agent) (5 6) “ AI: ” 2026

8 “ Android”(48-58 11) 9 “ iOS”(59-69 11): (Android Studio / Xcode) (Kotlin / Swift) (Jetpack Compose / SwiftUI) (+) (Google Play / App Store) (6 8) “ Android ” “ iOS ” 2026

:

- 13 6 ” ”: Qingzhuo National Anti-Fraud Center AdGuard ” / / ”
- : iOS, Xiaomi/Redmi, Huawei/Honor, OPPO/realme/OnePlus, vivo/iQOO Samsung
- 8: / SMS iOS 13
- : + + National Anti-Fraud Center

: / /

- 13 7: (+ +) (/) (≠)
- : Xiaomi, Huawei/Honor, OPPO, vivo, iPhone, Samsung

- 32 ” 0”: / SOS /
  13
- : ” ”

:

- 13 8: (Sunflower, ToDesk) AI (Trae / WorkBuddy / Dumate) Artificial Desktop miHoYo
- : LocalSend, Feige Chuanshu, AirDrop, KDE Connect, Universal Clipboard
- 2FA: Aegis, Microsoft Authenticator
- TickTick : /

—

F:

“ ” “ ”

?

:: “ ” “ ”

:

- ( ): : Huawei Mate X, Samsung Galaxy Z Fold, Xiaomi MIX Fold
- ( ): : Huawei Pocket, Samsung Galaxy Z Flip, OPPO Find N Flip

2024-2026: 500,000 (- 100 13), UTG , , 4,000-5,000

“ ” “ ” “ ”

?

“ ” :

1. / : PDF 7.5 PDF A4
2. / : PPT
3. ” ”: 90 (/) —
4. :

- : WeChat —
- : — 230g
- : 4 —

: 6

“ ” :

- : ( ) U ( )
- : UTG UTG
- : 2026 : ,
  ,
- : (6.4) (4)
- : 230-260g 230g 12-14mm 12mm
- : 2000-4000

1. - 10
2.
3. 30
4. : 90
5.

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/ : WeChat

(90)

“ ” :

- / : 90
- / :
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- : /
- + :

: / / WeChat ( )
( )

(Huawei, Samsung) “ ”

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- - : 0℃ UTG -
- / :
- :
- : UTG
- : IPX8
- - :

- : / —
- : —
- : - —
- : ” ” —

“ ”

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- : 2026 ” ” ” ”
- : 230g 12-14mm
- :
- : 50-60%
- : 2000-4000 500-1500
- : ” ”

“ ” “ ” “ ”

—

(2026)

( ) :

- **Huawei Mate X**: HarmonyOS
- **Samsung Galaxy Z Fold/Flip**: , ,Z Flip . .
- **Xiaomi MIX Fold/Flip**: , , . .
- **vivo X Fold** : , .
- **OPPO Find N** : , , .
- **Huawei Pocket** : , , .

: : HuaweiMate X; : SamsungZ Fold; : Xiaomi/vivo; HuaweiPocket SamsungZ Flip.

2026 : ,
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2 :

- : , AI , ,
  ” ” ” ” ” ” ” ” ”

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  ” ”

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“ ”

: **30** “ ” **?**

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H:

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AI 5 Context Requirements Engineering 2026 API Conversation Engineering — “ AI ” “ AI ” Android iOS: (Android Studio / Xcode) (Kotlin / Swift) (Jetpack Compose / SwiftUI) (+) (Google Play / App Store) —

(v1.9)

- : Apple Watch Huawei (Oura/Galaxy Ring)
- : eSIM
- :
- AI: CapCut AI, Kling AI, (Digital Human) AI
- :

2026 AI